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ADNOC

ADNOC GROUP PROJECTS AND ENGINEERING

DRAINAGE SYSTEMS - DESIGN AND CONSTRUCTION SPECIFICATION

Specification

APPROVED BY:

Abdulmunim Saif Al Kindy
NAME: Abdulmunim Al Kindy
TITLE: Executive Director PT&CS
EFFECTIVE DATE:

AGES-SP-01-007

ص ب 898، أبوظبي، الإمارات العربية المتحدة
PO Box 898, Abu Dhabi, UAE
adnoc.ae

شركة بترول أبوظبي الوطنية
Abu Dhabi National Oil Company

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GROUP PROJECTS & ENGINEERING / PT&CS DIRECTORATE

CUSTODIAN	Group Projects & Engineering / PT&CS
ADNOC	Specification applicable to ADNOC & ADNOC Group Companies

REVISION HISTORY

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In addition, Group Projects & Engineering is responsible for communication and distribution of any changes to this Specification and its version control.

This specification will be reviewed and updated in case of any changes affecting the activities described in this specification.

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INTER-RELATIONSHIPS AND STAKEHOLDERS

a. The following are inter-relationships for implementation of this Specification:

   i. ADNOC Upstream and ADNOC Downstream Directorates; and

   ii. ADNOC Onshore, ADNOC Offshore, ADNOC Sour Gas, ADNOC Gas Processing. ADNOC LNG, ADNOC Refining, Fertil, Borouge, Al Dhafra Petroleum, Al Yasat

b. The following are stakeholders for the purpose of this Specification:

   i. ADNOC PT&CS Directorate

c. This Specification has been approved by the ADNOC PT&CS is to be implemented by each ADNOC Group company included above subject to and in accordance with their Delegation of Authority and other governance-related processes in order to ensure compliance.

d. Each ADNOC Group company must establish/nominate a Technical Authority responsible for compliance with this Specification.

DEFINITIONS

"**ADNOC**" means Abu Dhabi National Oil Company.

"**ADNOC Group**" means ADNOC together with each company in which ADNOC, directly or indirectly, controls fifty percent (50%) or more of the share capital.

"**Approving Authority**" means the decision-making body or employee with the required authority to approve Policies & Procedures or any changes to it.

"**Business Line Directorates**" or "**BLD**" means a directorate of ADNOC which is responsible for one or more Group Companies reporting to, or operating within the same line of business as, such directorate.

"**Business Support Directorates and Functions**" or "**Non- BLD**" means all the ADNOC functions and the remaining directorates, which are not ADNOC Business Line Directorates.

"**CEO**" means chief executive officer.

"**Group Company**" means any company within the ADNOC Group other than ADNOC.

"Specification" means this Drainage Systems – Design and Construction specification.

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1 GENERAL

1.1 Introduction

This Specification establishes the minimum requirements for the design and engineering of gravity-based drainage systems for onshore facilities. The drainage systems addressed are Clean Water System (CWS), Accidently Oil Contaminated (AOC), Continuously Oil Contaminated (COC) and Sanitary Sewer (SS). It also includes the minimum material and installation requirements for these drainage systems.

This specification should be read in conjunction with the project specific drainage philosophy.

1.2 Purpose

This Specification contains the minimum criteria for the classification, design and engineering of site drainage runoff and collection systems including all hydraulic calculations for rainfall and firefighting capabilities. It also prescribes the requirements for the materials and installation of underground sewer and drainage pipes, with the necessary fittings, manholes, catch basins, sumps, pits, traps and lined ditches.

Excluded from this Specification are the design and construction requirements for pressurised drainage systems, closed drain systems, cryogenic drainage systems and offshore platform drainage systems. Also excluded from this Specification are requirements related to the secondary treatment facilities such as sewage treatment facilities, waste water treatment facilities, etc.

1.3 Definitions and Abbreviations

1.3.1 Definitions

The following defined terms are used throughout this Specification:

'**[PSR]**' indicates a mandatory Process Safety Requirement

"**COMPANY**" means ADNOC, ADNOC Group or an ADNOC Group Company and includes any agent or consultant authorized to act for, and on behalf of the COMPANY.

"**CONTRACTOR**" means the parties that carry out all or part of the design, engineering, procurement, construction, commissioning or management of the PROJECT. CONTRACTOR includes its approved MANUFACTURER(s), VENDOR(s), SUPPLIER(s) and SUBCONTRACTOR(s).

"**MANUFACTURER**" means the Original Equipment Manufacturer (OEM) or MANUFACTURER of one or more of the component(s) which make up a sub-assembly or item of equipment assembled by the main SUPPLIER or his nominated SUB-SUPPLIER.

'may' means a permitted option

'shall' indicates mandatory requirements

'should' means a recommendation

"**SUB-CONTRACTOR**" means any party engaged by the CONTRACTOR to undertake any assigned work on their behalf. COMPANY maintains the right to review all proposed SUB-CONTRACTORs; this right does not relieve the CONTRACTOR of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and the SUB-CONTRACTOR.

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"SUPPLIER" means the party entering into a Contract with COMPANY to provide the materials, equipment, supporting technical documents and/or drawings, guarantees, warranties and/or agreed services in accordance with the requirements of the purchase order and relevant specification(s). The term SUPPLIER includes any legally appointed successors and/or nominated representatives of the SUPPLIER.

"**SUB-SUPPLIER**" means the sub-contracted SUPPLIER of equipment sub-components software and/or support services relating to the equipment / package, or part thereof, to be provided by the SUPPLIER. COMPANY maintains the right to review all proposed SUB-SUPPLIERS, but this right does not relieve the SUPPLIER of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and any individual SUB-SUPPLIER.

1.3.2 Abbreviations

The abbreviations used throughout this Specification are shown in Table 1.1.

Table 1.1 List of Abbreviations

Abbreviations	
AOC	Accidentally Oil Contaminated
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
BS	British Standards
BSI	British Standards Institution
BWS	Boiler Water Sewer
CCTV	Closed Circuit Television
CD	Closed Drain
COC	Continuously Oil Contaminated
CS	Carbon Steel
CWS	Clean Water System
DIP	Ductile Iron Pipe
FFB	First Flush Basin
FFW	Fire Fighting Water
GD	Glycol Drain
GRE	Glass-fibre Reinforced Epoxy
HAZID	Hazard Identification
HAZOP	Hazard and Operability Study
HDPE	High Density Polyethylene
HSE	Health, Safety and Environment
ICSS	Integrated Control and Safety System
ISO	International Organisation for Standardisation

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Abbreviations	
ITP	Inspection and Testing Plan
JSA	Job Safety Assessment
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas
NBR	Acrylonitrile butadiene rubber
NPS	Nominal Pipe Size (inside diameter)
OD	Open Drain
P&ID	Process and Instrument Diagram
PVC	Poly Vinyl Chloride
PVC-U	Unplasticized Poly Vinyl Chloride
QC	Quality Control
RA	Risk Analysis
RCB	Reinforced Concrete Box
RCP	Reinforced Concrete Pipe
SDM	Solvent Drain MDEA (Amine)
SDR	Standard Dimension Ratio
SS	Sanitary Sewer
SW	Sour Water
TBC	To Be Confirmed
Tc	Time of Concentration
TDS	Total Dissolved Solids
UV	Ultra Violet
WMS	Work Method Statement

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SECTION A - GENERAL

2 REFERENCE DOCUMENTS

2.1 International Codes and Standards

The following Codes and Standards shall form a part of this Specification. When an edition date is not indicated for a Code or Standard, the latest edition in force at the time of the contract award shall apply.

AMERICAN ASSOCIATION OF STATE HIGHWAYS AND TRANSPORTATION OFFICIAL (AASHTO)

AASHTO	LRFD Bridge Design Specifications

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 224.3R	Joints in Concrete Construction
ACI 301	Specifications for Structural Concrete
ACI 302.1R	Guide for Concrete Floor and Slab Construction
ACI 318/ACI 318M	Building Code Requirements for Structural Concrete and Commentary
ACI 350/ACI 350M	Code requirements for Environmental Engineering Concrete Structures and Commentary
ACI 530/530.1-13	Building Code Requirements and Specification for Masonry Structures and Companion Commentaries

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A48/A48M	Standard Specification for Gray Iron Castings
ASTM A74	Standard Specification for Cast Iron Soil Pipe and Fittings
ASTM A105/A105M	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A106/A106M	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A123/A123M	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A746	Standard Specification for Ductile Iron Gravity Sewer Pipe
ASTM C76/C76M	Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe
ASTM C478/C478M	Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C497	Standard Test Methods for Concrete Pipe, Concrete Box Sections, Manhole Sections, or Tile

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ASTM C564	Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
ASTM C1433	Standard Specification for Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers
ASTM D1785	Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80,and 120
ASTM D2000	Standard Classification System for Rubber Products in Automotive Applications
ASTM D2241	Standard Specification for Poly(Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)
ASTM D2321	Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2564	Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
ASTM D2665	Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings
ASTM D2855	Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets
ASTM D2992	Standard Practice for Obtaining Hydrostatic or Pressure Design Basis for “Fiberglass” (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe and Fittings
ASTM D3034	Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
ASTM D3035	Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside Diameter
ASTM D3212	Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals
ASTM D3262	Standard Specification for “Fiberglass” (Glass-Fiber-Reinforced Thermosetting-Resin) Sewer Pipe
ASTM D3350	Standard Specification for Polyethylene Plastics Pipe and Fittings Materials
ASTM D3754	Standard Specification for “Fiberglass” (Glass-Fiber-Reinforced Thermosetting-Resin) Sewer and Industrial Pressure Pipe
ASTM D3839	Standard Guide for Underground Installation of “Fiberglass” (Glass-Fiber Reinforced Thermosetting-Resin) Pipe
ASTM D4976	Standard Specification for Polyethylene Plastics Molding and Extrusion Materials
ASTM D5421	Standard Specification for Contact Molded “Fiberglass” (Glass-Fiber-Reinforced Thermosetting Resin) Flanges
ASTM F477	Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe

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ASTM F667/F667M	Standard Specification for 3 through 24 in. Corrugated Polyethylene Pipe and Fittings
ASTM F714	Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Outside Diameter
ASTM F894	Standard Specification for Polyethylene (PE) Large Diameter Profile Wall Sewer and Drain Pipe
ASTM F1668	Standard Guide for Construction Procedures for Buried Plastic Pipe
ASTM F2487	Standard Practice for Infiltration and Exfiltration Acceptance Testing of Installed Corrugated High Density Polyethylene and Polypropylene Pipelines
ASTM F3371	Standard Specification for Polyolefin Pipe and Fittings for Drainage, Waste, and Vent Applications

BRITISH STANDARDS INSTITUTION (BSI)

BS 5911-1	Concrete pipes and ancillary concrete products-Part 1: Specification for unreinforced and reinforced concrete pipes (including jacking pipes) and fittings with flexible joints (complementary to BS EN 1916)
BS 5911-3	Concrete pipes and ancillary concrete products-Part 3: Specification for unreinforced and reinforced concrete manholes and soakaways (complementary to BS EN 1917)
BS 9295	Guide to the structural design of buried pipes
BS ISO 4427	Plastics piping systems for water supply and for drainage and sewerage under pressure — Polyethylene (PE)

EUROCODES (BS EN)

BS EN 124-1	Gully tops and manhole tops for vehicular and pedestrian areas Part 1: Definitions, classification, general principles of design, performance requirements and test methods
BS EN 124-2	Gully tops and manhole tops for vehicular and pedestrian areas Part 2: Gully tops and manhole tops made of cast iron
BS EN 681-1	Elastomeric seals. Material requirements for pipe joint seals used in water and drainage applications. Vulcanized rubber
BS EN 681-2	Elastomeric seals. Material requirements for pipe joint seals used in water and drainage applications. Thermoplastic elastomers
BS EN 682	Elastomeric seals. Materials requirements for seals used in pipes and fittings carrying gas and hydrocarbon fluids
BS EN 752	Drain and Sewer Systems Outside Buildings
BS EN 1295-1	Structural Design of Buried Pipelines Under Various Conditions of Loading – General Requirements

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BS EN 1401-1	Plastics piping systems for non-pressure underground drainage and sewerage. Unplasticized poly(vinyl chloride) (PVC-U). Specifications for pipes, fittings and the system
BS EN 1610	Construction and Testing of Drains and Sewers
BS EN 1796	Plastics piping systems for water supply with or without pressure. Glass-reinforced thermosetting plastics (GRP) based on unsaturated polyester resin (UP)
BS EN 1916	Concrete pipes and fittings, unreinforced, steel fibre and reinforced
BS EN 1917	Concrete manholes and inspection chambers, unreinforced, steel fibre and reinforced
BS EN 10204	Metallic products. Types of inspection documents
BS EN 12056-2	Gravity drainage systems inside buildings - Part 2: Sanitary pipework, layout and calculation
BS EN 12666-1	Plastics piping systems for non-pressure underground drainage and sewerage. Polyethylene (PE). Specifications for pipes, fittings and the system
BS EN 13101	Steps for underground man entry chambers. Requirements, marking, testing and evaluation of conformity
BS EN 14364	Plastics piping systems for drainage and sewerage with or without pressure. Glass-reinforced thermosetting plastics (GRP) based on unsaturated polyester resin (UP) - Specifications for pipes, fittings and joints
BS EN 14396	Fixed ladders for manholes
BS EN 14654-1	Management and control of operational activities in drain and sewer systems outside buildings Part 1: Cleaning
BS EN ISO 1452-1 to 5	Plastics piping systems for water supply and for buried and above-ground drainage and sewerage under pressure. Unplasticized poly(vinyl chloride) (PVC U).
BS EN ISO 9001	Quality Management Systems – Requirements
BS EN ISO 9004	Quality management – Quality of an organization – Guidance to achieve sustained success
BS EN ISO 14692	Petroleum and natural gas industries. Glass-reinforced plastics (GRP) piping. Parts 1 to 4.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4633	Rubber seals — Joint rings for water supply, drainage and sewerage pipelines — Specification for materials

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2.2 ADNOC Specifications

The following ADNOC general specifications, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

Electrical Engineering Design Guide	AGES-GL-02-001
Facility Layout and Separation Distances Guidelines	AGES-GL-03-001
Fire & Gas Detection and Fire Protection System Philosophy	AGES-PH-03-002
Automation and Instrumentation Design Philosophy	AGES-PH-04-001
Concrete Supply and Construction Specification	AGES-SP-01-001
Structural Steel Supply, Fabrication and Erection Specification	AGES-SP-01-002
Structural Design Basis – Onshore Specification	AGES-SP-01-003
Roads and Paving - Design and Construction Specification	AGES-SP-01-005
Site Preparation, Earthworks and Bund Construction Specification	AGES-SP-01-006
Instrumentation for Packaged Equipment Specification	AGES-SP-04-007
Centrifugal Pumps for General (Utility) & Fire water service Specification	AGES-SP-05-008
External Pipeline Coatings Specification	AGES-SP-07-002
Painting and Coating Specification	AGES-SP-07-004
Piping Material Specification Index	AGES-SP-09-002
Preservation and Export Packing Specification	TBC
Onsite Material Sampling and Testing Specification	TBC
HAZID/ENVID/OHID	HSE-RM-ST03
Hazard & Operability Study (HAZOP)	HSE-RM-ST04
HSE Design Philosophy	HSE-GA-ST07

2.3 Standard Drawings

The following ADNOC standard drawings, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

Culvert Details for Unlined Ditch	TBC
Headwall Details	TBC
Details of Ditches and Swales	TBC
Drainage Ditch and Road Crossings – Fire Traps	TBC
Tank Bund Outlet – Fire Trap	TBC
Sanitary Sewer & Storm Water Manhole and Catch Basin Details	TBC
Oily Water (COC) & Accidently Oily Water (AOC) Manhole Details	TBC

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Accidently Oil Contaminated (AOC) & Oily Water (COC) Double Compartment Manhole	TBC
Vents for Manholes	TBC

2.4 Other References

Model Code of Safe Practice Part 15, Area Classification for Installations Handling Flammable Fluids	EI-15

3 DOCUMENT PRECEDENCE

The specifications and codes referred to in this Specification shall, unless stated otherwise, be the latest approved issue at the time of contract award.

It shall be the CONTRACTOR's responsibility to be, or to become, knowledgeable of the requirements of the referenced Codes and Standards.

The CONTRACTOR shall notify the COMPANY of any apparent conflict between this specification, the related data sheets, the Codes and Standards and any other specifications noted herein.

Resolution and/or interpretation precedence shall be obtained from the COMPANY in writing before proceeding with the design/manufacture.

In case of conflict, the order of document precedence shall be:

a. UAE Statutory requirements

b. ADNOC HSE Standards

c. Equipment datasheets and drawings

d. Project Specifications and standard drawings

e. Company Specifications

f. National / International Standards

4 SPECIFICATION DEVIATION / CONCESSION CONTROL

Any technical deviations to the Purchase Order [or Sub-contract] and its attachments including, but not limited to, the COMPANY's General Specifications shall be sought by the CONTRACTOR only through technical deviation request format. Technical deviation requests require COMPANY'S review/approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to COMPANY approval are subject to rejection.

5 DESIGN CONSIDERATIONS / MINIMUM DESIGN REQUIREMENTS

Refer to Section B of this Specification.

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SECTION B – TECHNICAL REQUIREMENTS

6 ENVIRONMENTAL DATA AND SITE CONDITIONS

6.1 Meteorological Data

a. The meteorological data required for drainage design shall be in accordance with COMPANY / project basic engineering design data.

6.2 Topographical, Underground and Geotechnical Surveys

a. The drainage system shall be designed in accordance with the results and findings made in the topographical survey, underground detection survey and geotechnical investigation reports approved by the COMPANY.

b. The information contained in the existing Topographical survey data and the geotechnical investigation reports provided by COMPANY (if any) shall be considered for guidance only. All such information shall be validated by CONTRACTOR prior to start of design.

c. CONTRACTOR shall carry out any additional survey works deemed necessary to perform the work covered in this Specification.

7 DESIGN AND DRAWINGS

7.1 General

a. COMPANY review and/or approval of design and drawing shall neither relieve the CONTRACTOR from their obligations of meeting the project requirements / COMPANY Specifications nor from responsibility of defective work resulting from errors or omissions on the approved construction drawings.

7.2 Design

7.2.1 General

a. The CONTRACTOR shall prepare and submit for COMPANY approval the design criteria including the basic layout, design methodology and all assumptions prior to the start of the detailed design. Subsequently, the design calculation report shall be submitted and comprise the following as a minimum:

   i. Cover page with a report title, document number, project / contract No., date and revision mark, document prepared by, checked by and approved by columns.

   ii. The table of contents.

   iii. Introduction, assumptions if any, design philosophy and design criteria.

   iv. Applicable codes, formulas, graphs/tables, rainfall, fire water and other effluents considered.

   v. References to literature, etc., for subjects not covered by applicable codes.

   vi. The software used for analysis and design with native input and output files.

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vii. List of design parameters used.

viii. Analytical model of the drainage network used for computer analysis.

ix. Calculated flows / velocities / volumes / etc.

x. Design summary and conclusions.

xi. List of attachments/annexures in support of design calculations.

7.2.2 Drainage Structures

a. Structural design of the drainage structures shall be in accordance with AGES-SP-01-003 Structural Design Basis - Onshore Specification.

7.3 Drawings

7.3.1 General

a. Construction drawings shall comply with COMPANY CAD Manual and drafting procedure, as applicable.

b. Only drawings marked "Approved for construction" duly signed by COMPANY's representatives shall be used at the site for construction.

c. Drawings shall be submitted together with the relevant calculations under one transmittal, including those required for submission to local authorities.

d. CONTRACTOR shall provide to COMPANY native soft copy of all drawings along with PDF copy. All revisions shall be clearly highlighted and clouded for easy identification and / or review.

7.3.2 Underground Layout / Composite Drawings

a. CONTRACTOR shall prepare detailed composite civil work drawings covering all underground services. These drawings shall show all the gravity drainage lines to scale, giving all drain sizes, material, manhole positions, invert levels and gradient where applicable. The different drainage systems (CWS, AOC, COC and SS) shall be clearly identified. The direction of flow shall be indicated. Manholes shall be numbered for identification and should be scheduled to provide all relevant information.

b. These drawings shall also include all other underground items to ensure the design is clash free. These shall include the following

i. Electrical cable trenches and ductbanks

ii. Instrument cable trenches and ductbanks

iii. Pressurised process and utility lines such as potable water, cooling water and fire water (information to be provided by piping and process). This should include valves and associated pits.

iv. Process effluent streams such as sour water, amine discharge, glycol discharge. (Information to be provided by piping and process).

v. Outline of all underground closed drain drums labelled with equipment tag number (Information to be provided by piping and process).

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vi. Outline of foundations / pile caps / ground beams.

vii. Outline of any other existing underground features.

viii. Outline of all roads and shoulders.

8 DRAINAGE SYSTEM OVERVIEW

8.1 General

a. The drainage system conveys effluent from source to the treatment facility and / or discharge point. The drainage system shall be designed to protect atmosphere, soil, surface water and groundwater from contamination. The drainage system shall be designed to prevent the spread of fire.

b. Dependent upon source, type and level of contamination, effluents shall be classified into various categories. For each effluent category, the required drainage network and eventual treatment facility shall be determined.

c. The segregation of all effluent streams shall start at the source.

8.2 Effluent Sources

The following are typical sources of effluents:

a. Rainwater

b. Firefighting water

c. Wash/flush water

d. Cooling water blowdown / filter backwash

e. Equipment drains

f. Sampling points

g. Desalter drains

h. Accidental process spillages

i. Chemical drains (e.g. Laboratory Effluents, etc.)

j. Maintenance operations (e.g. exchanger bundle pulling)

k. Domestic sewers

l. Substation transformer pens/enclosures

m. Floor/roof drains

n. Amine network

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8.3 Effluent Classification

Atmospheric drainage effluents are classified as belonging to one of the following categories:

8.3.1 Clean Water System (CWS)

a. Clean rainwater draining to the Clean Water System (CWS) drain. This is applicable exclusively to areas where surface rainwater cannot be contaminated by hydrocarbons or chemicals.

8.3.2 Accidentally Oil Contaminated System (AOC)

a. Effluents from areas where spills or leakages of hydrocarbons or chemicals may be anticipated but not expected. This system collects storm water / fire water run-off from paved equipment areas divided into AOC catchment areas. These effluent streams normally meet the required water discharge specifications but on occasion may be contaminated.

8.3.3 Continuously Oil Contaminated System (COC)

a. Effluent collected underneath the oil process equipment such as pumps, compressors, vessels and manifolds that are likely to produce drips and spillage in routine operation. Such effluent shall be collected from a drip tray or bunded area. This also includes water collected underneath oil loading areas including loading arms, hose handling areas at jetties and filling points.

b. This system collects effluent water with possible hydrocarbon content that exceeds the required water discharge specifications.

8.3.4 Sanitary Sewer System (SS)

a. Domestic sewage draining to the sanitary sewer (SS). Sanitary sewer system mainly includes waste streams from toilets, showers, sinks, kitchens and canteen facilities. No hydrocarbons shall be allowed to enter the sanitary sewer.

8.3.5 Closed Drains (CD)

a. Additional underground hazardous Process Effluent Streams, comprising largely of pure hydrocarbons or chemicals which are routed to closed systems. The Closed Drain (CD) is defined as a fully contained drain – a hard pipe connecting the equipment to be drained to the collection system. The system connects to a collection vessel for recovery. Examples of Closed Drains include:

   i. Process water effluents such as Sour Water (SW) draining to a closed drain.

   ii. Hydrocarbon vessel drain - draining to a dedicated plant Closed Drain (CD) system.

   iii. Amine discharge draining to a closed Amine drain (SDM)

   iv. Glycol discharge draining to a closed Glycol drain (GD)

   v. Boiler blowdown and emergency water dumping (BWS)

b. These Process drains are dedicated buried closed drainage systems, carrying discharge collected via flanged closed drain hubs to buried collection vessel/s. These are designed as per respective piping specifications. The collected discharges are pumped via above ground drain headers to the designated units for processing. Closed drain hubs are flanged (and generally pressurized) systems which are distinct from atmospheric open drain hubs. These are listed here to ensure appropriate

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space allocation and integration with other underground services but are not otherwise addressed in this specification.

8.3.6 Chemical Effluent

a. Effluent streams that are contaminated with chemicals. For example, effluent from laboratories may be contaminated with chemicals. Chemical effluent shall be kept separate from other systems.

8.4 Effluent Routing

a. Effluents shall be routed by means of separate drainage networks to selected treatment facilities and / or discharge points.

b. The segregation of the drainage networks shall be done based on the contamination criteria as identified in Section 8.3. The selection of the routing of drainage networks shall be determined in conjunction with the plant layout requirements – refer to AGES-GL-03-001 Facility Layout and Separation Distances Guidelines. Effluent destinations shall be as follows:

   i. Oil-free run-off shall be collected in the CWS and directed to offsite discharge / soakaway ditches / site evaporation pond(s) / public water (sea, etc.).

   ii. AOC run-off collected within the AOC system Process and Utility Areas catch-basins / sumps / manholes / weir box shall be directed to AOC interceptor for first flush volume collection. The first flush volume is pumped to the waste water treatment units for further processing. Volume in excess of the first flush volume should be directed to site evaporation pond(s) / public water (sea, etc.).

   iii. COC run-off shall be collected in COC drain hubs / drip funnels (with strainer plate where required) / catch-basins and manholes shall be directed to oily water separator for hydrocarbon separation and further pumped / taken to the waste water treatment units and slop facilities for further processing.

   iv. Sanitary sewage shall be collected in the SS and directed to a sewage treatment facility. In case it is not feasible to provide sewage treatment facility, septic / holding down tank with provision for local removal / emptying by vacuum truck may be provided.

   v. Chemical effluent streams shall be directed to dedicated downstream treatment facilities. Chemical effluent streams contaminated with acids shall be routed to a neutralization pit/tank, located as near as possible to the source of effluent discharge, with dedicated downstream treatment. Method of neutralization is dependent on the type of chemical waste present and must be specified on a case-by-case basis and shall [PSR] be submitted for COMPANY approval.

c. All other effluent drainage networks – typically closed drains - are covered in the piping and process specifications.

d. For a diagrammatic summary of typical effluent classification and routing, see Figure 1 in Appendix A1.

e. The treated water from the secondary treatment facilities shall be handled as follows:

   i. If the treated water is high in TDS (total dissolved solids), it may be reinjected into the aquifer (where appropriate). Maximum particle size of suspended solids in such treated water shall [PSR] be agreed with COMPANY prior to aquifer injection.

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ii. If the treated water is low in TDS, (i.e., treated sewage and boiler blowdown), it may be discharged to a holding tank for irrigation purposes (where appropriate).

8.5 General Requirements for CWS Networks

a. Clean water collected from roads, permeable paved area, unpaved areas shall be discharged by means of an underground CWS network or via open channels / trenches to the appropriate discharge location (e.g., offsite discharge, soakaway, evaporation pond, public water). A water detention pond may be installed to check the quality of water before discharging to public water as per the approved effluent drainage philosophy.

b. Clean water is normally collected in unpaved areas (gravel, grass, sand, etc.), building roofs, roads, and paved areas permanently clear of process equipment / sources of contamination. The discharge area may be sloped to the surrounding offsite area (if the site conditions allow for) or the surface run-off may be collected in concrete or gravel lined swales / soakaways located at the perimeter of the relevant discharge areas. These swales shall be utilized as carriers to locations where uncontaminated run off can be discharged. Also, it shall be ensured that the water disposal shall not affect the integrity of adjacent facilities.

c. Gravel surfaced unlined earthen CWS channels are subject to COMPANY approval.

d. Where CWS drainage run off must cross roads between units a piped flooded culvert with reinforced concrete chambers and sand trap shall be constructed in accordance with the standard detail drawings (see typical detail in Figure 5 in Appendix A1).

e. The use of soakaways in CWS networks shall be subject to COMPANY approval.

f. When soakaways are implemented the following shall be ensured:

   i. Assessment of suitability of the geology;

   ii. The soakaway shall be located:

      1. in an area lower than the area being drained;

      2. at least 5 m away from any building;

      3. so that it will not saturate the foundations of any structure;

      4. so that the base of the soakaway is permanently above the water table;

      5. far enough away from other soakaways to ensure the capacity of those devices, and the ground itself;

      6. so that there is no risk of contamination from pollutants.

8.6 General Requirements for AOC and COC Networks

a. The networks shall comprise of paved areas which are impermeable and sloping to perimeter drain channels and/or catch basins which subsequently discharge the effluents through pipes and potentially manholes to buffer or primary treatment facilities.

b. The maximum (sub)catchment area to be served by one catch basin:

   i. shall [PSR] not exceed 600 m^2 for general areas;

   ii. shall [PSR] not exceed 400 m^2 for congested areas.

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iii. If the surface slope is not possible to establish due to congestion or other limitations, then additional catch basin to be provided irrespective of area limit.

c. In the event that no actual data on firefighting quantities is available, the design flow for the on plot underground drainage network serving a paved area of 400 m2 shall be taken as 250 m3/h.

d. Segregation of catchment areas shall be achieved by selection of high points.

e. These high points (e.g., bund, kerb, paving high point) shall be located between AOC, COC, CWS subcatchment areas within the overall catchment area as well as between AOC, COC, and CWS catchment areas.

f. Perimeter trenches shall slope to catch basins located at a maximum of 50 m apart and be provided with heavy duty grating covers suitable for the imposed loads during the design life of the facility.

g. Additional kerb areas to be provided and/or additional catch basins located such that accidental spill or spills of effluents which could be on fire, can be quickly collected in the underground system. Any spill shall not travel more than 15 m over the surface of the paved plant area.

h. All components as well as interfaces and joints between different components (e.g., pipe-to-pipe, pipes to catch basins/manholes, covers, channels, trenches, catch basins, manholes, paving to catch basins/manholes, joints in paving) shall be designed to ensure functional integrity during the design life of the facility.

i. In the COC system, cleanouts or rodding points may be required for small lines (e.g. from drip funnels) where there are bends in the line and there is limited space for manholes. These shall be reviewed and approved by COMPANY.

8.7 General Requirements for SS Networks

a. Atmospheric gravity type SS networks shall be:

   i. empty at no flow;

   ii. provided with liquid traps;

   iii. self-cleaning at design flow;

   iv. provided with clean outs and trap primers where required.

b. Pumped SS networks shall be provided with macerating pumps and sumps designed to cater for peak loads.

c. Sanitary pumps shall be suitable for solids handling etc. They shall be in accordance with AGES-SP-05-008 “Centrifugal Pumps for General (Utility) & Fire water service Specification" and provided with suitable support and lifting facilities. Also, the configuration shall include both duty and standby pumps as advised by COMPANY.

d. In case hazardous gas is expected to be generated in confined space such as sewage lifting stations, sewage pump shall be of suitable explosion classified type.

e. Grease trap shall be installed in the discharge from kitchens.

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9 SPECIFIC CATCHMENT AREAS

9.1 General

a. A catchment area is defined as an area in which several effluent streams have a common discharge point.

b. For calculation and identification purposes catchment areas shall be numbered and may be further divided into subcatchment areas.

c. The specific catchment areas in this section have been included as they have differences in approach to the standard design or they have features that must be included.

d. Catchment areas like Process Plants or Utilities shall follow the general requirements of Section 8. An overall guide to Process Plant drainage is shown in Figure 2 in Appendix A1. This figure indicates that most surface water effluents will drain over sloped paving towards drain channels / catch basins that connect to the AOC system. Effluents from areas where spills or leakages of hydrocarbons may be expected shall drain to the COC system.

9.2 Liquified Gas Facilities

a. Surface water effluents from liquefied gas processing, storage and loading areas shall be classified as AOC.

b. Subcatchment areas shall slope towards perimeter drain channels / catch basin. For slopes see Section 10.1.

c. Spills shall be drained away from equipment to a safe distance where the liquefied gas is allowed to evaporate in a collection basin or channel.

d. The collection basin or channel shall [PSR] be provided with a liquefied gas trap to prevent liquefied gas entering into the underground drainage system.

e. The minimum retention volume of the collection basin or channel upstream of the liquefied gas trap shall be 10 m^3.

f. The selection of the number of liquefied gas traps shall be submitted for approval to COMPANY.

g. Appropriate fire deflection walls or fire-breakers shall be installed between safety critical and production critical equipment.

h. In LNG storage areas an appropriate LNG impounding basin shall be provided as per the relevant safety code and / or project requirement.

i. Typical details for Liquified Gas Facilities are shown on Figure 6 and Figure 7 in Appendix A1.

9.3 Jetties

a. Jetty heads or quays shall be provided with subcatchment areas where CONTRACTOR identifies credible risk of spills or leakages of hydrocarbons or chemicals. Identified subcatchment areas for jetty heads or quays shall drain to a sump / manhole / vessel (typically integral with or below the jetty deck). For example, a dedicated kerbed area shall drain by means of sloped concrete paving to a collection sump or vessel (AOC or COC).

b. As a minimum the following dedicated subcatchment areas shall be identified:

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i. chemicals, crude oil and hydrocarbon product jetties:

1. loading arms, metering stations and manifolds in chemicals, crude oil and hydrocarbon product berths;

2. slops generated by draining of loading systems, activities during maintenance or sampling and by surge relief during operations shall be included with surface runoff within the COC collection areas in sizing jetty COC collection system.

ii. LPG/LNG jetties:

1. condensate, Paraffinic Naphtha or gasoline loading arms and associated metering stations and manifolds (to COC);

2. LPG and LNG loading arms (due to the potential spillage of the hydraulic system for operating the loading arm) (to AOC).

c. The buffer capacity and / or associated effluent discharge pump capacity for these areas shall be selected to allow for a drainage network design inclusive of downstream treatment capacity.

d. Discharge is typically done via a pumped system or truck

e. The effluent classification shall be according to section 8.3 and shall meet the associated network requirements.

f. All areas outside these subcatchment areas shall be considered CWS area:

i. these CWS areas shall slope away from COC and AOC catchment areas;

ii. these CWS areas should runoff directly to public water.

9.4 Electrical Transformers

a. All oil filled electrical transformers shall be surrounded by a gravel-filled or gravel-covered oil catchment pit, which is sized to contain the total oil content of the transformer (as per the requirements of AGES-GL-02-001 “Electrical Engineering Design Guide”) combined with the maximum daily design rainfall or firefighting water as specified by HSE department.

b. The catchment pit shall be isolated from the drainage network by means of a normally closed valve.

c. Transformer drainage pits shall have a collection sump in one corner connected to a diverter box located outside, via manholes. The diverter box shall have two valved outlets. One outlet from the diverter box shall be to the AOC system and the other to the oily water drainage system.

d. Discharge of leaked oil to the AOC network is not acceptable.

e. When selecting piping materials, it should be considered that the leaked oil may be at elevated temperatures.

9.5 Tank Compounds

a. Tank compounds shall be designed as AOC areas (inclusive of the bund walls).

b. Additional details on lining of tank compounds, bunds and protrusions are provided in AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction”.

c. A sand trap shall be provided at the discharge point of the main drain channels within the bunded area.

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d. The discharge of the effluent (storm water and FFW) within the tank compound shall [PSR] only run via a liquid filled underground valve-controlled drain line. Each compound shall have a dedicated drain line.

e. A practical valve size is 300 mm. If the design requires a larger outlet, additional outlet valves should be selected rather than increasing the valve diameter.

f. The penstock valve shall [PSR] meet all of the following:

   i. normally kept closed;

   ii. located outside the bunded area;

   iii. located to facilitate safe accessibility for operation in all conditions.

g. Fire propagation prevention between the compound and the outside network shall comply with Section 12.2.2.

h. The level-controlling height (at the sand trap) of the drainage network inside the tank compound shall always be 50 mm higher than the height of the level-controlling weir of the drainage network directly outside the tank compound which is receiving the compound discharge. Where a tank compound drainage valve discharges directly into an open channel a splash wall shall be provided.

i. A tank compound shall [PSR] not be drained via an adjacent tank compound.

j. A typical arrangement of tank compound drainage is shown in Figure 4 in Appendix A1.

9.6 Impounding Basins

a. Impounding basin(s) shall be used to provide additional secondary containment when the secondary containment within the tank compound does not meet the required secondary containment volume.

b. The connection between a tank compound and the associated impounding basin shall be designed with capacity to handle the larger of a major product spill or a FFW flow within the upstream compound.

c. Impounding basins shall be designed as AOC areas (inclusive of the bund walls). Additional details on lining of impounding basins, bunds and protrusions are provided in AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction Specification”.

d. The invert level of the sump within the impounding basin shall [PSR] be designed such that reverse flow into the upstream connected tank compound(s) is not possible.

e. Where pumping facilities are required to pump out any rainwater accumulated in the basins, the discharge from said pumps shall be directed to AOC drainage network.

f. Fire propagation prevention between tank compound and impounding basin as well as between impounding basin and the outside network shall comply with Section 12.2.2.

9.7 Tank Bottoms Drainage

a. Effluents that have ‘settled out’ in a storage tank are defined as tank bottoms effluent and shall be drained via a tank bottoms drainage network.

   i. These effluents are classified as either COC or CD.

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ii. If classified CD, the tank bottoms shall be drained through a closed funnel. Such tank bottoms drainage is not covered by this specification.

b. Tank bottoms drains that are classified COC shall drain through an open funnel or a closed funnel depending on regulatory requirements related to emissions associated with the effluent.

c. The tank bottoms drain of each tank shall discharge through a valve-controlled outlet that subsequently discharges into a funnel.

d. For tank bottoms draining through an open funnel an additional valve shall be installed directly under the funnel to enable shutting the funnel off from the system when the tank is not being drained.

e. A typical arrangement of a tank bottom drains open funnel is shown in Figure 19, in Appendix A1.

f. For tank bottoms draining through a closed funnel, hydrocarbon measurement equipment shall be installed in the tank bottoms drain line.

g. Where a branch of the system leaves the tank compound, a normally closed valve shall be installed at the outside of the bunded area.

h. The connection of the tank bottom drain to the funnel shall [PSR] be kept flooded with an upturned or downturned elbow.

i. Tank bottom drainage networks shall be located above ground to facilitate ease of inspection.

j. Automated tank bottom drainage systems shall be subject to COMPANY approval.

k. The capacity of the tank bottoms drainage network shall be designed considering the downstream buffer and/or treatment capacity.

l. Typically, the design should not allow for more than one crude tank and not for more than two product tanks to be drained at any one time. This is often based on the capacity of the oil separation sump / interceptor(s) installed downstream:

   i. the duration and quantities of drained effluents shall be determined in consultation with and submitted to COMPANY for approval;

   ii. the minimum capacity of the tank bottoms drainage network shall be as per Table 9.1.

Table 9.1 Minimum capacity of tank bottom drains

Product type	Capacity m^3/h
Crude oil tanks	100
Product tanks excluding low sulphur diesel tanks	50
Low sulphur diesel tanks	10

9.8 Tank External Floating Roof

a. The roof drains from an external floating roof shall discharge through a normally open valve into the tank compound drainage system (refer Section 9.5).

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9.9 Pipetracks (AOC and COC only)

a. All surfaces under a pipetrack shall slope away from the pipetrack. The minimum slope shall be 1:200.

b. The slope should be to one side only for pipetrack widths < 10 m.

c. The high point of the pipetrack finished grade level shall be at least 250 mm below the top of the pipe sleepers.

d. In areas where spills or leaks can be expected, concrete paving should be provided. This will facilitate cleaning, maintenance and washdown operations in the event of a spill having occurred. This paving shall be sloped to a side channel / trench (AOC or COC).

e. Controlled flooding of these pipetracks shall be as per Section 12.6.

f. Fire propagation prevention in long pipetracks shall be as per Section 12.2.2.

g. A typical detail of pipetrack drainage is provided in Figure 5, in Appendix A1.

9.10 Miscellaneous

9.10.1 Roads and Car Parks

a. Storm water collected from roads and car parks area is normally free from contamination and can be discharged directly to the CWS system.

b. For locations where possible contamination may occur, such as loading / unloading areas, AOC system shall be established to prevent the effluent collected from the area from directly entering the offsite discharge.

9.10.2 Slug catcher Area

a. Effluents collected from the slug catcher areas can be contaminated with hydrocarbons. Such effluents collected from the areas shall be routed via COC network(s) to ensure the effluents will receive proper segregated treatment.

9.10.3 Pump and Metering Stations

a. Surface runoff from pump and metering station areas can be contaminated with hydrocarbons and therefore shall be routed to AOC drainage system. For locations where connection to AOC drainage system is difficult, proper measures shall be taken to prevent soil, groundwater and water contamination. These measures shall be agreed with COMPANY.

9.10.4 Launcher and Receiver Area

a. Effluents collected from launcher and receiver area can be contaminated with hydrocarbons. Such effluents collected from the areas shall be routed via COC network(s) to ensure the effluents will receive proper segregated treatment.

9.10.5 Packaged Equipment

a. Condensate discharge from packaged equipment such as air compressor packages (from inter / after coolers, moisture separators, air filter drains etc) shall be drained to nearest available CWS system or to a soak pit.

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b. Drain points from packaged equipment which use forced lubrication systems, such as compressors, turbines, large capacity pumps etc shall be extended and connected to COC system.

9.10.6 Additional Catchment Areas

a. Plants and / or projects containing specific sub-catchment areas in addition to those mentioned in Section 9 shall address those areas explicitly in the drainage system design. These shall be reviewed and agreed with COMPANY prior to detail design.

10 DRAINAGE NETWORK COMPONENTS

a. The aim of the design is to achieve improved surface water run-off and provide better segregation of effluents at source, thus reducing the risk of contaminating cleaner effluent streams with dirtier streams. In line with this aim the following design parameters shall apply.

10.1 Surfaces

a. All CWS catchment areas (typically finished with soil, gravel, or paving) shall have a free runoff slope towards a CWS channel or trench, subject to the following:

   i. where ponding (with subsequent soakaway or evaporation) of CWS is acceptable (taking into account HSE risks with respect to standing water) such sloping to channel or trench is not required;

   ii. the design shall take into account the erosion and sedimentation control due to transportation of silt and sand.

b. AOC and COC paving slopes shall [PSR] be directed away from equipment to reduce the risk of escalation of fire.

c. CWS, AOC and COC area surface slopes shall be a minimum 1:100, with a maximum level difference of 200 mm in paved areas. Shallower slopes require COMPANY approval.

d. All paving shall comply with AGES-SP-01-005 “Roads and Paving - Design and Construction Specification”.

10.2 Catch Basins and Manholes

a. Refer to applicable standard drawings as listed in Section 2.3 and see typical details in Figures 12 to 16 in APPENDIX A1.

10.2.1 Definitions

a. Catch basin: Catch basins collect effluent from the connected sub-catchment paved area and discharge the effluent via gravity through underground pipes.

b. Manhole: Chamber that connects two or more underground pipes and is designed to allow for inspection and maintenance access to the undergrounds.

10.2.2 General

a. Catch basins shall be located such that they collect surface water from the surrounding areas.

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b. Catch basin to catch basin connection is not allowed however in cases of congested areas and subject to COMPANY approval a maximum of 2 catch basins may be connected.

c. Catch basin shall not be directly located under equipment, piperacks or equipment supporting structures.

d. Manholes shall be located so that they are completely clear of any foundations.

e. Manholes shall be located at adequate spacing intervals to facilitate maintenance, inspection and cleaning. The manholes shall be provided at all changes in horizontal direction. For straight length portions, the manholes / catch basins shall be provided such that the maximum manhole - manhole, catch basin – manhole distance is limited to:

   i. 60 m for pipe diameter ⩽ 300 mm

   ii. 100 m for pipe diameter >300 mm to ≤ 1000 mm

   iii. 300 m for pipe diameter > 1000 mm

f. Manholes shall not be located within areas confined by firewalls. Catch basin shall be used in these areas but in general, the manholes shall not be designed to collect surface water runoff.

g. For sealing of pipes in manholes and catch basins see Section 11.9.3.

10.2.3 Dimensional Requirements

a. The minimum diameter of the circular opening of the manhole shall be 600 mm.

b. Catch basins shall have minimum internal plan dimensions of 600 mm x 600 mm.

c. Manholes in CWS service shall have minimum internal plan dimensions of 900 mm x 900 mm.

d. Manholes in AOC / COC service shall have minimum internal plan dimensions of 1200 mm x 1200 mm.

e. Manhole dimensions shall allow access space for maintenance and inspection.

f. The internal plan dimensions for a manhole shall not be less than the largest pipe size plus 500 mm.

g. The top of the manhole cover should be at the same level as the top of the pavement in paved areas and at least 200 mm above ground level in non-paved areas.

10.2.4 AOC and COC Requirements

a. In order to transport the free oil and to avoid accumulation of hydrocarbons, self-skimming buckets or upturned elbow at the downstream side should be considered for all catch basins / manholes. Alternatively, stainless steel weir box at pipe outlet may be used if approved by COMPANY (see Figure 17 & Figure 18 in Appendix A1 for details).

b. Wherever possible manholes shall be provided with open steel or ductile iron grating cover in order to have proper ventilation and prevent the accumulation of hydrocarbon vapours in the manholes. Where sealed manholes are used a vent pipe shall be provided – see Section 12.2.3.

c. Entry points in open manholes in COC drainage networks to be designed to prevent clean or AOC surface water run-off entry into the COC network.

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d. The invert level of each manhole shall be at least 200mm below the bottom level of the lowest connected pipe.

e. Sealed manholes shall [PSR] be designed with seals that are leak tight with an internal pressure of at least 2 barg and where the integrity (leak tightness) is verifiable by means of field tests.

10.3 Underground Piping

10.3.1 General

a. All non-hydrocarbon drainage network pipes (CWS & SS) should be free draining gravity flow systems. For much of the time they will be dry (empty) but should only be partially filled when flowing.

b. Underground piping in drainage service shall have a minimum diameter of 150 mm.

c. For soil cover above the underground piping see Section 12.5.3.

d. Underground piping shall not be installed underneath foundations.

e. No pipe work shall pass through a foundation. If any existing / new foundation interferes with new / existing underground pipe work, the pipe work shall be re-routed to clear the foundation.

f. In case underground pipe is placed nearby and below foundation level the pipe shall not be placed in load distribution area (30 degree from foundation bottom edge with horizontal line).

g. Where pipes cannot be re-routed to avoid foundations, then the pipes shall be installed within pipe sleeves.

h. In instances where differential settlement or differential expansion between pipe and manhole can be expected, a flexible connection shall be applied.

i. Pipe connections and materials shall be selected such that the underground pipes and sewers in plant areas can easily be installed and will not give rise to leakage or collapse due to:

   i. settlement of subsoil,

   ii. shock loads of accidently released hydrocarbons/chemicals or during plant shutdown (polluted effluent),

   iii. cleaning of pipes by rodding, the use of low-pressure steam and/or the insertion of high-pressure jetting water.

j. Pipes passing through walls of manholes, catch basins, etc., shall be provided with a suitable seal. Such seals include puddle flange, hydrophilic sealant and modular as indicated on the standard drawings.

k. Underground piping layout requires multi-discipline co-ordination to ensure the requirements of civil, electrical, piping and drainage are all combined into an optimal solution.

10.3.2 AOC and COC Requirements

a. AOC / COC drainage network, underground pipes shall be laid horizontally and designed such that they are kept flooded (liquid-filled) at all times. A minimum liquid head of 150 mm, measured from the inside top of the pipe, shall be maintained.

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b. The required liquid head in the piping system shall be maintained by the weir in the inlet chamber of first flush interceptor / oily water separator. If required as per the plant design, separate weir box shall be provided at each area to maintain the liquid seal in the system.

c. Special attention shall be applied to pipe joints – these should be welded type joints to ensure water tightness.

10.3.3 Concrete Cradles

a. Concrete cradles shall be required when crossing over or under other sewer pipe or utility lines.

10.4 Open Channels and Trenches

a. The size of open drain channels shall be determined by hydraulic calculation for the actual design condition applicable.

b. The cross-sectional profile of open drain channels may be rectangular or hydraulically shaped.

c. All permanent ditches shall have a trapezoidal cross section. Side slopes shall be 1(V):1.5(H) for lined ditches and 1(V):3(H) for unlined ditches unless soil conditions permit otherwise. Gravel surfaced unlined ditches require COMPANY approval.

d. A minimum slope of 1:300 shall be used for drains up to 600 mm wide and a minimum slope of 1:500 shall be used for drains up to 900 mm wide. For oil free networks slopes of up to 1:1000 may be used if it can be executed without any low spots and possible ponding.

e. Open channels may be constructed in-situ or by utilizing precast sections. All channels shall be designed to prevent flotation.

f. The minimum width of rectangular channels shall be 300 mm. All open channels shall be designed with a minimum freeboard of 150 mm.

g. No piping and/or pipe supports shall be installed directly over or across open channels.

h. Open channels shall be covered with grating only at those places where operator/maintenance access is required – such grating shall be designed to support the expected traffic. Guard rails may be necessary in certain areas to allow personnel safe egress from area.

i. Open channels located at the toe of an embankment shall be provided with a 'walking strip' (minimum 350 mm wide) for maintenance access.

10.5 Culverts

a. A culvert is a relatively short conduit to convey water, normally run within an open ditch or channel, passing under a road or other physical obstruction. Drainage piping shall be selected as necessary to meet the requirements for storm water drainage, vehicular loads and depth of fill above the culvert.

b. Culverts under roads shall be designed to support the earth pressures on the culvert and the maximum wheel load that will be imposed over it through its design life, plus applicable impact loading, as defined in AASHTO Standard Specifications for Highway Bridges.

c. The minimum size of pipe culvert shall be 450 mm diameter for lengths of 9.0 m or less and 760 mm diameter for lengths over 9.0 m. Temporary culverts during construction are excluded from these requirements.

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d. Culvert minimum gradients shall be determined using the design flow rate. Culverts shall be sized to prevent headwater depths that will cause flooding within the plant facilities. In general, a minimum gradient of 1 in 300 shall be provided where possible. Otherwise, the slope of the upstream or downstream channel should be matched.

e. The culvert at a road intersection should be aligned with the ditch and located as perpendicular to the road as possible.

f. Culvert headwalls shall be used where required for flow and erosion control. Plain or flared-end culverts may be used with consideration given to traffic and hydraulic design.

10.6 Fire and Sand Traps

a. Fire traps and sand traps shall [PSR] be installed at road crossings and/or fire breakers within pipetracks and as required to isolate the inlet separator units from the rest of the Process Area.

b. The minimum liquid head in fire traps shall [PSR] be 150mm.

c. The minimum dimensional requirements for fire trap sumps shall be the same as those specified for the drainage sumps. The minimum distance between two interconnecting fire trap sumps shall be 9 m and the connecting underground pipe shall always be liquid filled.

d. Sand traps shall [PSR] be provided upstream of a fire trap and at the lowest point in an open drain channel system where this enters a sump. Sand traps shall have the same width as the drain channel, shall have a minimum length of 2 m and shall have a minimum depth of 0.50 m.

e. Typical details of a sand/fire trap together with the recommended level difference between drain channel inverts and sand trap overflow are shown in Figure 5, in Appendix A1.

10.7 Neutralisation Pit / Tank

a. Neutralisation pits shall be located as close to the source of chemical effluent as is practically possible. They shall be constructed in accordance with AGES-SP-01-001 “Concrete Supply & Construction Specification”, including all chemical resistant lining requirements. Sizing shall be determined by Process group.

10.8 AOC First Flush Interceptor

a. The effluent from AOC drainage networks, collected in catch basins and drainage channels, shall be routed by underground liquid sealed drain lines to the AOC First Flush Interceptor.

b. The AOC First Flush Interceptor should consist of the following:

   i. Inlet chamber: to maintain the water seal (to ensure the AOC drainage system remains flooded).

   ii. Dry weather basin: to control flows in dry periods that may arise from routine maintenance activities on site or from hosing down paved areas. These flows would be limited, and the designed dimensions of the chamber are small.

   iii. First Flush Basin (FFB): shall be sized to retain the quantity of the first flush rainfall / FFW after a relatively dry period and is expected to be contaminated. This initial flow shall be pumped to the wastewater treatment unit for further treatment. The FFB should be separated into at least two compartments (separated by an underflow overflow) to maximize separation in the influent. Skimming facilities should be installed in the first flush compartment to allow management of floating oil at the first opportunity.

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iv. Outfall chamber: Water in excess of the volume of FFB will overflow to this chamber for routing to the evaporation pond / public water outfall (sea, etc.).

c. The FFB shall be sized to retain the first flush resulting from the greater of the following conditions:

i. Design rainfall intensity (corresponding to Tc) x AOC catchment area x Tc

ii. Design FFW intensity x critical zone catchment area x Tc

d. Time of concentration (Tc) = time required for rainfall water / FFW to travel from the farthest AOC catchment area / critical zone to the AOC First Flush Interceptor, subject to minimum of 10 minutes.

e. FFB to be emptied before the next first flush arrives – either pumped or gravity-based discharge.

f. A conceptual sketch is shown in Figure 22 and Figure 23, in Appendix A1.

g. For requirements specific to ADNOC Onshore refer to APPENDIX A2.

10.9 Oily Water Separator

a. The effluent from COC drainage networks, collected in catch basins, drain hub / drip funnels shall be routed by underground liquid sealed drain lines to the Oily Water Separator, for hydrocarbon removal.

b. The sump / separator is to be designed like a simplified atmospheric oil-water gravity separation facility, consisting of various compartments such as:

i. Oil / water receiving and separation compartment,

ii. Oil compartment,

iii. Effluent water compartment.

c. The bottom of each compartment shall be sloped to facilitate sediment or sludge removal.

d. The following design parameters apply:

i. The oil / water separation compartment shall be designed to receive the incoming drain flow (by gravity) and to maintain the water seal (to ensure the COC drainage system remains flooded). The volume of the compartment shall allow a minimum retention time of 12 minutes based on the maximum continuous incoming flowrate (as per process data).

ii. An adjustable oil skimmer shall be located in the separation compartment, to skim the separated oil from the surface and route the skimmed oil to the oil compartment. The oil compartment shall be provided with pumps to route the skimmed oil to the oil treatment/slop facilities.

iii. The baffle / weir configuration should be designed to allow basic clean water to flow under the baffle and over the weir into the effluent water compartment. The effluent water compartment shall be provided with pumps to route the separated water to the secondary treatment facility.

iv. The effluent water compartment shall also be provided with an emergency overflow to the evaporation pond. The invert level of the overflow pipe should be located as high as possible to maximise the storage volume within the separator as well as to ensure no reverse flow is possible.

v. The separator arrangement shall be prepared / designed based on the process data sheet.

e. See in Figure 24 and Figure 25 in Appendix A1 for typical layout sketches.

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10.10 Supporting Infrastructure

a. Designs of primary treatment facilities (such as AOC First Flush Interceptor and Oily Water Separator) shall address the following:

   i. safe personnel access for operation and maintenance activities including:

      1. access to skimmers, pumps, plate packs and filters;

      2. access for gulley sucker, vacuum truck and mobile cranes;

      3. appropriate ventilation.

   ii. utilities and lighting:

      1. perimeter lighting;

      2. water supply for flushing and/or pump seal cooling.

   iii. tagging (e.g., unit number, capacity);

   iv. instrumentation, including field instrumentation and automation shall be designed and engineered with control systems, safety, fire and gas systems as an integrated part of the ICSS. For details refer to AGES-PH-04-001 “Automation and Instrumentation Design Philosophy” and AGES-SP-04-007 “Instrumentation for Packaged Equipment Specification”.

b. If the SS network includes septic / holding down tank, the following shall be addressed:

   i. provision of necessary level detecting sensors / level gauge.

   ii. Provision of double seal heavy duty cover and access ladder (if required) for facilitating inspection / maintenance works.

   iii. Provision of sump at one corner inside the tank for emptying.

10.11 Hydrocarbon Recovery

10.11.1 General

a. Floating hydrocarbons from effluents at gravity separators shall be skimmed. The selection of type and location of the skimming device required shall consider the following factors:

   i. type of treatment facility (e.g., Oily Water Separator, first flush interceptor), and requirements for downstream treatment;

   ii. the quantity of floating hydrocarbon expected;

   iii. the economics (costs versus recovery);

   iv. operability (manual versus automatic, inclusive of maintainability).

10.11.2 Skimmers

a. Skimmers shall be designed to prevent trash being skimmed (e.g. provided with trash rack or bar screen).

b. Fixed pipe skimmers shall only be used when the water level in an interceptor is always constant.

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c. Rotating disc or drum skimmers should be used where continuous oil skimming operation is required. Continuously operating disc or drum skimmers shall be powered by hydraulic or pneumatic power packs.

d. Floating skimmers should be considered for continuous/non-continuous operation where high fluctuations in the water level are expected.

e. Rotating trough skimmers with hand wheel operation:

   i. should be used in interceptors where the fluctuation in water level does not exceed 75 mm;

   ii. shall not be operated during 'rain';

   iii. shall always be kept in an upright position when unattended.

f. A typical application of a rotating trough skimmer (hand wheel operated) is shown in Figure 20, in Appendix A1.

10.11.3 Oil Sumps

a. Sumps intended to collect skimmed hydrocarbons shall incorporate the following design features:

   i. the bottom shall be sloped to facilitate removal of sediment or sludge;

   ii. sumps shall be sized to provide personnel access for inspection purposes and sumps with vertical centrifugal cantilever type pumps shall be accessible without removal of the pumps.

   iii. sump capacity shall be based on a minimum pump running time of 10 minutes.

10.11.4 Pumps

a. The design of pumps to discharge skimmed hydrocarbons from oil sumps shall consider the following:

   i. the pump type shall be vertical centrifugal or vertical screw (low shear) type which is capable of handling sludge etc. as required and selected to suit the expected viscosity of the oil slops;

   ii. the location of the pumps shall be adjacent to or above the sump;

   iii. pump capacities shall be selected based on the capacity and / or design of the downstream facility;

   iv. where manual skimming operations are present, low viscosity hydrocarbon with high water content may be present.

   v. by default the installation of a spare pump is required.

   vi. all pumps installed shall have the following:

      1. level control set for automatic start and stop;

      2. Low-Low (LL) level control with alarm and switch off;

      3. High-High (HH) level control switching on the pumps.

   vii. Pump requirements shall be as per AGES-SP-05-008 “Centrifugal Pumps for General (Utility) & Fire water service Specification” or relevant ADNOC Business Unit specification.

b. A typical arrangement and line-up of vertical centrifugal pumps at an oil collection sump is shown in Figure 21, in Appendix A1.

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10.11.5 Operational Instructions and Requirements

a. It is important that regular skimming of the free oil is carried out, recommended to do at least twice a week. Setting the adjustable skimmer in a continuous skimming position is an alternative but is not recommended as it often results in skimming more water than oil.

b. Pump start and stop functions (level switches and manual) to be tested once a week.

c. Since the interceptor configuration is recommended to be an open structure, dirt and sand will find its way into the sump and will settle at the bottom. The bottom profile is to be designed with a slope towards a cleaning channel, where the collected dirt can be removed by means of a vacuum truck / gulley sucker.

10.12 Evaporation Pond

a. Where local regulation prohibits discharge of clean rainwater or treated water into public water then an evaporation pond shall be considered.

b. The Evaporation Pond shall be sized to provide appropriate area and depth to contain and evaporate the greater of the following:

   i. the peak fire water demand as per project specific credible FFW case

   ii. Storm water run-off collected via AOC, COC and CWS drainage systems (as applicable) for one-hour peak intensity rainfall based on Tc (minimum 10 minutes) or 12-hour average rainfall with a 5-year return period.

   iii. Project specific storm water event / effluent drainage philosophy.

   iv. A minimum freeboard of 0.6 m shall be considered.

   v. For design an average evaporation rate of 0.4 mm/hr shall be considered.

c. The Evaporation Pond shall be of single compartment or double compartments (with a dividing wall having an overflow section) as per maintenance and operation philosophy.

d. Evaporation pond shall be lined with impervious material (HDPE membrane). See AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction Specification” for further details.

e. Specific requirements for ADNOC Onshore alone refer to APPENDIX A2.

10.13 Housekeeping Instructions and Requirements

a. Preventive maintenance and good housekeeping shall be practiced at the source of all drainage networks and shall continue through the systems and treatment facilities up to and including the outfall to the public waters.

b. Regular maintenance, housekeeping and inspection activities and water flushing (to maintain liquid seal at all times) shall be considered as a safety and environmental care item.

c. The responsibility for housekeeping and maintenance of drainage networks, treatment facilities and any other related disposal activities shall be assigned to plant personnel or departments qualified and trained for these particular duties. The overall coordination shall be allocated to a supervisor, who shall maintain a complete record of the overall networks and treatment facilities updating these as necessary to include as-built situations.

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d. Records shall be kept of all cleaning and other related activities (spill control), to establish upgrading and cleaning / maintenance schedules.

e. Recommended maintenance and housekeeping intervals are as follows:

Table 10.1 Recommended maintenance and housekeeping intervals

Item	Recommended Intervals
Sumps, manholes, catch basins	6 months
Channels and underground piping	6 months
Inspection for sludge cleaning	3 months
Tank compound (valve controlled)	Every 6 months & after every major rainstorm

f. However, it is also to be noted that sludge build-up in the sump is a gradual process and so a proper condition-based maintenance program should be developed to address the issue of preventive maintenance for sump cleaning.

10.14 Area Classification

a. For area classification, primary water treatment facilities (e.g. AOC Interceptor and Oily Water Separator) shall be considered as hydrocarbon containing equipment and their location shall be subject to the minimum safety distances noted as follows:

   i. Distance to the edge of public roads: 30 m

   ii. Distance to the edge of main roads: 15 m

   iii. Distance to a fixed source of ignition: 15 m (unless otherwise ascertained by a Fire Risk Assessment study)

b. Prevailing winds and meteorological conditions shall be taken into consideration when choosing the location and orientation of the treatment facility.

c. For sumps / manholes that may contain flammable liquid or hydrocarbon, minimum safety distance shall be 15 m away from any ignition source. The distance maybe reduced for manholes with tight seal cover, but hydrocarbon gases need to be vented to a classified safe area (in relation to emissions).

10.15 Closed Drain Trenches

a. Where required based on type of service and process / HSE requirements, concrete trench shall be provided for closed drain pipes. Concrete trenches housing closed drain pipes shall be backfilled with clean, dry sand to prevent the collection of gas within the trench and shall have removable covers.

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11 MATERIALS

11.1 General

a. Drainage component materials should meet the design criteria (including the effluent chemical composition, effluent temperature and the environmental conditions) while being the most economical selection based on availability and facility lifecycle costs.

b. Drainage networks and components should be of leak-proof design. System tightness shall be controllable and measurable during the design life of the facility.

c. All materials used in the drainage networks (including surface finishes) and exposed to effluents shall be resistant against the effluent constituents and duration of exposure in relation to corrosion and/or chemical attack. Material selection shall be based on the most aggressive medium upstream of the connection.

d. All materials used in the construction of the drainage networks shall be resistant to erosion from the effluent velocities to which they are exposed.

e. All materials used shall conform to the current edition of the appropriate standard / code / specification. Certificates of compliance shall be provided.

f. Products manufactured to equivalent standards require the approval of the COMPANY.

g. Materials shall be stored in such a manner as to preserve their quality and condition to the standards required by the specification.

h. Materials and components shall be handled in such a manner as to avoid any damage or contamination.

i. Material for all other drainage networks (not included in this specification) shall be as per Process and Piping drawings and specifications.

j. Piping material for various drainage networks shall be as shown on the project drawings and documents.

11.2 Materials for Clean Water Systems (CWS)

11.2.1 Glass-Fibre Reinforced Epoxy Pipe (GRE)

a. GRE pipe systems shall be in accordance with BS EN ISO 14692, Parts 1 to 4 ‘Petroleum and natural gas industries. Glass-reinforced plastics (GRP) piping’. MANUFACTURER’S recommendations and limitations shall comply with the requirements of ISO 14692 that shall cover requirements such as jointing (considering bell and spigot adhesive joint system), Temperature limitations, UV protection liner, 1000 hr regression test, Fire safe requirement for installation in process area etc.

11.2.2 High Density Polyethylene (HDPE) pipe

a. HDPE piping materials shall conform to ASTM F894.

b. Joints shall be spigot and socket with “O” ring Nitrile Rubber gaskets or butt fusion jointed. Installation shall be in accordance with ASTM D2321 and pipe manufacturer’s recommendations.

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c. Minimum wall thickness shall be based on the following:

NPS	SDR
100 – 400 mm (4 – 16 inches)	17
450 – 1200 mm (18 – 48 inches)	32.5

11.2.3 Reinforced Concrete Pipe (RCP)

a. RCP and fittings shall comply to BS 5911-1 and BS EN 1916 and shall be Class 120 as per Table 8 in BS 5911-1 or ASTM C76 equivalent and shall be bell and spigot type only.

b. Joint rings shall be Type WG seals in accordance with BS EN 681 with a hardness of 46 to 55. The material shall be acrylonitrile butadiene rubber (NBR). The size and shape of jointing shall be specified by the MANUFACTURER.

c. Pipes, fittings and joints shall comply with all test requirements in BS 5911-1 and BS EN 1916. The MANUFACTURER shall supply a certificate of compliance with the requirements of this standard.

d. The products shall be covered by a third-party certification scheme such as BSI certification or a recognized equivalent. If an equivalent is used it shall be subject to approval by COMPANY.

11.3 Materials for culverts

11.3.1 Pipe Culverts

a. Material for pipe culvert shall be in accordance with Section 11.2 above.

11.3.2 Reinforced Concrete Box (RCB)

a. RCB sections shall be in accordance with ASTM C1433 unless otherwise specified in the contract documents.

b. The appropriate table of section requirements from the listed ASTM standard shall be used for the design loading specified in the contract documents.

c. Joints shall be pre-formed plastic gasket-type joint sealant, unless specified otherwise in the contract documents.

d. The size of RCB shall be shown on the construction drawings. The concrete material and reinforcement shall be in accordance with AGES-SP-01-001 “Concrete Supply & Construction Specification”.

e. Protective coatings shall be in accordance with AGES-SP-01-001 “Concrete Supply & Construction Specification”.

11.4 Materials for AOC and COC Systems

11.4.1 Glass-fibre Reinforced Epoxy (GRE)

a. GRE pipe systems shall be in accordance with BS EN ISO 14692, Parts 1 to 4 ‘Petroleum and natural gas industries. Glass-reinforced plastics (GRP) piping’. MANUFACTURER’s recommendations and limitations shall comply with the requirements of ISO 14692 that shall cover requirements such as jointing (considering bell and spigot adhesive joint system), temperature

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limitations, UV protection liner, 1000 hr regression test, fire safe requirement for installation in process area etc.

b. Seals between GRE piping and concrete pits shall be watertight and flexible. It is recommended that a modular seal be used (see Section 11.9.3). All components, including pipe sleeve and seal shall be compatible with the effluent and the external conditions.

11.4.2 High Density Polyethylene (HDPE)

a. HDPE pipe and fittings shall be manufactured from virgin high-density polyethylene resin, in accordance with ASTM D3350, Cell Class PE345434C minimum, with a standard dimension ratio (SDR) as specified in the contract documents.

b. Joints shall be either butt fusion welded or socket fusion welded conforming to ASTM F894. Installation shall be in accordance with ASTM D2321 and pipe manufacturer's recommendations.

c. Minimum wall thickness shall be based on the following:

NPS	**SDR**
100 – 400 mm (4 – 16 inches)	17
450 – 1200 mm (18 – 48 inches)	32.5

d. Seals between HDPE piping and concrete pits shall be watertight and flexible It is recommended that a modular seal be used (see Section 11.9.3). All components, including pipe sleeve and seal shall be compatible with the effluent and the external conditions.

e. When selecting HDPE material for AOC / COC systems due consideration should be given to temperature and chemical sensitivity – especially to aromatic hydrocarbons.

11.5 Materials for Sanitary Sewer (SS) Systems

11.5.1 Ductile Iron Pipe (DIP)

a. DIP shall conform to requirements of ASTM A 746, and fittings shall be cast iron conforming to ASTM A 74.

b. Joint material for DIP shall be rubber gasket conforming to ASTM C 564 for compression joints.

c. Ductile Iron Pipe shall be Polybond lined for sewage services and the material shall comply with ASTM D 4976.

11.5.2 Unplasticized Polyvinyl Chloride (PVC-U)

a. PVC-U material used in SS shall conform to Table 11.1

Table 11.1 PVC-U Pipe

Service	Material	Standard	Joints
Non pressure flow	PVC-U	BS EN 1401 - 1	Bell and spigot.
Pumped Sanitary sewer system	PVC-U	BS EN ISO 1452 (Parts 1 to 5)	Bell and spigot with solvent welded joints

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11.5.3 High Density Polyethylene (HDPE) pipe

a. HDPE material used in SS shall conform to Table 11.2

b. HDPE material used shall be able to withstand effluent temperature as stated in Section 12.4

Table 11.2 HDPE Pipe

Service	Material	Standard	Joints
Non pressure flow	HDPE	ASTM F894	Spigot and socket with "O" ring Nitrile Rubber gaskets
Pumped Sanitary sewer system	HDPE (solid wall)	ASTM F894 / BS ISO 4427	Butt fusion jointed

c. Minimum wall thickness shall be based on the following:

NPS	**SDR**
100 – 400 mm (4 – 16 inches)	17
450 – 1200 mm (18 – 48 inches)	32.5

11.6 Materials for Chemical Effluent

a. Piping material for chemical effluents shall be as shown on project P&ID.

11.7 Materials for Other Systems

a. Piping materials for all pressurised services and closed drain systems shall be as defined within project Piping Line Class specifications (see also 'Piping Material Specification Index', AGES-SP-09-002).

11.8 Materials for Manholes, Catch Basins, Sumps & Neutralisation Pits

11.8.1 Concrete Manholes, Catch Basins and Sumps

a. Manholes, catch basins and sumps shall be constructed in-situ with reinforced concrete or precast units.

b. Precast concrete manholes and catch basins shall conform to BS EN 1917 / ASTM C478 and shall be constructed as shown on the construction drawings. The bottom slab and first wall section shall be monolithic.

c. All manholes, sumps and catch basins in hydrocarbon service shall be designed as water retaining structures.

d. Concrete and protection for drainage structures including, but not limited to manholes, catch basins, sumps, pits and support blocks shall conform to the requirements of AGES-SP-01-001 "Concrete Supply & Construction Specification".

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11.8.2 Neutralisation pits

a. Neutralisation pits shall be of reinforced concrete construction and shall be lined in accordance with AGES-SP-01-001 “Concrete Supply and Construction Specification”.

11.9 Accessories for Manholes and Catch Basins

11.9.1 Gratings

a. All gratings for drainage systems shall be designed to support the project loading criteria for their location (in accordance with AGES-SP-01-002 “Structural Steel Supply, Fabrication and Erection Specification” and AGES-SP-01-003 “Structural Design Basis – Onshore Specification” for material and design respectively).

b. Gratings (gulley tops) and frames for catch basins may be of either grey or ductile iron castings in accordance with BS EN 124-2. They shall be heavy duty D400 in accordance with BS EN 124.

c. Channel gratings and frames shall be hot dipped galvanised in accordance with AGES-SP-07-004 Painting and Coating Specification.

11.9.2 Manhole Steps, Frames and Covers

a. Manhole frames and covers shall be grey or ductile iron castings in accordance with BS EN 124-2 / ASTM A48.

b. Manhole steps / rungs may be castings or bent steel rod in accordance to BS EN 13101. Steps shall be plastic encapsulated or galvanised to protect from corrosion. Alternatively, fixed ladders may be used and shall be in accordance with BS EN 14396.

c. The manhole cover and frame shall be heavy duty D400, medium duty C250 or light duty B125 in accordance with BS EN 124, as shown on the construction drawings and on COMPANY’s standard drawing, “Sanitary Sewer and Storm Water Manhole & Catch Basin Details”. As a minimum the following shall apply:

   i. Heavy duty D400 shall be used in road carriageways, road hard shoulders, heavy duty paved areas, heavy vehicle parking areas and other comparable areas.

   ii. Medium duty C250 shall be used in light duty paved areas, light vehicle parking areas, areas inaccessible to heavy vehicular traffic and other comparable areas.

   iii. Light duty B125 shall only be used in pedestrian areas and areas where no vehicular access is envisaged.

d. The CONTRACTOR shall paint or dip the frames and covers in accordance with the requirements of AGES-SP-07-004 “Painting and Coating Specification”.

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11.9.3 Pipe Penetration Seals

a. Pipes that penetrate the walls of manholes and catch basins shall be adequately sealed against leaks. Refer to COMPANY standard drawings for specific details on seals. Pipe penetration seals include:

   i. Puddle flange: where the pipe is provided with a puddle flange and cast into the wall. A single strip of hydrophilic sealant around the pipe may be required.

   ii. Puddle flange in precast wall: additional strips of hydrophilic sealant shall be used when casting the boxed-out hole in a precast wall.

   iii. Modular seals shall be mechanical type, consisting of interlocking synthetic rubber links shaped to continuously fill the annular space between pipe and wall opening. The elastomeric element shall be sized and selected per the MANUFACTURER's recommendations and shall be Model O, oil-resistant, nitrile rubber in accordance with ASTM D2000 M1BF510. Each individual modular seal shall be clearly and permanently identified with the name of the MANUFACTURER and the model number.

11.10 Pipe Bedding Materials

a. Bedding materials for Underground pipes shall be granular material as per details in construction drawings and in accordance with AGES-SP-01-006 "Site Preparation, Earthworks and Bund Construction Specification".

12 DESIGN PARAMETERS FOR DRAINAGE NETWORKS

12.1 General

a. The following minimum requirements shall be applicable to designs in order to achieve functional drainage networks.

12.2 Safety and Environment

12.2.1 Plant Safety

a. Drainage systems at operating facilities shall be designed to ensure that plant safety is not impaired by the drainage systems. The design shall ensure that the drainage systems do not contribute to the possibility of propagation of fire from one area to the next should a plant fire break out. Design of the drainage system shall be subject to Hazard Identification (HAZID) and HAZOP reviews in accordance with HSE-RM-ST03 and HSE-RM-ST04, respectively.

b. Operation and maintenance effort for drainage system may involve work in confined spaces. Human access into underground drainage components shall be considered as the last option. Wherever possible, unmanned operation, e.g., by means of CCTV / inspection camera shall be adopted for inspections or facilities shall be designed to enable maintenance activities to be accessible from above ground like provision of adequate clean-out or rodding points.

c. Operational and maintenance staff responsible for primary treatment facilities such as interceptors need adequate access and walkways to undertake their tasks in a safe manner. CONTRACTOR shall ensure adequate walkways, handrails and access points are provided within the design.

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d. CONTRACTOR shall consult with COMPANY's operations and maintenance personnel on the specific requirements of safety in design.

e. Use of nitrogen for purging into the confined spaces should be avoided due to asphyxiation hazards.

12.2.2 Prevention of fire propagation

a. To prevent the propagation of fire, detailed elements of the drainage networks shall [PSR] be in accordance with the following:

i. AOC and COC Headers are to be laid horizontally and be designed as liquid-filled with a minimum head of 150 mm.

ii. Firetraps shall be provided to isolate one plant area from another and shall always be provided at road crossings. Sand traps in AOC and COC networks shall [PSR] be provided upstream of a firetrap and at the lowest point in a trench or channel.

iii. The liquid head in firetraps shall be 150 mm.

iv. Typical details of a sand and firetrap are shown on Figure 5 in Appendix A1.

12.2.3 Prevention of hydrocarbon vapour build-up

a. Hydrocarbon vapor build-up in the drainage system shall [PSR] be relieved by the following:

i. Sealed sumps or manholes shall be provided with a vent pipe with a minimum diameter of 75 mm for those supported by structures and 100 mm for the self-standing type. Vent pipes shall be carbon steel.

ii. Vent pipes shall be vented to a safe location in order to avoid potential ignition sources, and the volume around the vent outlet shall be classified in accordance with EI-15.

iii. For a sealed manhole with more than one compartment, vent pipes shall not result in any direct connection between the different compartments.

iv. Sumps and manholes that may contain flammable liquid, or hydrocarbon shall maintain a minimum safety distance of 15 m from any ignition source or a distance determined by application of EI-15, whichever is greater.

12.3 Flow Velocities

a. Velocities shall be kept within a range that prevents damage to the pipes and fittings and allows self-cleansing.

b. Maximum flow velocities shall be:

i. CWS pipes and culverts: 2.5 m/s

ii. AOC and COC pipes: 0.9 m/s (to avoid emulsification).

iii. Concrete lined channels: 3.0 m/s

iv. Other open channel linings, such as riprap, or grass: the maximum velocity shall be limited to avoid erosion.

v. Unlined Ditches: 0.9 m/s (subject to COMPANY approval).

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c. These velocities may be exceeded only under the firefighting design condition. The impact of higher velocities on upstream surcharging shall be identified by design calculation.

d. Minimum Velocities for all systems:

   i. shall be 0.4 m/s.

   ii. Sanitary sewers shall be designed to keep solids suspended and transported to the treatment facility without stagnant pockets where solids can settle out and collect.

12.4 Effluent Temperature

a. Network components shall be designed for the greater of the maximum ambient temperature or the following effluent temperatures:

   i. 30 °C for CWS system

   ii. 45 °C for AOC and COC systems

   iii. 50 °C for SS system

12.5 Design of Underground Pipes

12.5.1 Pipe Design

a. Branch connections shall be at 45° angle with the flow direction. Exceptions can be allowed to give branch connections at an angle of 90° if enough rodding-out points are provided.

12.5.2 Slopes

a. The preferred minimum slope for CWS systems shall be 5 mm per meter (1:200). This may be relaxed where hydraulic calculations dictate otherwise and provided the required minimum velocity is maintained to avoid deposition of solids.

b. AOC and COC pipes shall be laid flat (flooded system).

c. The preferred slope for sanitary sewer headers should be approximately 10 mm per meter (1:100).

12.5.3 Burial Depth

a. The minimum soil cover above the underground piping shall be 0.6 m (from surface to top of pipe).

b. When no load carrying protection is provided above underground pipes, the minimum burial depth shall consider the strength of pipe in relation to the live loads (e.g., vehicles / craneage) and dead loads surcharging the pipe. Refer to BS EN 1295-1 'Structural design of buried pipelines under various conditions of loading' and BS 9295 'Guide to the structural design of buried pipes'.

c. All types of crossings and external loads that act on GRE / HDPE / PVC piping of drain system during different phases such as construction, operation and maintenance shall be considered by CONTRACTOR / SUPPLIER in the design and related calculations shall be submitted for COMPANY approval.

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d. Typical burial depths are:

i. 1.2 m underneath roads and channels;

ii. 1.0 m underneath pedestrian walkways, cycle paths;

iii. shallower burial depths typically require structural protection of the pipe.

12.5.4 Segregation

a. Vertical segregation between pipes shall be in place to ensure no point loads can be exerted between the pipes during the lifetime of the plant.

b. Potable water lines shall always be installed above non-CWS piping.

c. Horizontal segregation between drainage piping shall be based on the results from the constructability review.

d. Horizontal segregation between other utilities and drainage piping shall be based on constructability as well as specific separation requirements related to the utility and/or cathodic protection interference.

12.6 Controlled Flooding

a. Temporary storage of AOC effluent in pipetracks is allowed, provided that the safety and environmental requirements are met. The maximum allowable effluent level during any surcharging condition shall not be higher than 150 mm below the top of the pipe sleeper.

b. Flooding is not permitted in any other area – except as detailed in Sections 12.10.2 and 12.11.

12.7 Rainfall Peak Flow / Accumulated Flow

a. Peak flow calculations shall determine the design of drainage systems.

12.7.1 Rainwater Peak Flow Design Parameters

a. Rainfall 'Intensity-Duration -Frequency' curves shall form the basis of all hydraulic calculations.

b. The following rainfall data shall be considered for the design of drainage systems (based on Figure 8, Rainfall Depth–Duration-Frequency Curves for Abu-Dhabi International Airport, in Appendix A1):

i. the overall drainage system rainfall intensity shall be based on a 5-year return period (frequency).

ii. Peak intensity for individual catchment areas shall be calculated from rainfall 'Intensity-Duration-Frequency' curves using the catchment Tc (time of concentration). See example in Figure 10, in Appendix A1.

iii. The hourly rainfall intensity to be used to obtain discharge rate for design of drainage system shall be calculated based on the computed Tc. Refer example below for guidance:

For Tc = 15 minutes

Rain fall depth = 12 mm (Figure 8, Appendix A1)

Rainfall intensity, i = 12 x (60/15) = 48 mm/hr (value to be used to calculate discharge, Q_{rain})

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c. In case detailed local meteorological data is made available, this shall be assessed, and the relevant location-specific rainfall intensity to be used shall be agreed with COMPANY.

12.7.2 Run-off Values - Rational Method

a. Calculations for establishing the run-off values shall be carried out using established and proven methods such as the Rational Method. Another acceptable method would be the Colebrook-White equation. The information below is from the Rational Method:

Q_{rain} = C i A

Where:

Q_{rain} = quantity of rainwater run-off in m^3/hr

C = run-off coefficient

i = design rainfall intensity (m/hr) based on the time of concentration (T_c)

A = catchment area under consideration in m^2.

b. The Rational Method assumptions and design parameters are as follows:

i. The maximum run-off resulting from a particular rainfall intensity occurs if the duration of rainfall is equal to or greater than the time of concentration (Tc). The time of concentration is defined as the time required for water to flow from the most distant point of the catchment area under consideration to the point of flow measurement (investigation). See Section 12.9.

ii. The maximum run-off resulting from a particular rainfall intensity, whose duration is equal to or greater than the time of concentration, is directly proportional to the rainfall intensity.

iii. The peak discharge per unit area decreases as the drainage area increases, and the intensity of rainfall decreases as its duration increases.

iv. The coefficient of run-off remains constant for all storms on a given watershed.

v. The method may only be applied for surface water run-off by gravity for an area not larger than approximately 150 hectares (1,500,000 m2). For areas greater than 150 hectares, the total area should be sub-divided into smaller catchment areas. For each individual catchment area, the Rational Method may then be applied.

12.8 Run-Off Coefficient

a. The permeability of surface finishes and in some instances the evaporation of rainwater affect the quantity of rainwater/firewater which enters a drain system.

b. Table 12.1 lists some typical run-off coefficients which have been selected for drainage network design for various types of surface finishes.

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Table 12.1 Run-off Coefficients

Label	Area	Coefficient
A	Plant area/utility area Roofs, concrete, asphalt and other paved areas	1.0
B	Gravelled and other unpaved areas	0.25
C	Manifold and pump slab areas	1.0
D	Roads and road shoulders	0.95
	Brick roads/tiled areas	0.75
E	Pipe tracks (general)	0.50
	Pipe track concrete finish	0.90
F	Bundwalls (average)	0.55
	Bundwalls with seepage prevention	0.90
G	Tank roofs	1.0
H	Tank compounds areas (unpaved)	0.3
	Tank compound areas with seepage prevention	0.9
	Grassed areas (sandy soil, flat)	0.1

12.9 Time of Concentration / Entry / Flow

a. The time of concentration (Tc) shall be used to determine the applicable rainfall intensity which, when used in the Rational Formula, will determine the peak flow values at the point of measurement in the system. The minimum time of concentration (Tc) shall be 10 minutes. See Appendix A1 Figure 10 Example of Rainfall Intensity / Duration Curves for Various Periods.

b. The time of entry or 'overland flow'(Te) shall be taken as the time required for one rain droplet to enter a drain channel in the drainage network. The time of overland flow within typical plant facility areas normally varies between 5 and 25 minutes. Figure 9 (in Appendix A1) shows typical Te values for various surface slopes and finishes.

c. The time of flow (Tf) shall be taken as the time required for the effluent to flow through the drainage network or an element thereof.

d. For open drain channels and for non-liquid-filled pipes, the time of flow (Tf) is the time required for the effluent to flow through the relevant channel length, i.e. length divided by the velocity.

e. For liquid-filled pipes the time of flow should be taken as the time required for the system to reach stable flow conditions and is referred to as Tdf. Typically, this value can vary between 2 and 5 minutes.

f. Thus, for drainage networks the following T_c values should be used:

   i. For open drain channels and for non-liquid-filled pipes:

$$T_c = T_e + T_f$$

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ii. For liquid-filled pipes only:

$$T_c = T_e + T_{df}$$

iii. For a combination of liquid-filled pipes and open drain channels the overall network calculations should be simplified using the formula:

$$T_c = T_e + T_f$$

12.10 Fire Fighting Water (FFW)

12.10.1 General Design Parameters

a. FFW flow / intensity and duration shall be determined from data and scenarios prepared by CONTRACTOR's Safety department and approved by COMPANY.

b. The quantity of FFW to be collected and discharged through the drainage system shall be based on firewater calculations performed in line with AGES-PH-03-002, Part 4, Fire & Gas Detection and Fire Protection System Philosophy.

c. Under the FFW design condition all available drainage networks may be used to discharge firefighting water and the parameters governing the velocities in the individual drainage network elements are not applicable.

d. Normally, FFW design condition will govern design of upstream networks, whereas the rainfall design condition will govern downstream networks.

e. Used FFW shall [PSR] not be discharged directly to public water (sea, etc.) / evaporation pond. Retention facilities shall be provided to allow inspection and analysis for decision making with respect to treatment requirements or discharge to public water.

12.10.2 Tank Compounds

a. Where automatic spray water systems are installed, these will be used to cool adjacent tanks in the event of a tank on fire. The drainage network and outlet valve shall be designed such that the water level inside the tank bund area will not exceed a height of 0.3 m above the lowest tank pad during a fire fighting condition. A practical valve size is 300mm. If the design requires a larger outlet, additional outlet valves should be selected rather than increasing the valve diameter.

12.11 Indirect Flows

a. The design of drainage networks shall consider the maximum use of indirect flows to reduce peak treatment capacity.

b. Indirect flows can be achieved by buffering of rainwater at defined areas during a rainstorm and releasing the buffered effluents only after treatment of the direct run-off effluents has been completed.

c. The buffering of rainwater can be achieved by keeping the valve-controlled outlet of the tank compound closed during a rainstorm. Sequential release of buffered rainwater shall take place within a maximum period of 6 hours after a given storm. The compound shall have adequate buffer volume such that the level of buffered rain shall not reach the level of the underside of the tank base.

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12.12 Design Flow

12.12.1 CWS

a. CWS drainage networks and their components shall be designed for the maximum rainfall flow to which they are likely to be exposed.

12.12.2 AOC

a. AOC drainage networks shall be designed to convey without flooding the maximum flow resulting from the greater of the following load combinations:

   i. Rainfall (Q_{rain}) + any other known contributing continuous 'dry weather flow'.

   ii. Fire water (Qfire) + any other known contributing continuous 'dry weather flow'.

12.12.3 COC

a. The flow requirements for COC shall be provided by Process / P&ID documents / Project Drainage Philosophy.

12.12.4 Sanitary Sewer SS

a. Design flows for the Sanitary Sewer System shall be based on personnel numbers as specified by the 'Project Design Data" in conjunction with building architectural drawings showing plumbing fixture counts. Corresponding flow rates are recommended in 'Table 2 – Discharge units (DU)' of BS EN 12056-2.

b. If septic / holding tank is provided, it shall have seven days capacity corresponding to the number of persons considered in the design flow.

12.13 Provision for Future Expansion

a. The drainage system shall be designed to allow for the identified future expansion where applicable / specified in the project specific drainage philosophy.

13 HYDRAULIC CALCULATIONS

13.1 General

a. Hydraulic calculations shall be prepared to achieve an economical and effective drainage network design. All drainage piping, sewers and open drain channels shall be designed in accordance with accepted standard formulae and calculation methods, including sump influences (losses), weir overflow losses, and friction coefficients which consider the anticipated future condition of the drainage network.

b. All calculations shall be submitted to COMPANY for review and approval.

c. Although there are many 'empirical' formulae which can be used, the formulae given in Section 13.2, should be used in order to simplify and standardize the design method and presentation. As an alternative the formulae (based on Manning's equation) given in APPENDIX A3 may be used.

d. The design of drainage networks should be based on gravity flow where the installed depth of the underground pipes relates directly to the weir level at the main outlet point, e.g. treatment facility.

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See Figure 3 in Appendix A1. At locations where the available positive head is insufficient to achieve a gravity flow condition, pumping facilities shall be provided.

13.2 Hydraulic Gradients

a. In both open channel and pipe flow the fall of the energy gradient for a given length of channel or pipe represents the loss of energy by friction.

b. The hydraulic gradient (I) can be determined by using Chezy's formula:

$$I = \frac{V^2}{C^2 R}$$

Where:

V = velocity [m/s]

C = Chezy coefficient [$m^{0.5}/s$]

R = hydraulic radius, or hydraulic mean depth [m]

I = hydraulic gradient or incline [m/m]

R is the relationship between the amount of liquid being conveyed and the contact area between this liquid and the inside of the channel.

R = cross sectional area of flow divided by the wetted perimeter

c. The Chezy coefficient can be calculated using the simplified Colebrook formula:

$$C = 18 \log_{10} \left[\frac{12\,R}{k} \right]$$

Where:

C = Chezy coefficient [$m^{0.5}/s$]

R = hydraulic radius [m]

k = Nikuradse wall roughness factor [m]

d. The value k is independent of the diameter of the drainage pipeline. Table 13.1 gives k values for various materials. Note that for GRE material the influence of the boundary layer in the calculations of the Chezy coefficient has been considered. The influence of the boundary layer can be disregarded for k values of 0.5 mm and greater.

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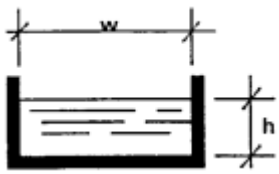
Table 13.1 Nikuradse Wall Roughness Factor "K" For Various Materials

Material	k (mm)
Concrete	2
Steel (non-corroded)	0.5
Steel (corroded)	5
GRE / HDPE / PVC-U	0.2 (includes influence of boundary layer)

e. Open Channels

i. For a rectangular drain channel, R is given by:



$$R = \frac{(h\ w)}{2h + w}$$

ii. C values for open drain channels, based on the Colebrook formula, are shown in Figure 11 in Appendix A1.

iii. If precast drain channels are used, Manufacturer's design charts may be used to establish I, V and C.

iv. All reference documentation shall be enclosed in any design calculation packages prepared by CONTRACTOR.

f. Liquid-filled pipes

i. For 100% liquid filled drainage pipes, Chezy's formula can be simplified as follows:

$$(1) \qquad I = \frac{V^2}{C^2 R}$$

ii. For liquid filled drainage pipes, the hydraulic radius (R) will be:

$$(2) \qquad R = \frac{\pi D^2 / 4}{\pi D} = \frac{D}{4}$$

iii. V is derived as follows:

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(3) $$V = \frac{Q}{\pi (D^2 / 4)}$$

iv. Inserting (2) and (3) into (1) gives:

$$I = \frac{64\, Q^2}{\pi^2 \times D^5 \times C^2}$$

v. The relationship between I and Q can be written as:

$I = \alpha \times Q^2$

Where

I = incline (slope) [m/m]

Q = flow [m^3/s]

α = alpha

$$\alpha = \frac{64}{\pi^2 \times D^5 \times C^2}$$

vi. α is related to the pipe material. The α factors are shown in Table 13.2, using C values calculated with Colebrook's formula.

Table 13.2 "C" and "alpha" Values for Liquid-Filled Pipes (based on Chezy's formula)

Diameter (m)	GRE/HDPE/PVC-U K = 0.2 mm		Concrete k =2 mm		Steel k = 0.5 mm		Corroded steel k = 5 mm	
	C	alpha $*10^{-3}$	c	alpha $*10^{-3}$	c	alpha $*10^{-3}$	c	alpha $*10^{-3}$
0.10	57.2	198 403.47	39.2	422 650.13	50.0	259 312.5	32.0	632 991.41
0.15	60.3	23 454.36	42.3	47 636.18	53.2	30 198.59	35.2	69 011.59
0.20	62.6	5 173.04	44.6	10 192.73	55.4	6 596.52	37.4	14 467.73
0.25	64.3	1 604.42	46.3	3 093.20	57.2	2 031.65	39.2	4 327.94
0.30	65.8	617.13	47.8	1 170.00	58.6	777.24	40.6	1 619.31
0.35	67.0	275.34	49.0	515.00	59.8	345.25	41.8	706.62
0.40	68.0	136.92	50.0	253.23	60.8	171.06	42.8	344.99
0.50	69.8	42.65	51.8	77.48	62.6	52.97	44.6	104.37
0.60	71.2	16.46	53.2	29.49	64.0	20.35	46.0	39.39
0.70	72.4	7.36	54.4	13.05	65.2	9.07	47.2	17.30
0.80	73.4	3.67	55.4	6.44	66.3	4.51	48.3	8.50
0.90	74.3	1.99	56.3	3.46	67.2	2.43	49.2	4.54

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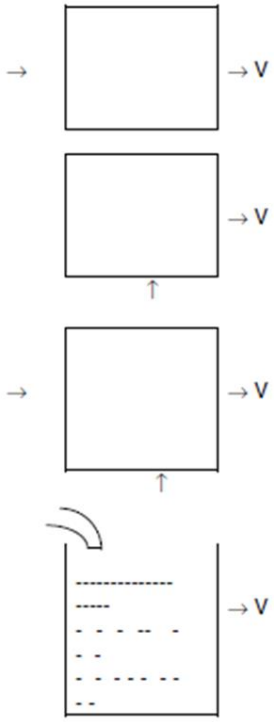
Diameter (m)	GRE/HDPE/PVC-U K = 0.2 mm		Concrete k =2 mm		Steel k = 0.5 mm		Corroded steel k = 5 mm	
	C	alpha *10^{-3}	c	alpha *10^{-3}	c	alpha *10^{-3}	c	alpha *10^{-3}
1.00	75.2	1.15	57.2	1.98	68.0	1.40	50.0	2.59
1.10	75.9	0.70	57.9	1.20	68.8	0.85	50.8	1.56
1.20	76.6	0.44	58.6	0.76	69.4	0.54	51.4	0.99
1.30	77.2	0.29	59.2	0.50	70.1	0.36	52.1	0.64
1.40	77.8	0.20	59.8	0.34	70.6	0.24	52.6	0.44
1.50			60.3	0.23	71.2	0.17	53.2	0.30
1.60			60.8	0.17	71.7	0.12	53.7	0.21
1.70			61.3	0.12	72.2	0.09	54.2	0.16
1.80			61.8	0.09	72.6	0.07	54.6	0.12
1.90			62.2	0.07	73.0	0.05	55.0	0.09
2.00			62.6	0.05	73.4	0.04	55.4	0.07

13.3 Manhole / Sump Losses and Weir Losses

a. Manhole / Sump losses can be determined by using the following simplified formulae:



$$P = 0.3\frac{V^2}{2g} = \frac{V^2}{66}$$

$$P = 1.5\frac{V^2}{2g} = \frac{V^2}{13}$$

$$P = (0.3+0.6)\frac{V^2}{2g}$$

$$P\ max = \frac{V^2}{2g} = \frac{V^2}{20}$$

$$P = \frac{V^2}{2g} = \frac{V^2}{20}$$

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Where:

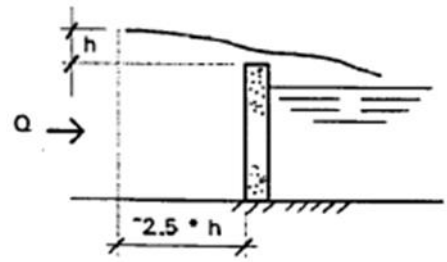
V = velocity (m/s)

P = sump head loss (m)

g = acceleration due to gravity (10 m/s^2)

b. Weir overflow losses can be determined by using the following simplified formulae:

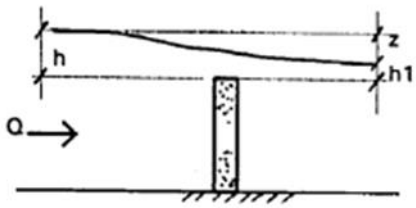
a) Free weir



$$Q = 1.84Lh^{3/2}$$

$$\text{or} \quad h = \left(\frac{Q}{1.84\ L}\right)^{2/3}$$

b) Submerged weir



$$Q = \left(1.84\ L\ z^{3/2}\right) + \left(2.8\ L\ h_1\sqrt{z}\right)$$

Where

Q = flow [m^3/s]

L = length of the weir [m]

h = liquid depth over the weir [m]

z = liquid depth over the weir [m]

13.4 Computer Programs

a. Computer programs used for design purposes shall consider that AOC and COC underground piping networks as required by this Specification are principally laid horizontally and liquid filled.

b. Programs and calculation methods proposed by CONTRACTOR shall be subject to review and approval by COMPANY. The programs shall be identified at an early stage and all required documentation shall be supplied to demonstrate their accuracy and applicability.

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14 INSTALLATION AND CONSTRUCTION

14.1 HSE in Construction

a. Construction of drainage systems and treatment facilities is particularly exposed to safety hazards involving confined space work, excavations, working at height (construction of deep basins and manholes) and lifting.

b. Work Method Statements (WMS) and Job Safety Assessments (JSA) with Risk Analysis (RA) for drainage and treatment activities shall be prepared and shall address these safety risks. These shall be submitted to COMPANY for approval.

c. For excavation work in general, measures shall include but are not limited to:

   i. calculations supported by WMS to ensure safe excavation and related work;

   ii. proper shoring (if required);

   iii. solid barriers and signage;

   iv. exclusion zones around excavations to prevent surcharge from vehicles and spoil heaps which may cause collapse;

   v. safe access and egress. In long excavations, access points shall be provided at 10 m intervals;

   vi. confined space permitting;

   vii. supporting for exposed cables.

d. CONTRACTOR's construction methods shall fully consider the dangers of excavation work in brownfield areas, specifically for the locating and identification of existing underground piping, cabling, and foundations. Work methods shall fully consider COMPANY permit requirements whilst working in a brownfield environment.

e. Excavation shall only commence on demonstration that project HSE and permitting requirements have been met.

14.2 Temporary Drainage and Settling Ponds

a. Temporary drainage networks necessitated during plant construction shall be undertaken as per ADNOC specification AGES-SP-01-006 "Site Preparation, Earthworks and Bund Construction".

b. All rainfall runoff that may occur during construction shall be collected by means of temporary drainage networks in order to facilitate access and safe construction.

c. CONTRACTOR shall provide temporary drainage diversion / pumping for tie-in into existing system where required.

d. Evaporation ponds and / or soakaways may be used for disposal of effluents collected as per COMPANY approval.

14.3 Excavation, Bedding, Backfilling, and Compaction

a. Unless otherwise specified or indicated on the construction drawings, trench excavation, pipe bedding, trench backfilling and compaction shall be executed in accordance with AGES-SP-01-006 "Site Preparation, Earthworks and Bund Construction".

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b. Excavation for underground pipes shall conform to the specified locations, elevations and slopes as indicated on the approved construction drawings, allowing for installation works. The trench depth and width shall be sufficient to allow inspection of the pipe, pipe joints and proper compaction of bedding material at bottom sections and sides of the pipes.

c. All excavated material shall be stacked at least 1.0 m from the edge of the trench and the size of the spoil bank shall be limited as determined by calculation so as not to cause instability of the trench sides.

d. Water shall not be allowed to accumulate in excavations.

e. Bedding and backfill material shall be placed symmetrically around and above pipes and structures such as culverts, manholes, catch basins to prevent eccentric loading.

f. The trench shall be backfilled and compacted to a density equal to that specified on the construction drawings or AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction Specification.

g. As soon as possible after pipe joints are made, backfill material shall be placed along the sides of the pipe with care to prevent movement of the line or grade.

h. Joints shall be left exposed until completion of leak testing.

i. Backfilling of trenches shall be completed within 24 hours after testing.

j. Backfilling shall be compacted in a manner that avoids displacement or damage to the pipe.

14.4 Pipe laying

a. All pipes in between manholes shall be laid in straight lines.

b. The installation sequence shall commence at the downstream end of the system.

c. Each length of pipe shall be inspected for defects and internal debris before installation.

d. Pipe defects that do not fall within the required specification shall be removed from location.

e. Pipe laying and joining shall be carried out in a dry excavation.

f. Pipes are to be carefully lowered into the trench, piece by piece.

g. Pipe shall be carefully placed to the line and grade specified and nested in the bedding in such a manner as to provide uniform support throughout its entire length.

h. Pipes with spigot or tongue ends shall be laid with spigot ends pointing in the direction of flow.

i. The joint faces shall be maintained clean and undamaged at all times.

j. Pipes shall be laid on a minimum bedding depth of 150mm in accordance with AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction”.

k. Open ends shall be protected with a stopper to prevent earth or other materials from entering the pipe during construction. The interior of the pipe shall be free from dirt, excess water and other foreign material as the pipe laying progresses, and up to the completion of the installation.

l. The method of installation and materials used for jointing of the different pipe material types shall be in accordance with the relevant MANUFACTURER's recommendations and instructions and shall be suitable for the expected conditions and temperature range within the system.

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m. GRE pipes are vulnerable to damage during installation and backfilling. Therefore, WMS, Inspection and Testing Plans (ITPs) and site controls shall be developed to carefully control handling of pipe, conditioning of bedding and backfill material and backfilling activities.

n. The MANUFACTURER of GRE, HDPE, etc. pipe shall provide training to the installation / erection team, supervision and QC personnel. Training shall cover general installation practices and particularly the integrity of joints. Training records shall be maintained by CONTRACTOR and shall be signed, stamped and submitted to the COMPANY.

o. UV protection for GRE piping shall be provided for storage, handling and installation.

p. Installation of HDPE and uPVC pipes shall be in accordance with ASTM D2321 and pipe MANUFACTURER's recommendations.

14.5 Joints

a. Joints shall be constructed as described herein, with the intent that they be made watertight. Inspection and testing procedures are described under Section 19.

b. Joint types for all pipe materials shall be as per the construction drawings and in accordance with this specification.

c. Joint installation procedures for all pipe materials shall be in accordance with the MANUFACTURER's instructions and recommendations.

d. Joint installation shall be performed by experienced, qualified personnel.

e. Bell and spigot joints shall be clean and dry. A lubricant recommended by the pipe MANUFACTURER shall be applied to the joint materials. The spigot shall have full bell penetration.

f. After the joints have been made, the pipe shall not be disturbed in any manner.

14.6 Pipe culverts

a. The type and class of pipes shall be as stated in Section 11.3. The sizes of pipes used shall be as shown on the construction drawings or as indicated herein. All inlet/outlet and junction/transition structures shall be constructed to the required line and grade as shown on the construction drawings.

14.7 Box culverts

a. Box culverts of reinforced concrete shall be constructed as indicated on the construction drawings. Bases, walls and roofs shall be smooth, plumb and true. Design of the culvert shall be in accordance with AGES-SP-01-003 "Structural Design Basis – Onshore Specification" and supporting calculations shall be submitted for COMPANY review/approval prior commencement of the work.

14.8 Manholes, Sumps, Catch Basins, Fire and Sand Traps

a. These shall be constructed in-situ with reinforced concrete or precast units. Where precast units are used, the bottom slab and first wall section shall be monolithic.

b. All concrete shall be installed as shown on the construction drawings and in accordance with AGES-SP-01-001 "Concrete Supply & Construction Specification".

c. Precast manholes shall be installed as per the MANUFACTURER's instructions and recommendations.

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d. All drainage structures and pits shall be of watertight construction.

e. Waterstops shall be provided at all joints in concrete construction in accordance with AGES-SP-01-001 “Concrete Supply & Construction Specification”.

f. Waterproofing materials for internal and external concrete surfaces shall be applied in accordance with AGES-SP-01-001 “Concrete Supply and Construction Specification”, and in strict conformance with the MANUFACTURER‘s instructions. Precast sections shall be coated prior to invert placement.

g. Manholes for various pipe sizes shall be as shown on the construction drawings.

h. Pipe openings shall be aligned to that of the pipe entering and leaving the manhole. Pipe shall be properly aligned with connections to manholes, sumps or catch basins as shown on the construction drawings. Where connections are not shown on construction drawings, pipe shall terminate flush with the inside of the structure wall.

i. Where pipes intersect the walls of manholes, sumps and catch basins they shall be sealed appropriately in accordance with this Specification and related standard details. This shall provide a watertight seal.

j. Frames and covers shall be set to the proper elevation. The frames shall be firmly embedded in mortar approximately 25 mm thick and aligned to fit the top section of the structure.

k. Partly constructed manholes shall be provided with timber or metal temporary covers.

l. The CONTRACTOR shall provide and hand over to COMPANY at completion of the works, two sets of lifting/locking keys to suit each specific pattern of installed manhole cover/grating. The lifting/locking keys shall be made of hot dip galvanized steel.

14.9 Concrete cradles

a. Concrete cradles shall be constructed / installed as shown on the construction drawings.

14.10 Channel Lining

a. Excavation of open drainage channels shall be as per ADNOC specification, AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction”.

b. Channels lined with rip-rap should be constructed as per ADNOC specification, AGES-SP-01-006 “Site Preparation, Earthworks and Bund Construction”.

c. Concrete lining shall be constructed in a smooth and symmetric pattern to the required thickness, line and grade as shown on the construction drawings.

d. Channel surfaces should be lined with a minimum 100 mm thick, grade 30F fiber-reinforced concrete layer, as per standard detail drawing, “Details of Ditches and Swales”.

e. All concrete shall be installed as shown on the construction drawings and in accordance with AGES-SP-01-001 “Concrete Supply & Construction Specification”.

14.11 Treatment Facilities

a. These facilities shall be constructed as per the construction drawings. All concrete work shall be in accordance with AGES-SP-01-001 “Concrete Supply & Construction Specification”. All steelwork shall be in accordance with AGES-SP-01-002 “Structural Steel Supply, Fabrication and Erection Specification”.

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14.12 As Built Drawings

a. CONTRACTOR shall prepare as-built drawings including all relevant details of the drain system. These shall be submitted to COMPANY in pdf and native format.

b. The relevant details for inclusion on as-built drawings (which shall be measured in the field) shall as a minimum include:

   i. the invert levels of drainage pipes at manholes, sumps and basins;

   ii. top of weir and other level controlling elevations;

   iii. the bottom inside level of sumps and manholes;

   iv. horizontal distance between sumps, bends, fittings, etc.;

   v. position and rotation of the kinks, bends, etc.; and

   vi. incorporate all technical queries, site modifications and approved changes to construction drawings.

15 ADDITIONAL SPECIFIC REQUIREMENTS

Not Applicable.

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SECTION C – OTHER REQUIREMENTS

16 QUALITY PLANS

16.1 General

a. Quality plans shall address all aspects related to local conditions, such as climatic conditions, backup facilities, spare parts, transport possibilities, storage facilities, quarries, local MANUFACTURER/SUPPLIERs, test facilities (field laboratory).

b. Quality Management Systems shall comply with the applicable requirements of BS EN ISO 9001 "Quality Management Systems - Requirements" and BS EN ISO 9004, "Quality management - Quality of an organization - Guidance to achieve sustained success". Written quality plans and procedures shall be submitted to the COMPANY for review and approval.

c. Materials should be obtained from the same source(s) throughout the work (including both trial and production mixes).

16.2 Inspection, Testing and Reporting

a. Inspection and Test Plans (ITPs) identifying the tests, frequencies, acceptance criteria, and responsibilities shall be prepared as part of the quality plan.

b. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.

c. The reporting system shall record all results obtained during the testing.

d. Weather conditions shall be recorded in the quality records.

e. The proposed ITPs and QCRs shall be included in the Quality Plan.

f. Standard forms of inspection and test plans shall be used as a basis for the development of the ITPs and QCRs supplemented by the requirements of the Specifications for the work to be executed.

g. The COMPANY reserves the right to make inspections of the source of supply of materials.

h. Prior to supply of any material to site the CONTRACTOR shall obtain COMPANY approval by submitting necessary documentation such as test certificates, source of supply, date of manufacture, etc.

16.3 Quality Control and Assurance

a. CONTRACTOR shall be solely responsible for quality control of all materials and workmanship. The quality system shall provide for the planned and systematic control of all quality-related activities performed during design and construction of all components of the drainage system. Implementation of the quality system shall be in accordance with the Project Agreement, CONTRACTOR's approved Quality Manual and Project Specific Quality plan.

b. CONTRACTOR shall ensure that SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERs have in place a QA/QC program which clearly establishes the authority and responsibilities of those responsible for the quality system. Persons performing quality functions shall have sufficient and well-defined authority to enforce quality requirements that they initiate or identify and to recommend and provide solutions for quality problems and thereafter verify the effectiveness of the corrective action.

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c. Quality System and Quality Control requirements shall be identified and included in CONTRACTOR's Purchase documentation. Based on these requirements, the SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERS will develop QA/QC programs which shall be submitted to COMPANY and CONTRACTOR for review and concurrence. The SUB-CONTRACTOR's, SUPPLIER's and MANUFACTURER's QA/QC programs shall extend to SUB-SUPPLIERS.

d. CONTRACTOR shall submit certified reports of all tests as soon as the tests are completed satisfactorily.

e. COMPANY reserves the right to inspect or to conduct a quality audit of materials, installation and workmanship standards and shall have unrestricted entry to the shop of the fabricator at all stages of manufacture and to witness any or all tests. SUB-CONTRACTORs, SUPPLIERs and MANUFACTURERs, within 30 days after award but prior to the pre-inspection meeting, shall provide CONTRACTOR and COMPANY with a copy of its Manufacturing and Inspection Plan for review and inclusion of any mandatory CONTRACTOR and COMPANY witness or hold points.

f. COMPANY may reject improper, inferior, defective, or unsuitable materials, installation and workmanship. All materials and workmanship rejected shall be repaired or replaced by CONTRACTOR as directed by the COMPANY.

17 SUB-CONTRACTORS, SUB-SUPPLIERS

a. CONTRACTOR shall assume unit responsibility and overall guarantee for the drainage systems.

b. CONTRACTOR shall transmit all relevant Purchase Order documents including COMPANY specifications to his SUB-CONTRACTORs and SUPPLIERs.

c. It is CONTRACTOR'S responsibility to enforce all Purchase Order and Project Specification requirements on its SUB-CONTRACTORS and SUPPLIERS.

d. CONTRACTOR shall submit all relevant SUB-CONTRACTOR and SUPPLIER engineering data to COMPANY.

e. CONTRACTOR shall obtain and transmit all SUB-CONTRACTOR and SUPPLIER warranties to COMPANY.

f. All drawings and engineering data submitted to COMPANY shall be in the English language, and in metric units.

18 MATERIAL CERTIFICATION

a. CONTRACTOR is responsible for the supply of all materials conforming to the relevant codes and standards, including all supply of material certificates/testing.

19 INSPECTION AND TESTING REQUIREMENTS

19.1 General

a. Drainage systems, upon completion or at such time as directed, shall be cleaned, inspected and tested. The system shall have a true grade and line. Actual elevations shall be within 25 mm of the elevations given on the construction drawings. Departure from and return to established grade and alignment shall not exceed 10 mm per meter.

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b. All underground drainage networks (piping, sumps, manholes, basins etc.) as described in this specification (AOC, COC, CWS, SS and chemical effluent) shall be tested for water tightness before and after backfilling. The CONTRACTOR shall submit to the COMPANY, for their prior approval, a testing programme for all underground drainage and sewerage systems. Inspection and testing should be carried out in general accordance with BS EN 1610 – Construction and Testing of Drains and Sewers. Smoke or air tests shall not be permitted.

c. CONTRACTOR shall notify COMPANY at least two (2) working days prior to his intention to test a section of drainage system. Items failing any testing requirements shall be corrected immediately and re-tested prior to the commencement of further work.

d. Unless otherwise stated in the Contract document, CONTRACTOR shall be responsible for providing all materials and apparatus required for testing purposes and for their removal and proper disposal upon completion of testing.

19.2 Visual Inspection

a. Visual inspection shall be carried out before backfilling in order to identify faults in construction or material not shown up under test. A careful record shall be kept of such inspections. On external pipelines, the following shall be visually inspected:

   i. Pipe bed.

   ii. Pipe line and level.

   iii. Joints.

   iv. Pipe protective coating.

   v. Any pipeline appurtenance.

19.3 Cleaning - Piping Systems

a. CONTRACTOR shall furnish an inflatable rubber ball of the size that will inflate to fit snugly into the pipe to be cleaned and tested. The ball may, at the option of the COMPANY, be used without a tag line, or a rope or cord may be fastened to the ball to enable the CONTRACTOR to know and control its position. The ball shall be placed in the highest cleanout or manhole on the pipe to be cleaned, and water shall be introduced behind it. The ball shall pass through the pipe with only the pressure of the water impelling it. All debris flushed out ahead of the ball shall be removed at the first manhole where its presence is noted. In the event of cemented or wedged debris or a damaged pipe stopping the ball, CONTRACTOR shall remove the obstruction and repair the damaged pipe at no cost to COMPANY.

b. Alternatively, for pipes of diameter 600 mm and above, inspection cameras can be used to ensure proper installation and internal cleanliness.

19.4 Alignment Test - Piping Systems

a. After backfilling has been placed and compacted to a depth not less than 300 mm above top of pipe, the system shall be filled with water. Filling shall be to a head not less than 1.5 m above the maximum head which the pipe will normally sustain or to the top of the sump, whichever is lower. Once filled, the system shall be drained to remove all water and CONTRACTOR shall make a visual inspection by flashing a light between manholes. Any displacement or misalignment of piping shall be corrected by CONTRACTOR.

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19.5 Leak Test for Drainage Systems

a. The bedding and back fill shall be sufficiently applied to support the pipe. The joints shall be left exposed until the completion and acceptance of the test by COMPANY.

b. The section of pipe to be tested shall be sealed between the upstream and downstream manholes of the pipe section.

c. The pipe shall be filled with water to a head of not less than 1.5 m above the maximum head which the pipe will normally sustain or to the top of the sump, whichever is lower.

d. After topping up and allowing for water absorption by the pipes and joints, the head shall be held for 24 hours to the satisfaction of COMPANY. Any visible leaks detected at the joints shall be repaired and the section shall be retested until compliance is achieved.

e. Leakage shall not be permitted in any circumstances. CONTRACTOR shall demonstrate that all underground piping systems are watertight.

f. All sumps, manholes and basins shall be hydraulically tested, by filling up to the top of the structure with water and visually show no leakage for at least 24 hours.

19.6 Record of Tests

a. CONTRACTOR shall keep a complete record of all tests carried out after installation. CONTRACTOR shall supply COMPANY with a copy of all the records as the testing proceeds and at the end of the Contract, or as otherwise requested by COMPANY. CONTRACTOR shall submit for approval a test record sheet and proposal for an underground drainage and sewerage Construction Records File.

20 PAINTING, PRESERVATION AND SHIPMENT

20.1 Packaging and shipping

a. Preparation of shipment shall be in accordance with COMPANY specification, “Preservation and Export Packing” and as noted herein.

b. CONTRACTOR shall be solely responsible for the adequacy of the preparation for shipment provisions with respect to materials and application.

c. Preparation for shipment and packing will be subject to inspection and rejection by COMPANY’s inspectors. All costs occasioned by such rejection shall be to the account of the CONTRACTOR.

d. One complete set of the installation, operation, and maintenance instructions shall be packed with the shipment.

20.2 Preservation and Storage

a. Piping materials shall be protected to withstand ocean transit and extended period of storage at the jobsite. Materials shall be protected to safeguard against all adverse environments, such as: humidity, moisture, rain, dust, dirt, sand, mud, salt air, salt spray and seawater.

b. Manufacturers’ recommendations for handling and storage of materials shall be followed. Care shall be taken in the storage of GRE piping which is subject to deterioration under long term exposure to sunlight, as well as fracturing and cracking under load.

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21 DOCUMENTATION / MANUFACTURER DATA RECORDS

a. CONTRACTOR shall submit the type and quantity of drawings and documentation for COMPANY's authorization or information as listed in the Construction Drawing Register.

b. CONTRACTOR shall maintain a complete up-to-date set of construction drawings at the jobsite.

c. Mutual agreement on scheduled submittal of drawings and engineering data shall be an integral part of any formal Purchase Order.

d. Comments made by COMPANY on drawing submittal shall not relieve CONTRACTOR of any responsibility in meeting the requirements of the COMPANY General Specifications. Such comments shall not be construed as permission to deviate from requirements of the Purchase Order unless specific and mutual agreement is reached and confirmed in writing.

e. Each drawing shall be provided with a title block in the bottom right hand corner incorporating the following information:

   i. Official trade name of the company.

   ii. CONTRACTOR's drawing number.

   iii. Drawing title giving the description of contents whereby the drawing can be identified.

   iv. A symbol or letter indicating the latest issue or revision.

   v. Purchase Order number and item tag numbers.

f. Revisions to drawings shall be identified with symbols adjacent to the alterations, a brief description in tabular form of each revision shall be given, and if applicable, the authority and date of the revision shall be listed. The term "Latest Revision" shall not be used.

g. Prior to commencement of construction CONTRACTOR shall submit an "Affidavit of Compliance" that confirm that all materials meet the requirements of this Specification.

h. Where applicable, the following documents shall be submitted to COMPANY prior to the start of any fabrication and construction activities:

   i. Inspection and Test Plan (ITP).

   ii. Inspection procedures document that provide details of compliance with the requirements in this Specification, the design drawings, and construction drawings.

   iii. Test Procedure Qualification and Test Acceptance Criteria.

   iv. Jointing procedure Method statement.

   v. Jointing personnel qualification and certification records.

   vi. Quality control program, quality plan, quality control procedure, including Quality Control Records (QCR's), records of quality inspection test reports, and test results.

   vii. Procedure Qualification Record (PQR).

   viii. Material Handling Procedure.

   ix. Materials and workmanship specifications.

   x. One set of final construction drawings.

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xi. Method statements covering construction methods, including temporary facilities and Health Safety Environment (HSE).

xii. Repair procedures.

xiii. Material Traceability Procedure.

xiv. Engineering calculations for each drainage system.

i. The following documentation shall also be provided to COMPANY:

i. MANUFACTURER's certifications shall be submitted to COMPANY for pipe, joint materials, and accessories, showing compliance with this Specification prior to installation.

ii. MANUFACTURER's certifications shall be submitted to COMPANY for precast manholes, sumps and catch basins including all accessories, showing compliance with this Specification prior to installation.

iii. All product data sheets shall be provided to COMPANY for review.

iv. All the inspection reports and test results or documents required by this specification shall be submitted to COMPANY prior to backfilling and within three days of testing.

v. The CONTRACTOR is to supply a full and final set of as-built Underground Services Layout drawings.

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SECTION D – STANDARD DRAWINGS & DATASHEETS

22 DATASHEET TEMPLATES

Not applicable.

23 STANDARD DRAWINGS

Not applicable.

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SECTION E - APPENDICES

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APPENDIX A1. FIGURES – TYPICAL DETAILS, SKETCHES AND GUIDES

All the sketches included under this appendix are conceptual and only for guidance on the general arrangement. For more details, reference should be made to COMPANY Standard Drawings as applicable.

Figure 1 Effluent Classification and Routing

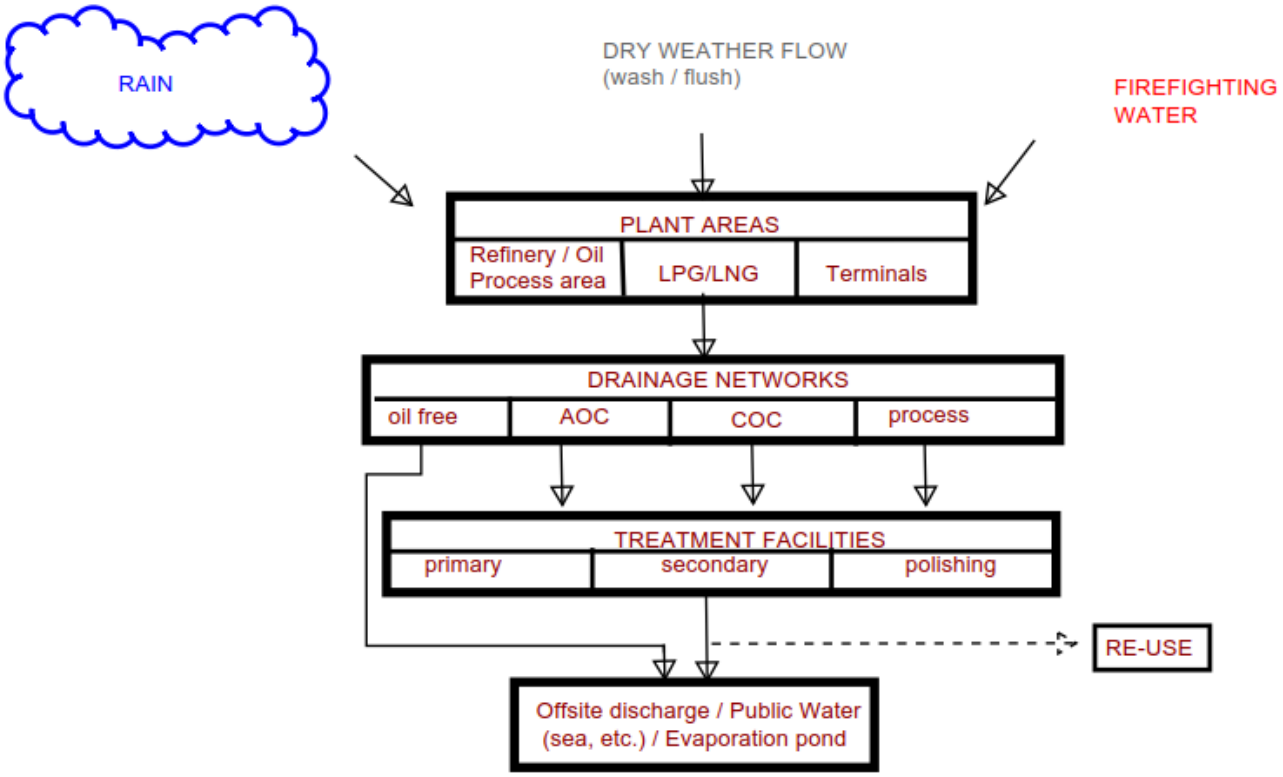

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Figure 2 Typical Process Area Surface Water Drainage

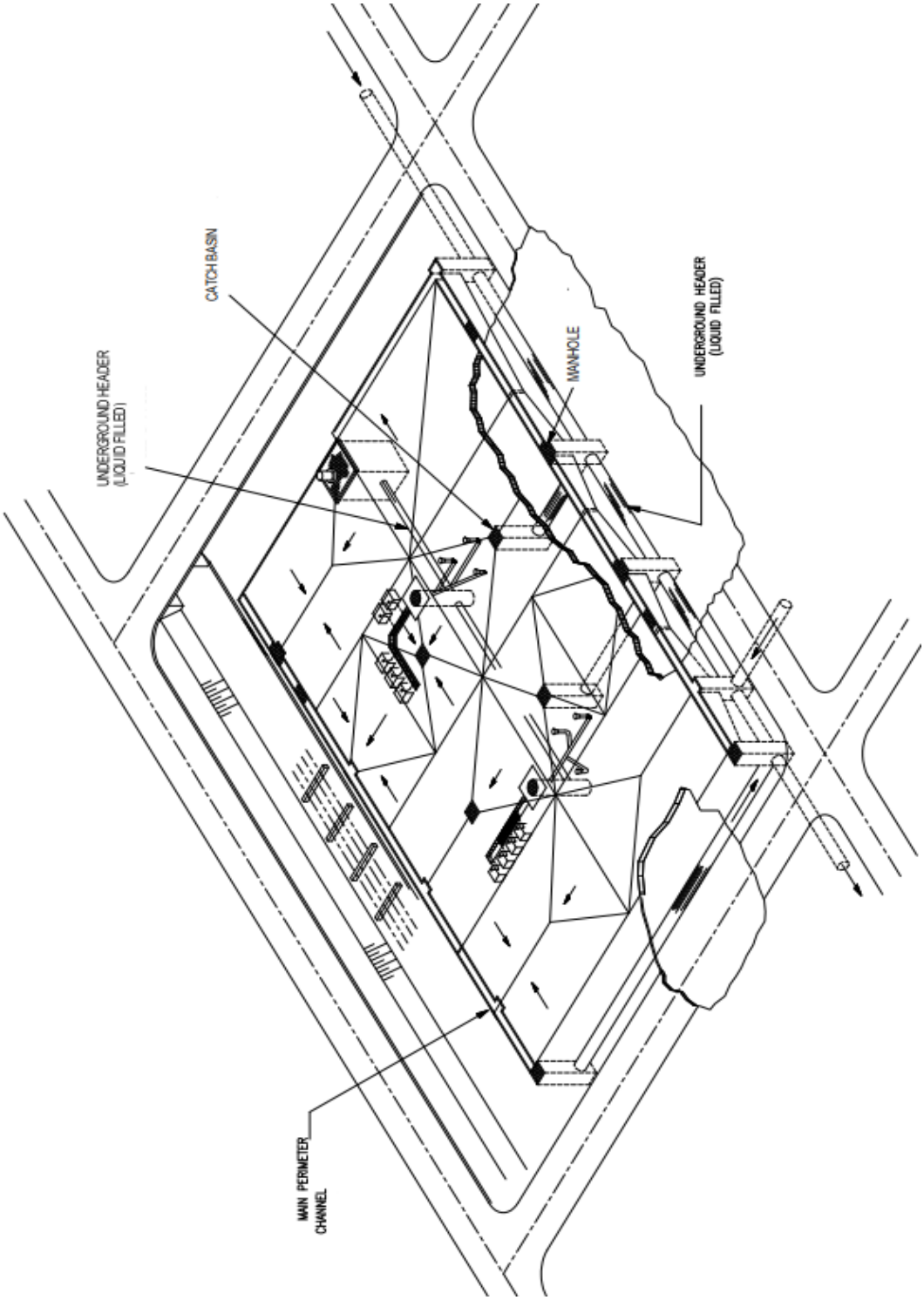

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Figure 3 Schematic Layout Liquid Filled Underground Drainage Network (AOC & COC)

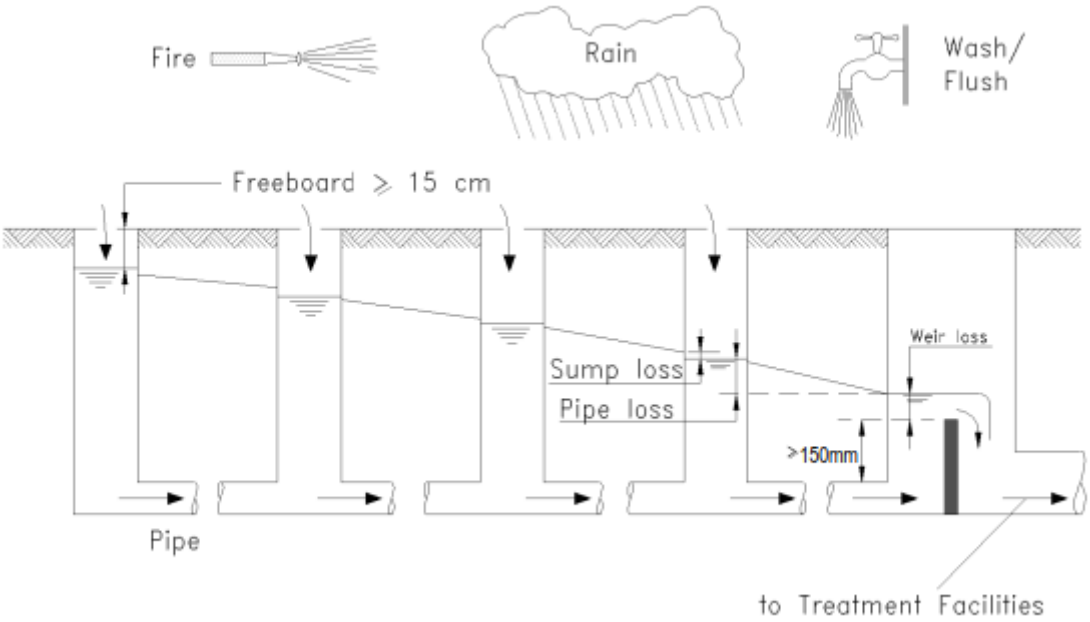

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Figure 4 Typical Tank Compound Drainage

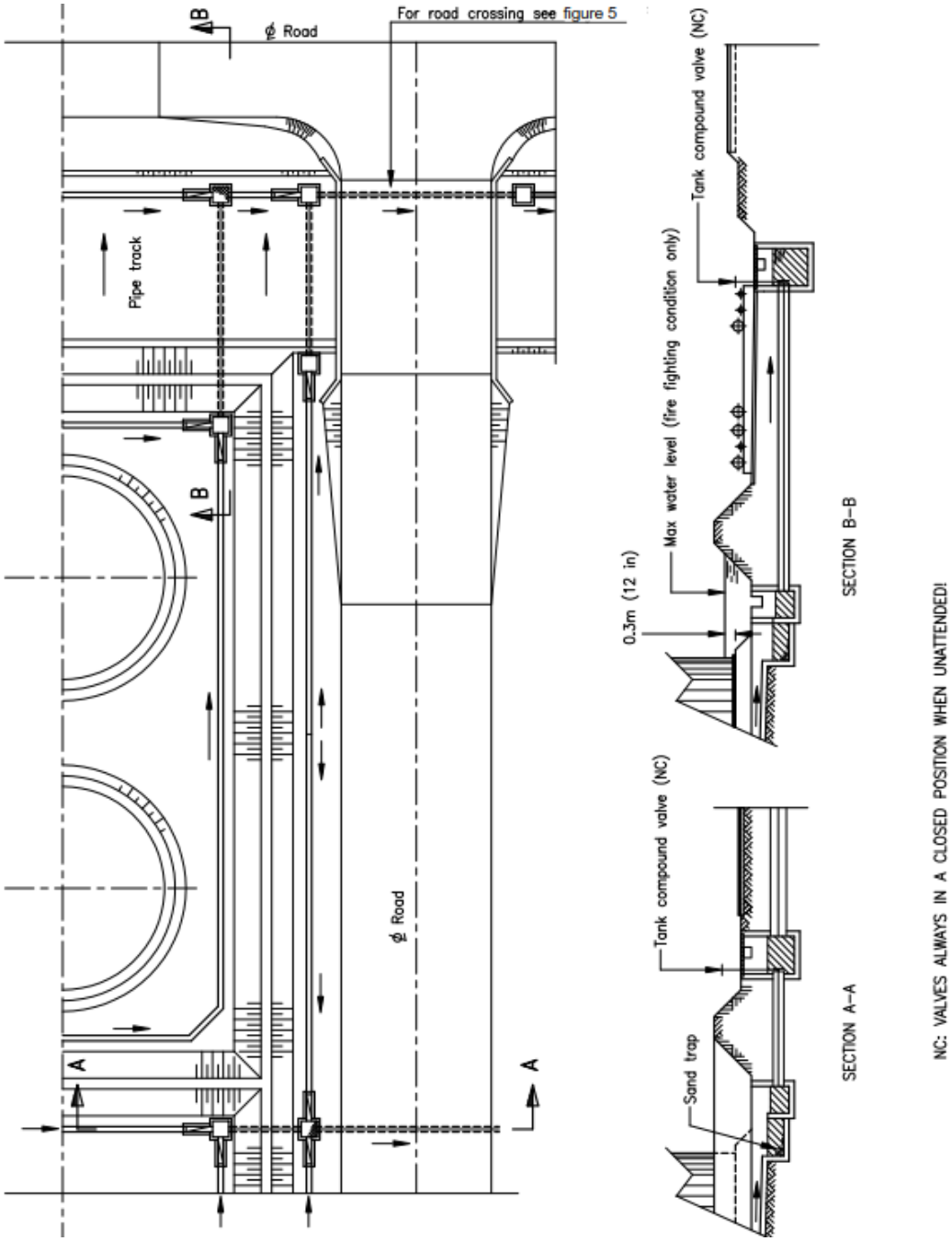

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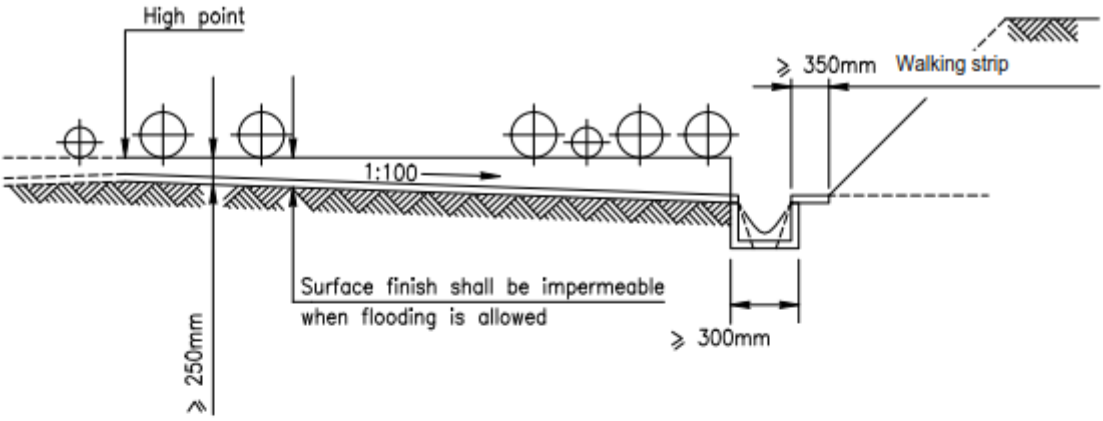
Figure 5 Typical Pipetrack Drainage / Sand Trap and Fire Trap Details



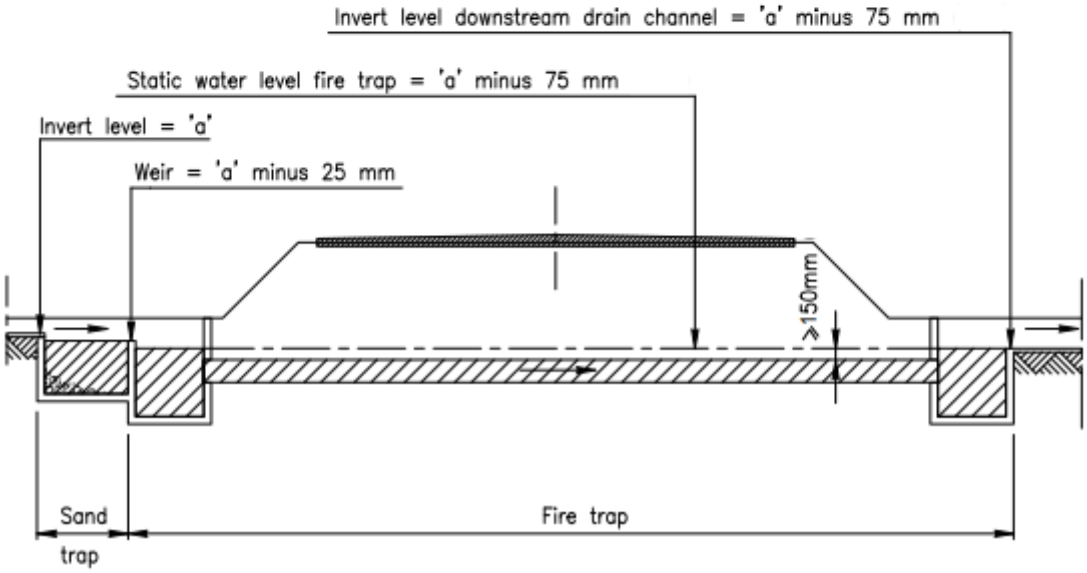


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Figure 6 Schematic Drainage Layout of Storage Areas in Liquified Gas Facilities

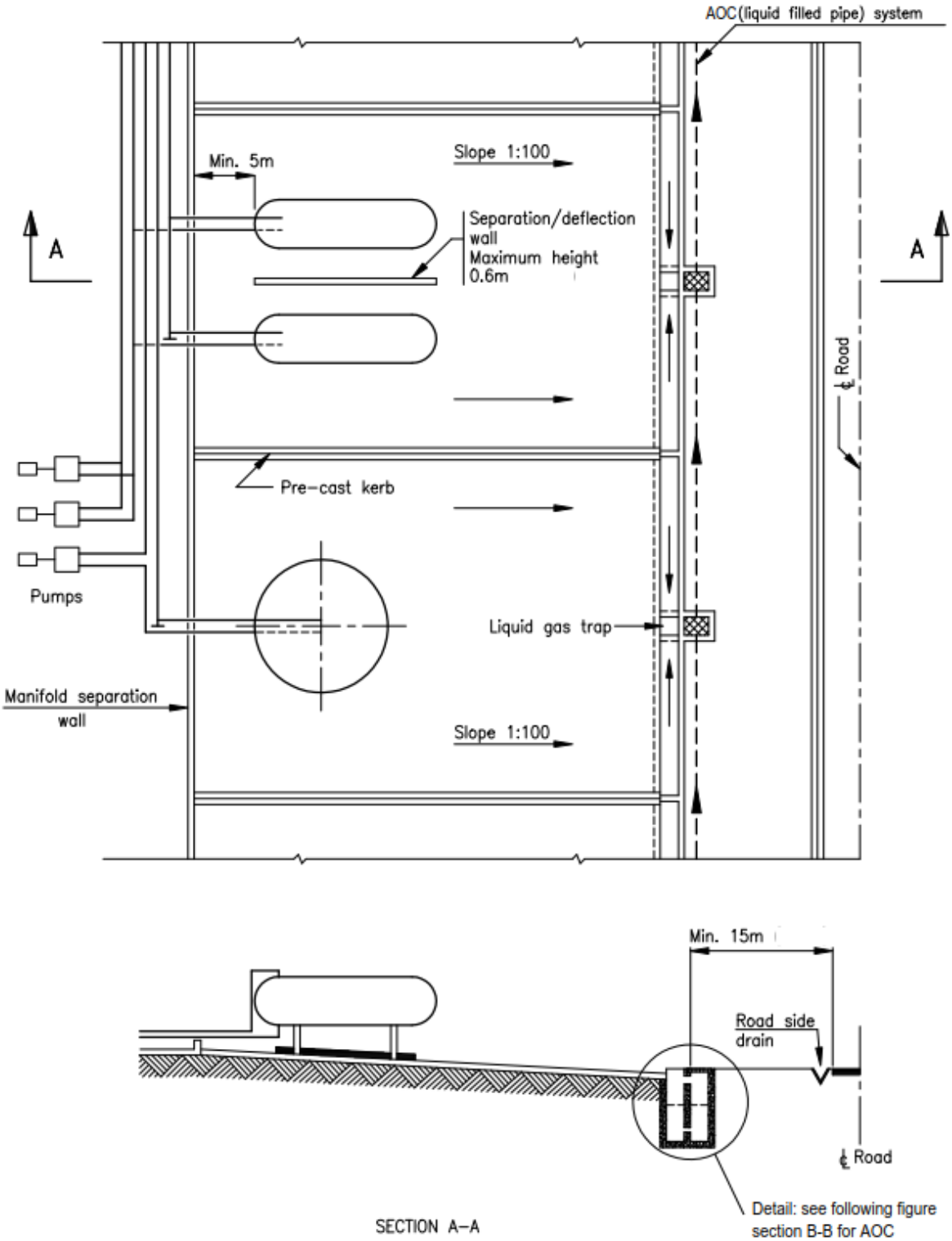

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Figure 7 Details for Liquified Gas Processing Area

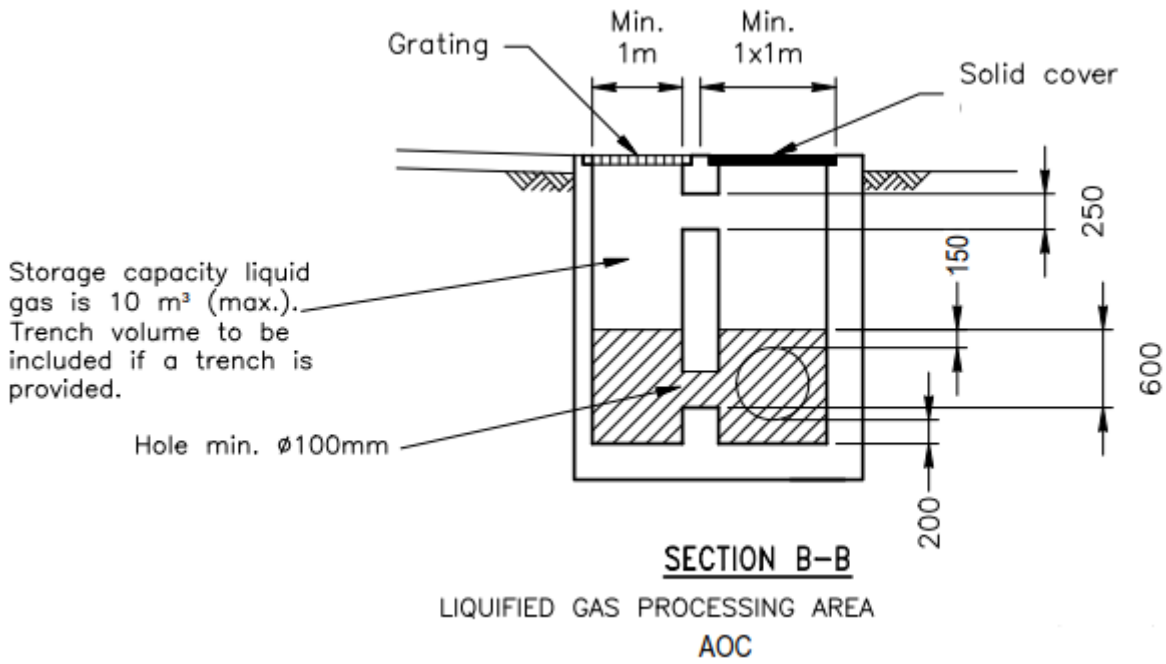

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Figure 8 Rainfall Intensity Duration (IDF) Curves for Various Recurrence Periods

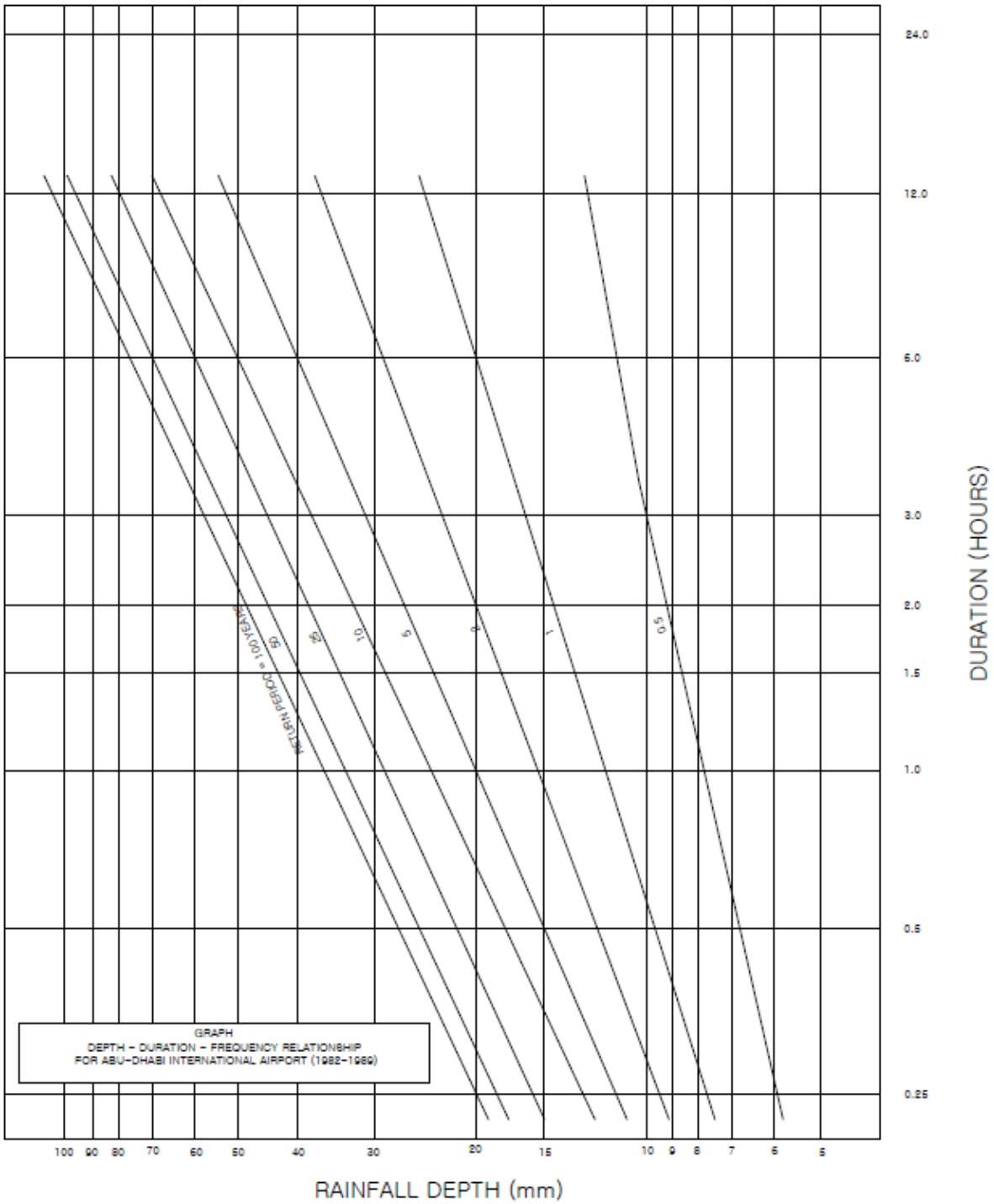

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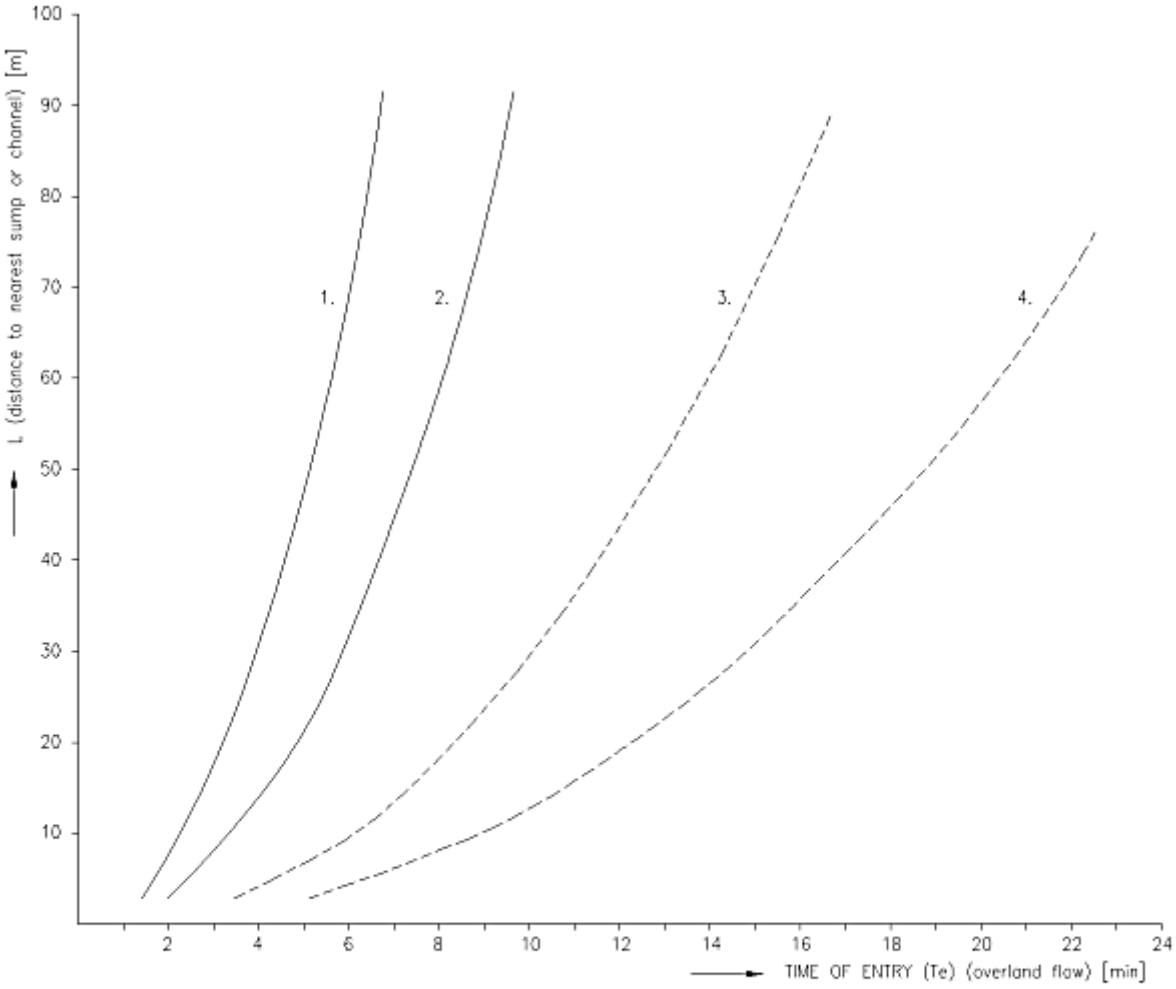
Figure 9 Time of Entry (Te) (Overland Flow)



1. = paved surface (n = 0.02), slope 1:200
2. = paved surface (n = 0.02), slope 1:1000
3. = bare surface (n = 0.15), slope 1:200
4. = bare surface (n = 0.15), slope 1:1000

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Figure 10 Example of Rainfall Intensity / Duration Curves for Various Periods

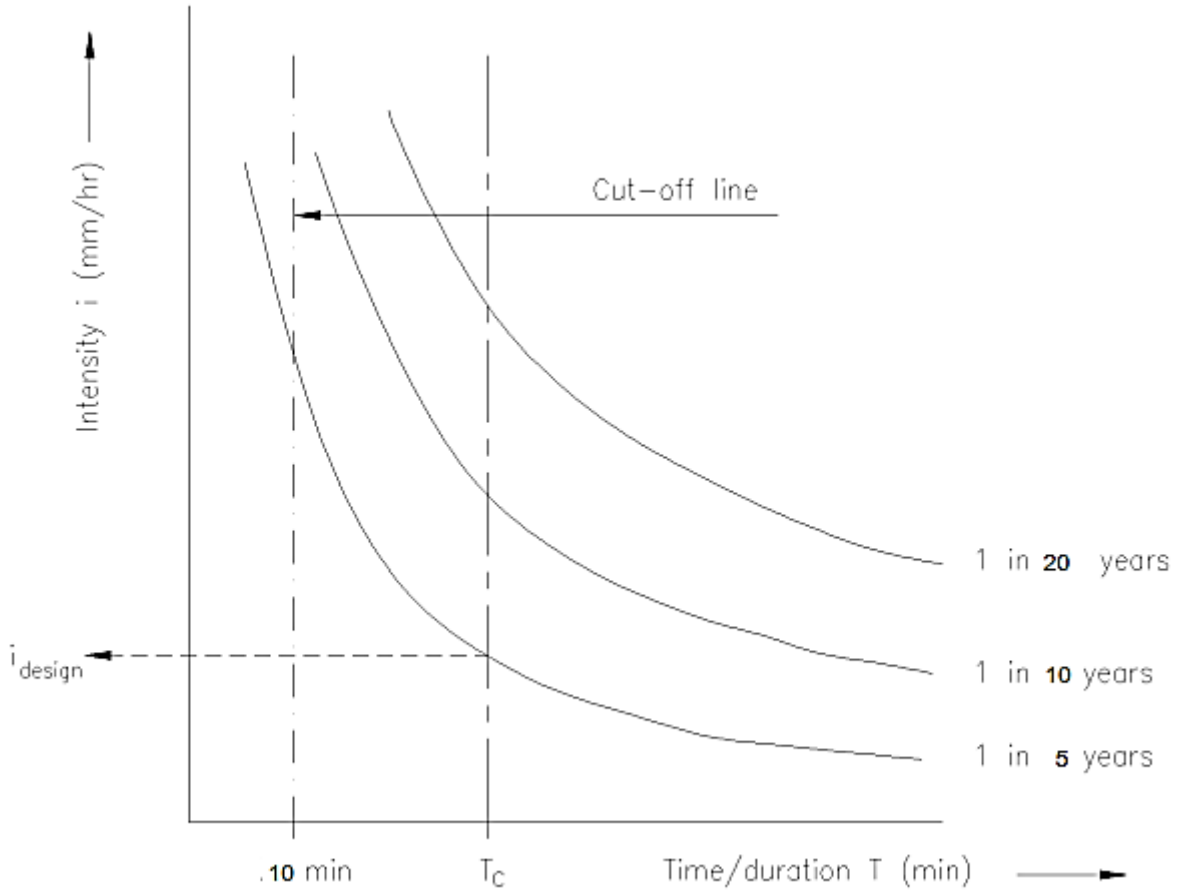

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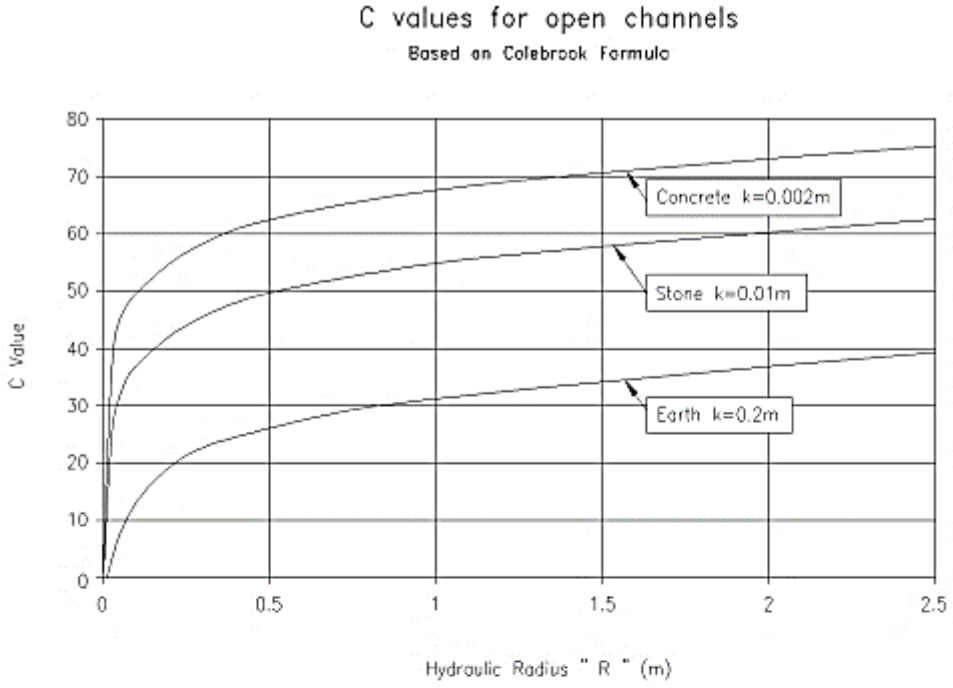
Figure 11 C Values for Open Channels (Colebrook)



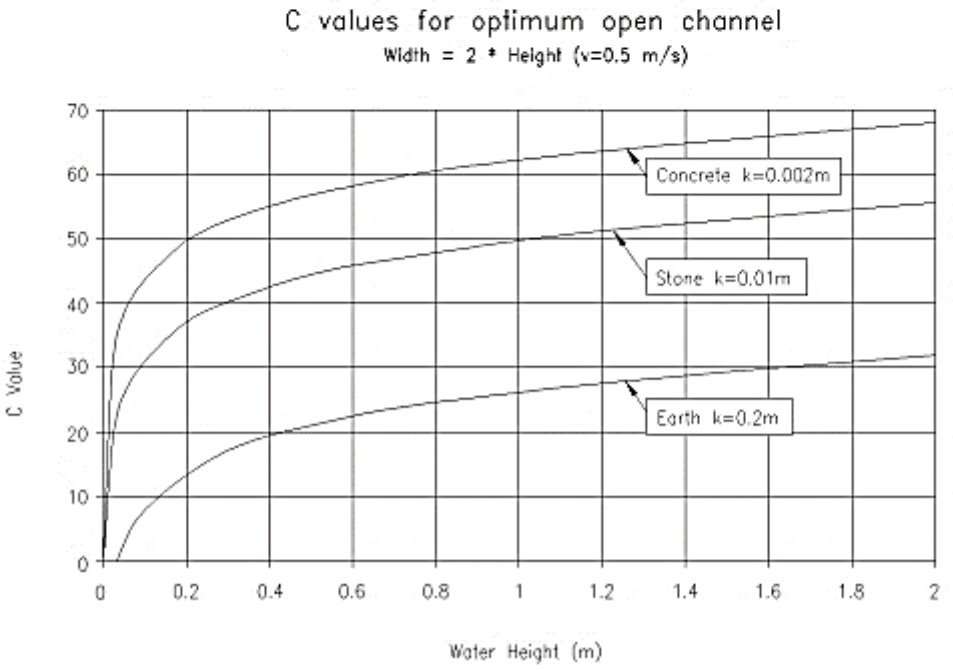


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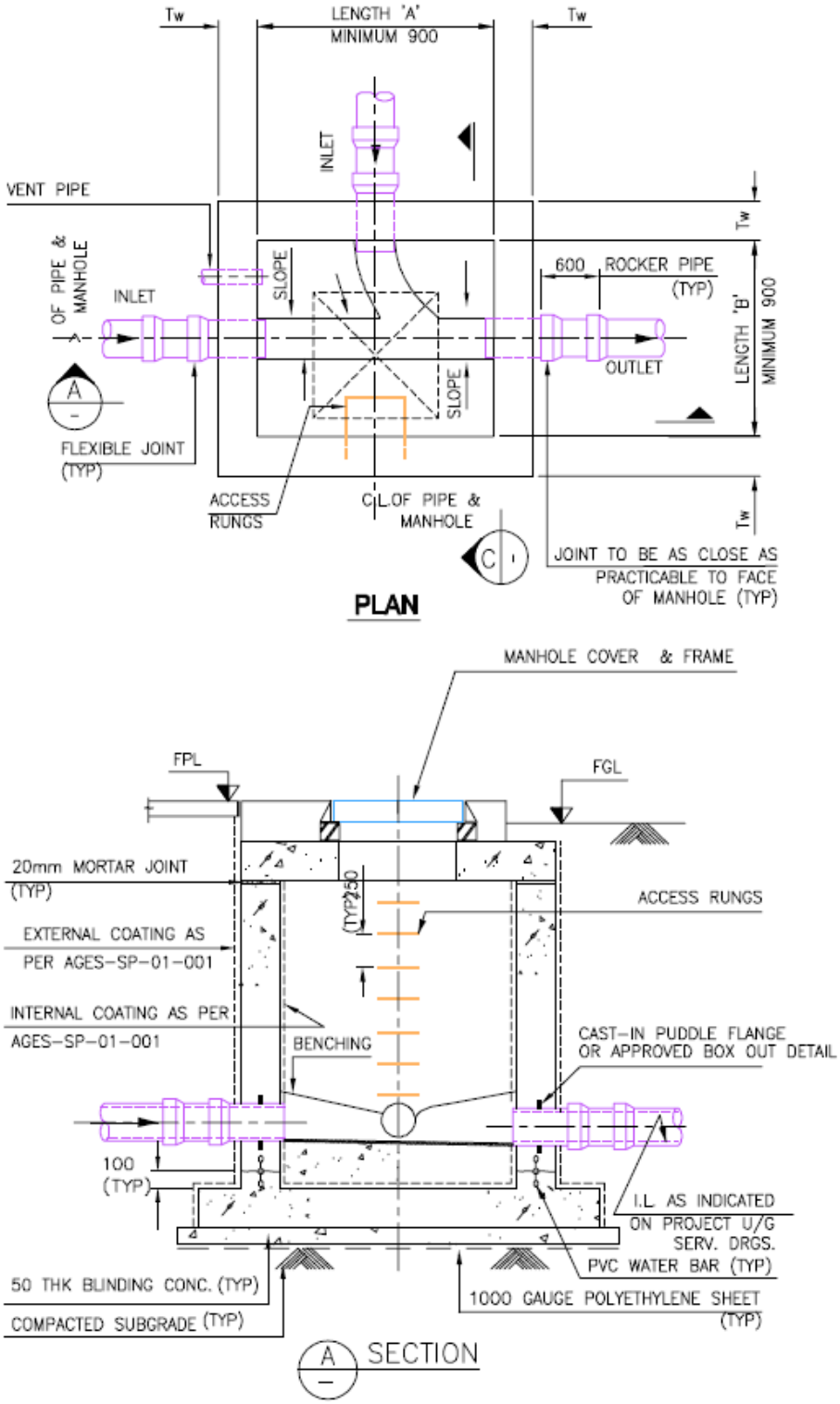
Figure 12 Typical Detail: Sanitary Sewer Manhole



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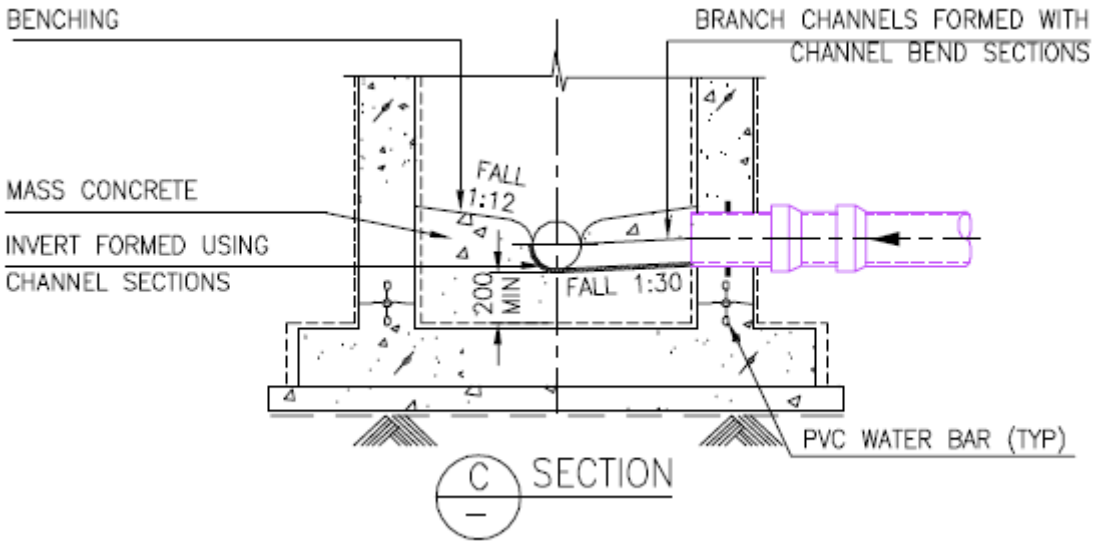

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Figure 13 Typical Detail: Storm Sewer (CWS) Manhole

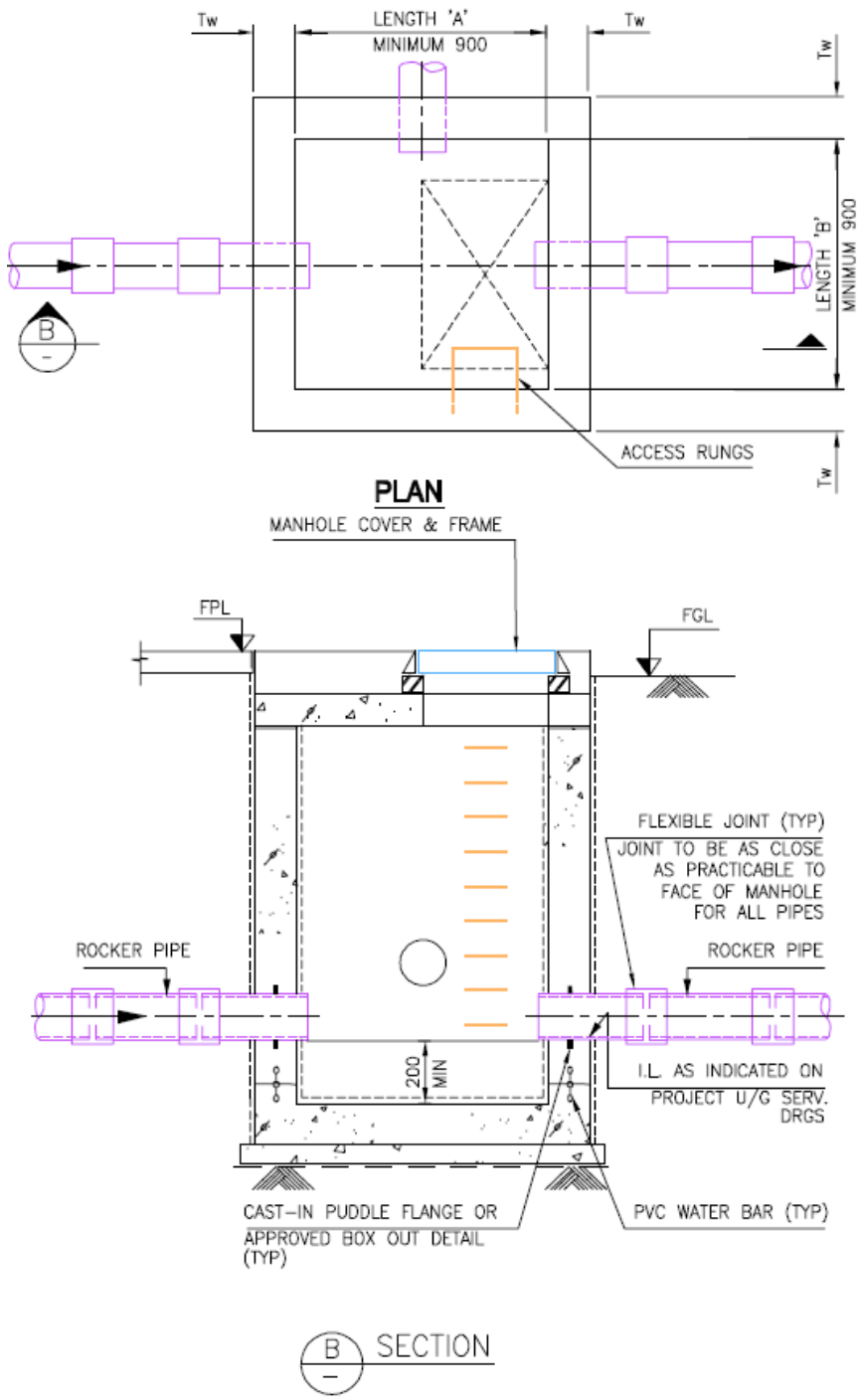

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Figure 14 Typical Detail: AOC/COC Manhole

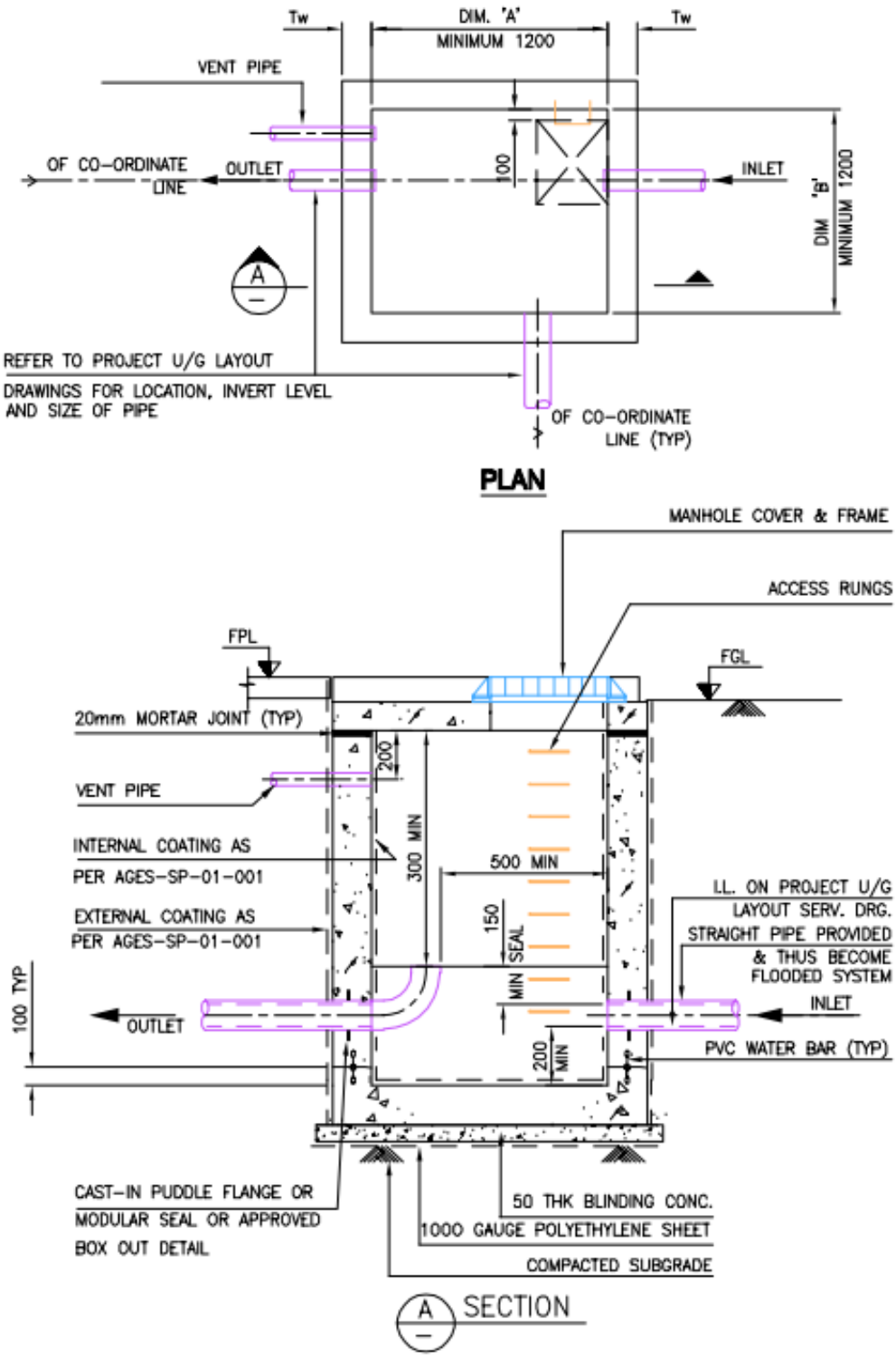

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Figure 15 Typical Detail: Catch Basin

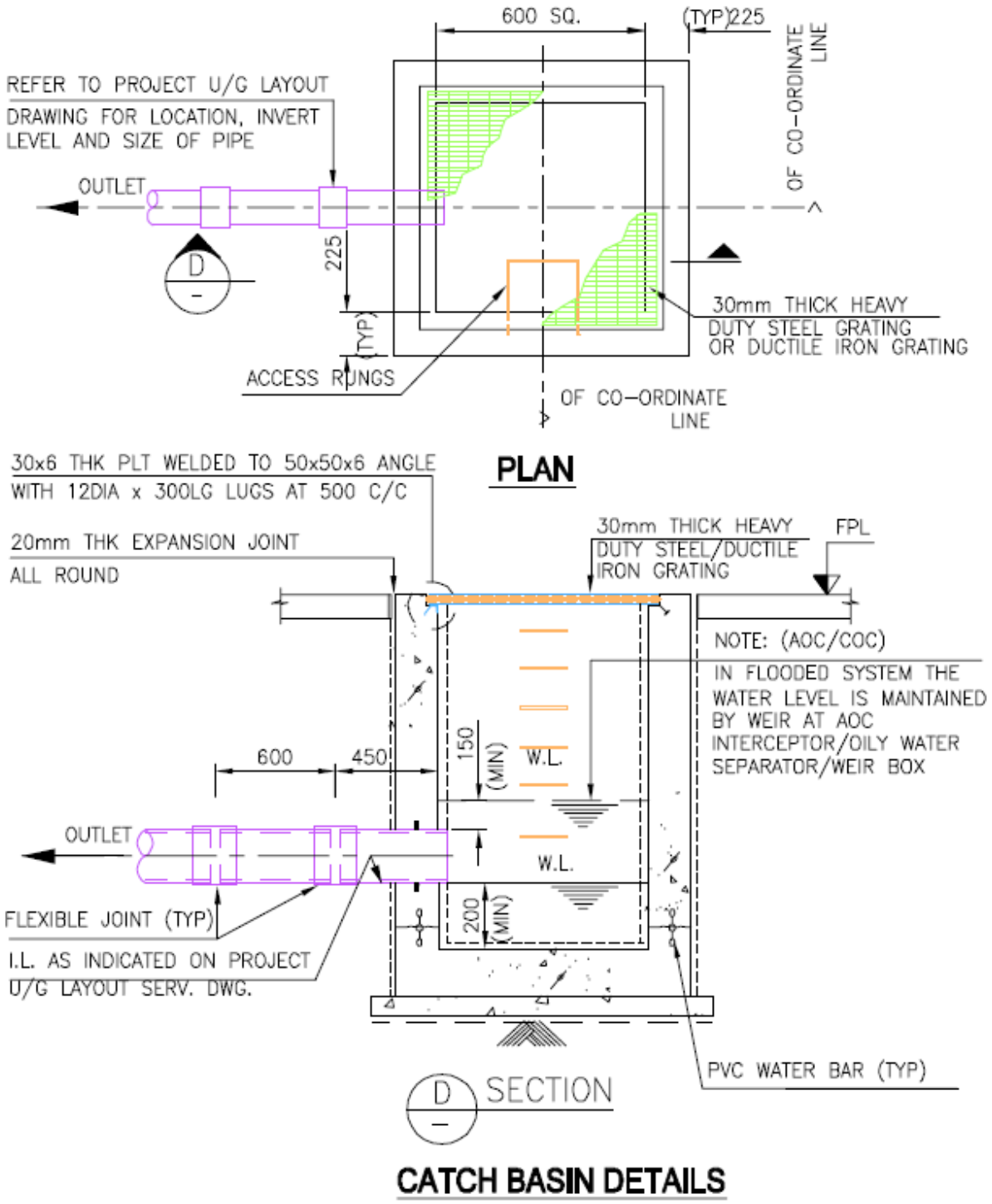

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Figure 16 Typical AOC Weir Box

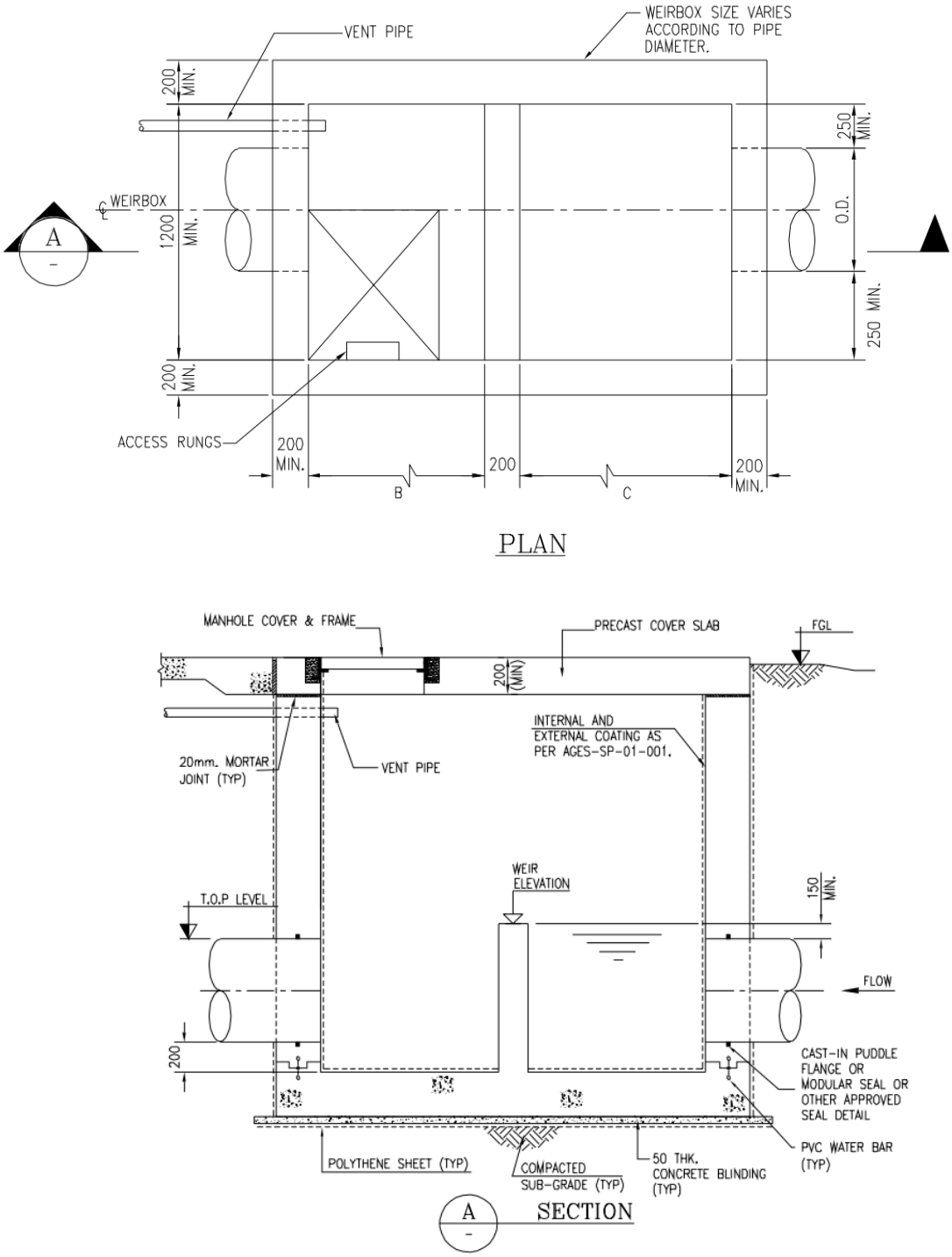

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Figure 17 Typical Detail: AOC / COC Manhole with Stainless Steel Weir Box

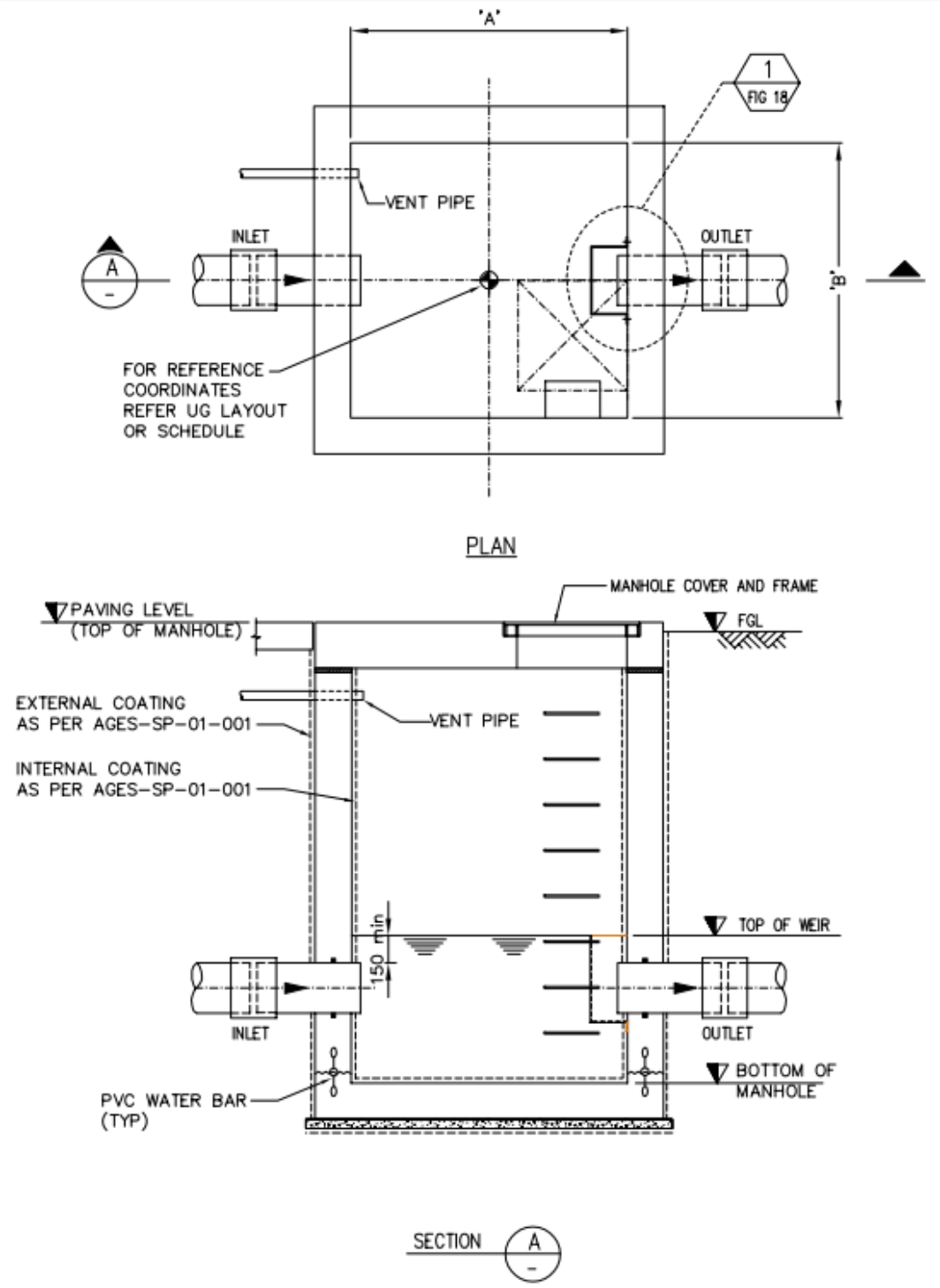

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Figure 18 Stainless Steel Weir Box details

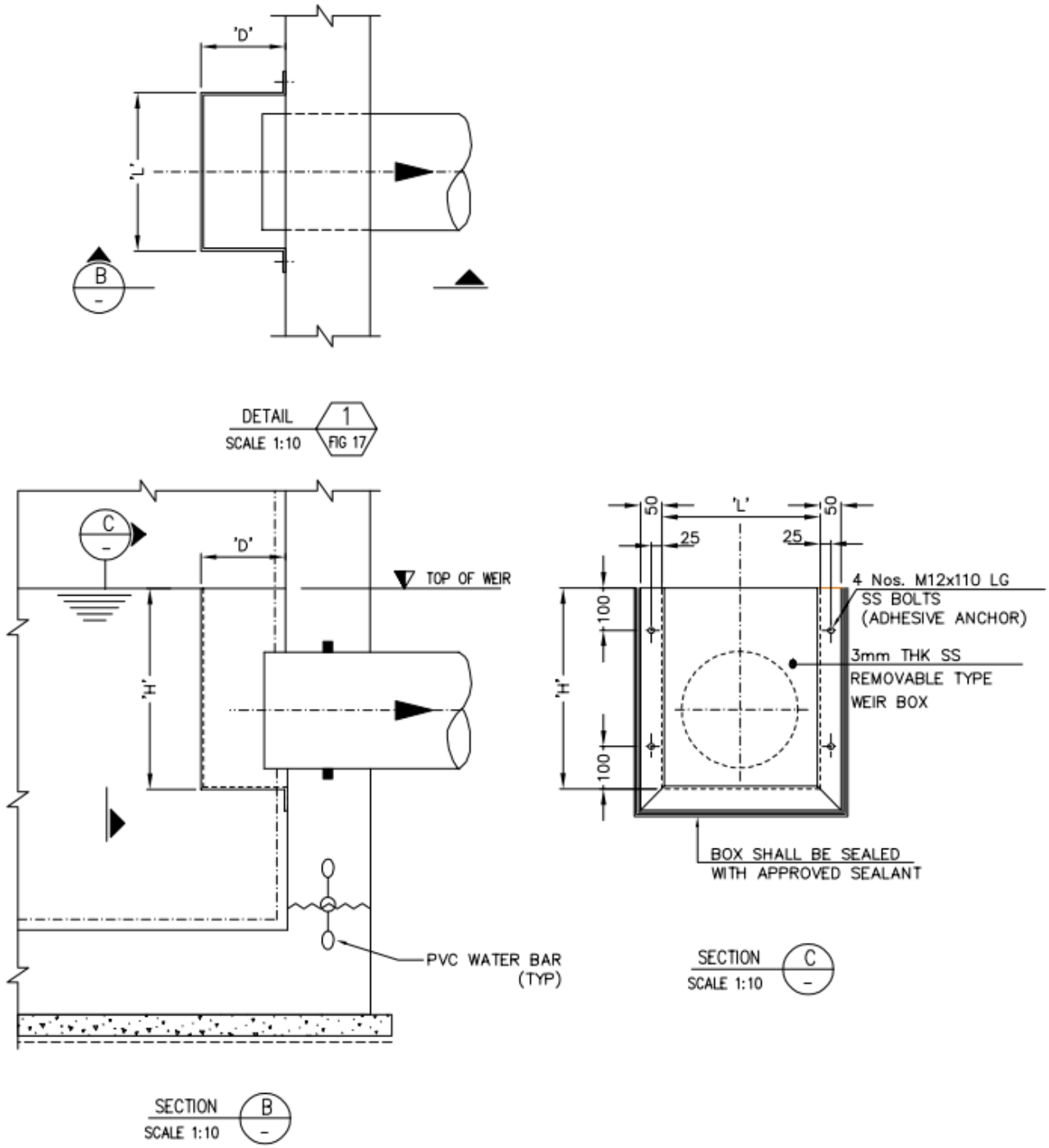

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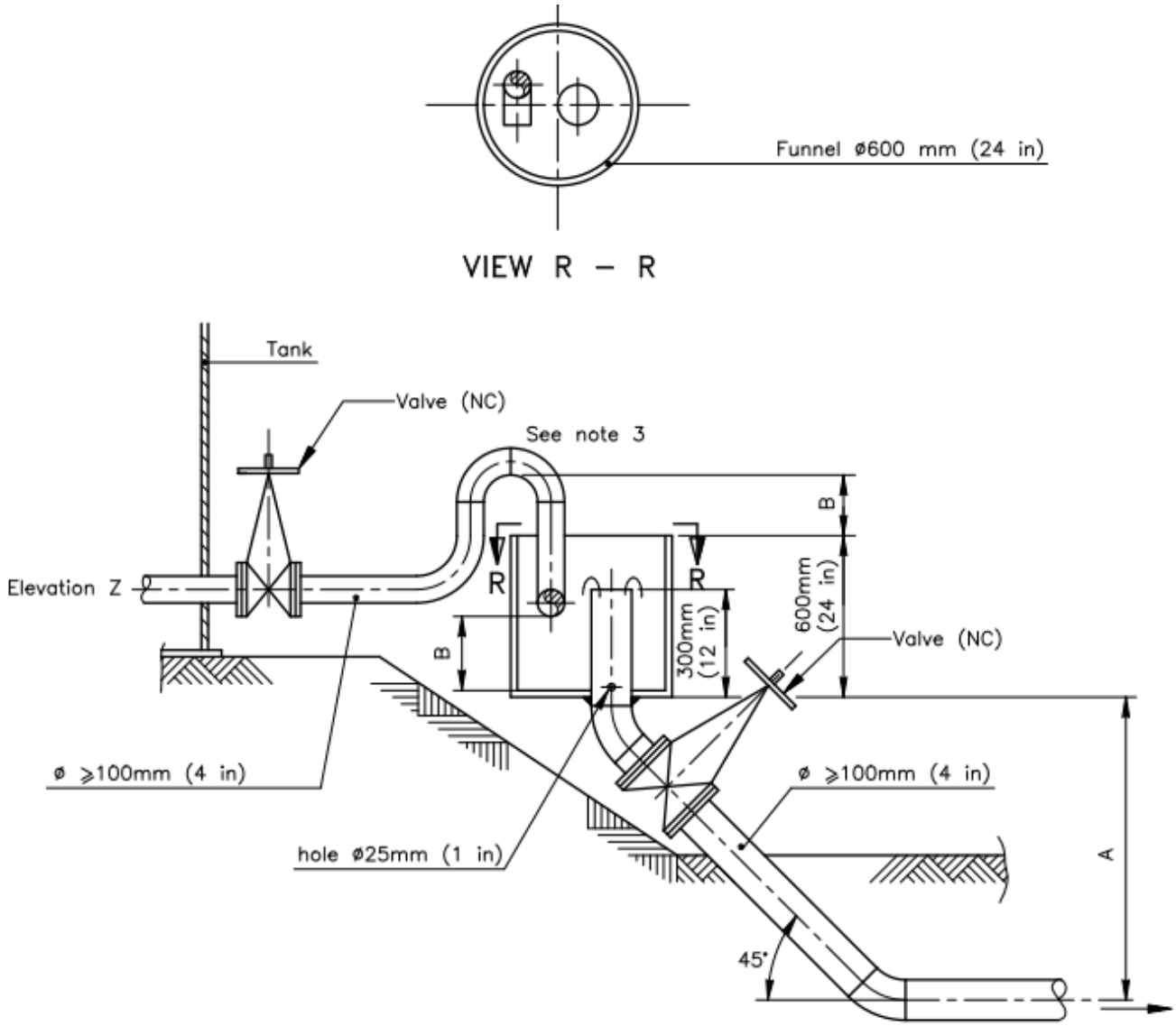
Figure 19 Typical Tank Bottom Drain Funnel



NOTES:

1. Dimension 'A' is dependent upon pressure drop in system.
2. Dimension 'B' is dependent upon maximum expected settlement of tank and height of tank foundation.
3. This turned up elbow is required if the minimum operable control level inside the tank is higher than the elevation Z. If the minimum operable control level inside the tank is lower than the elevation Z the line on the interior of the tank is turned down to maintain a liquid seal between inside and outside the tank if the valve is accidentally left unattended.

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Figure 20 Typical Handwheel Operated Rotating Trough Skimmer

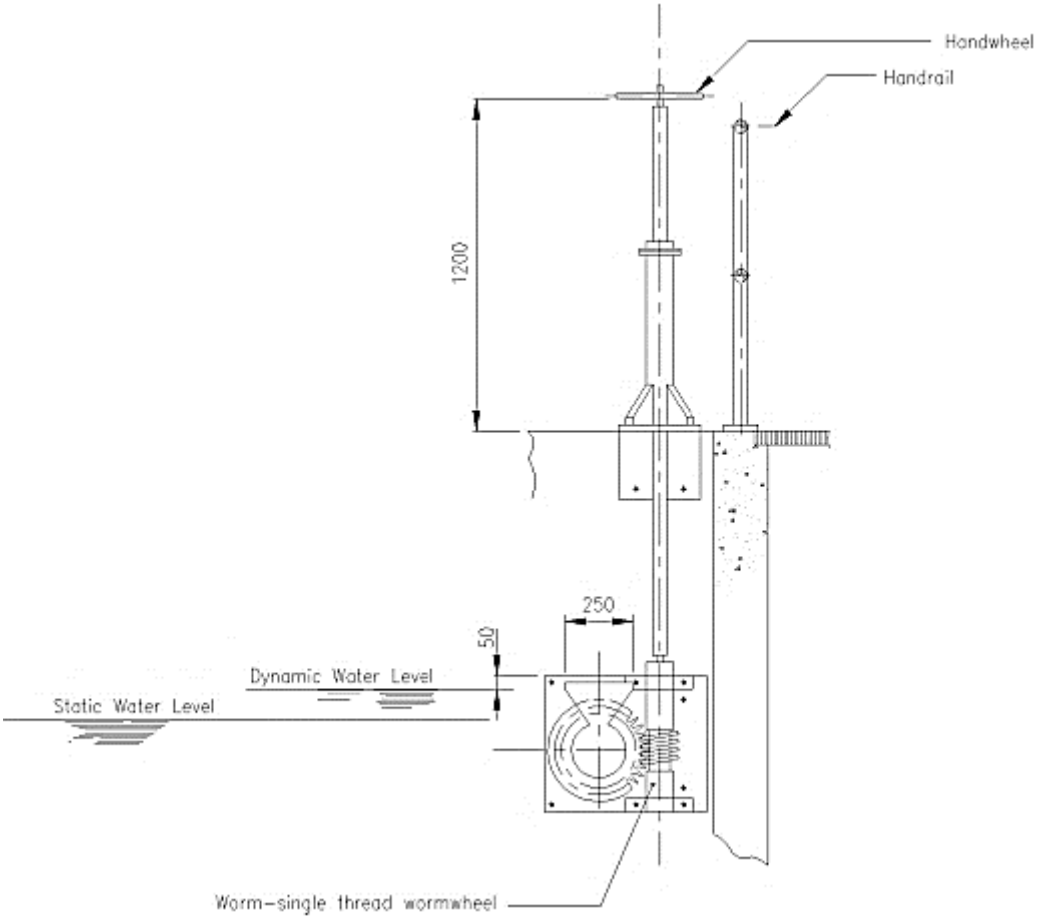

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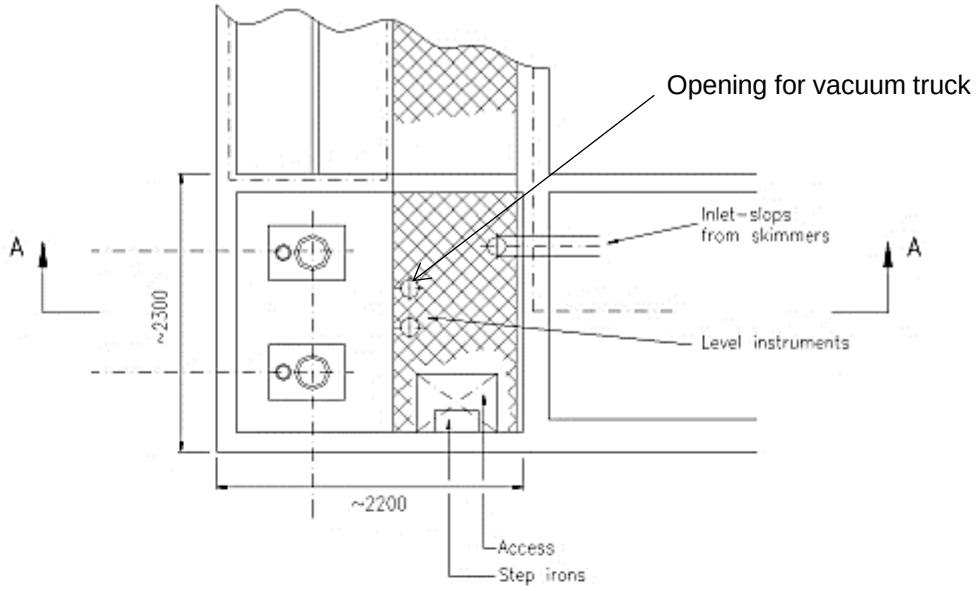
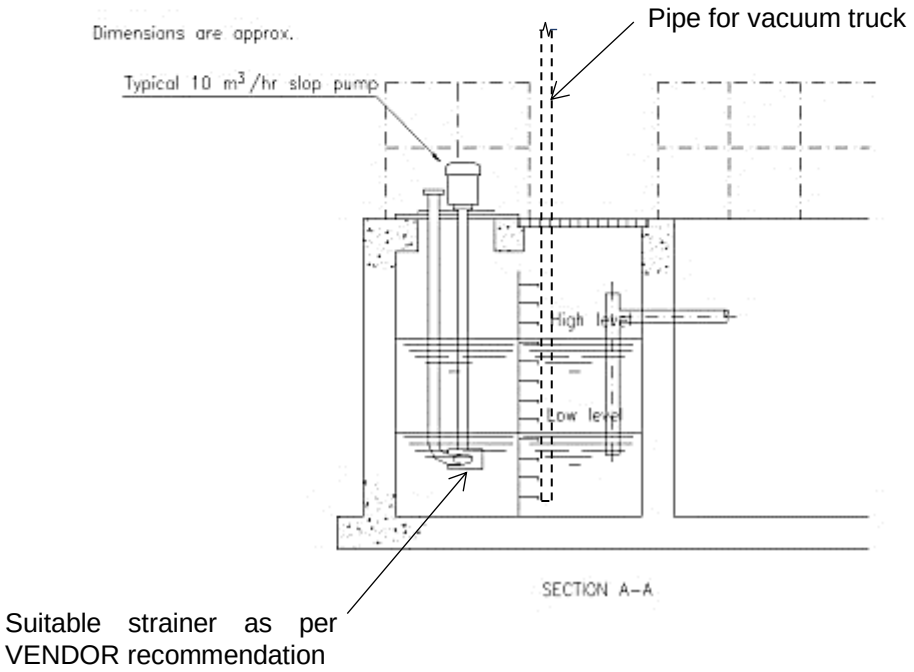
Figure 21 Typical Arrangement Vertical Centrifugal Pumps at Oil Sump





Note: - Number of pumps shown are indicative only and shall depend on collection / evacuation rates.

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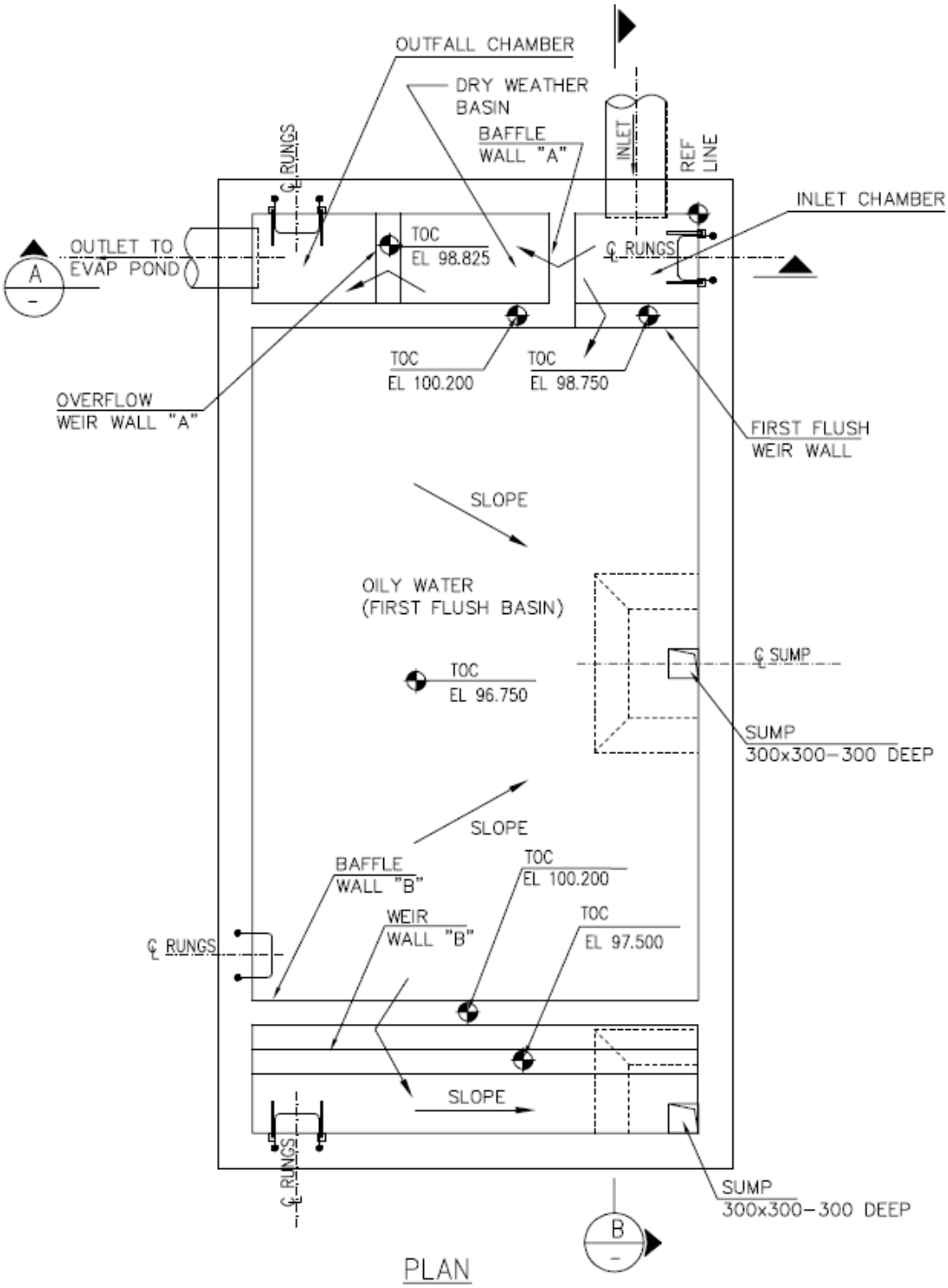
Figure 22 AOC First Flush Interceptor 1 of 2



Note: - The Elevation (TOC, etc) shown are indicative and are for guidance/reference purpose only.

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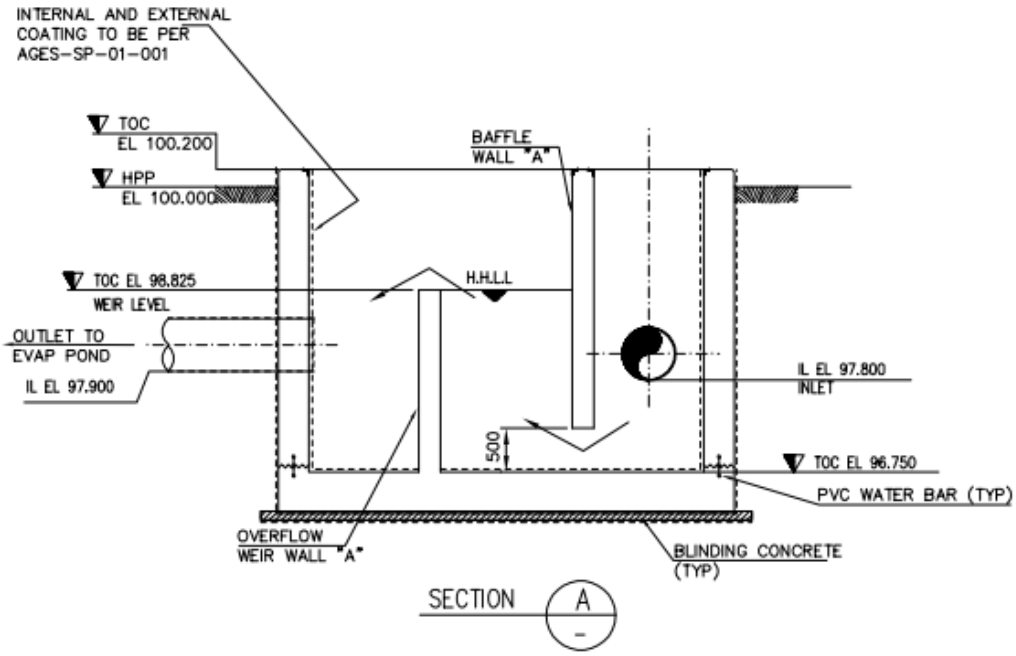
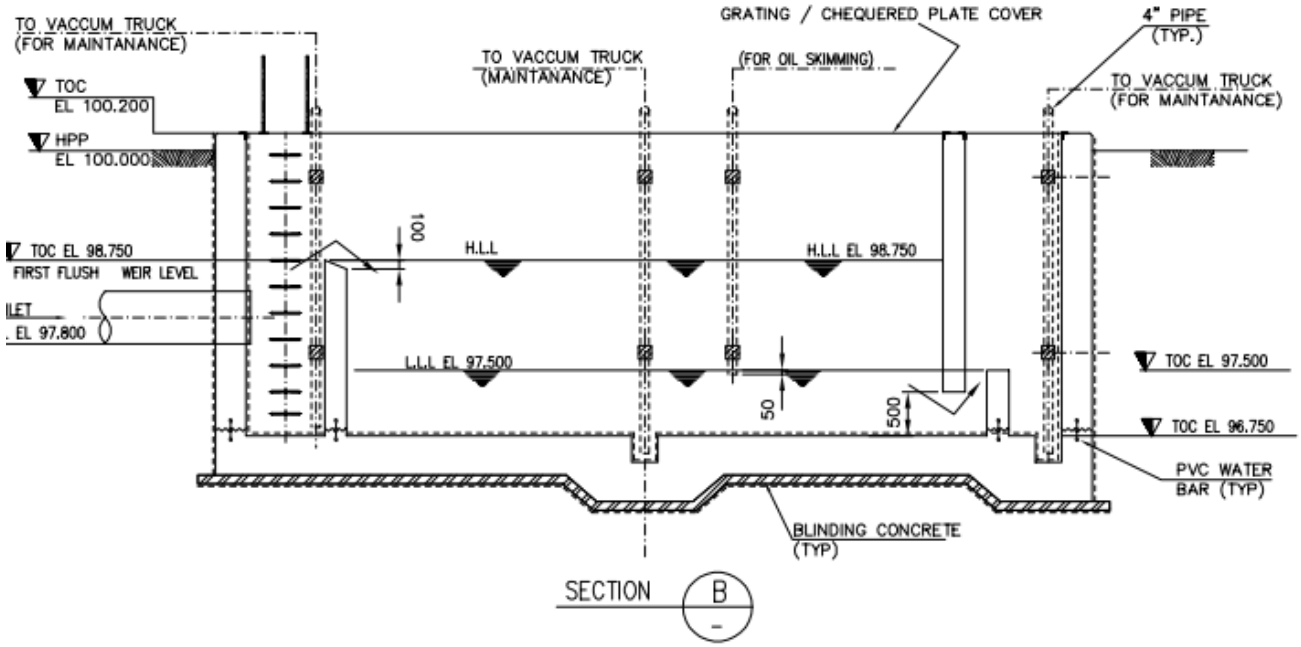
Figure 23 AOC First Flush Interceptor 2 of 2





Note: - The Elevation (TOC, etc) shown are indicative and are for guidance/reference purpose only.

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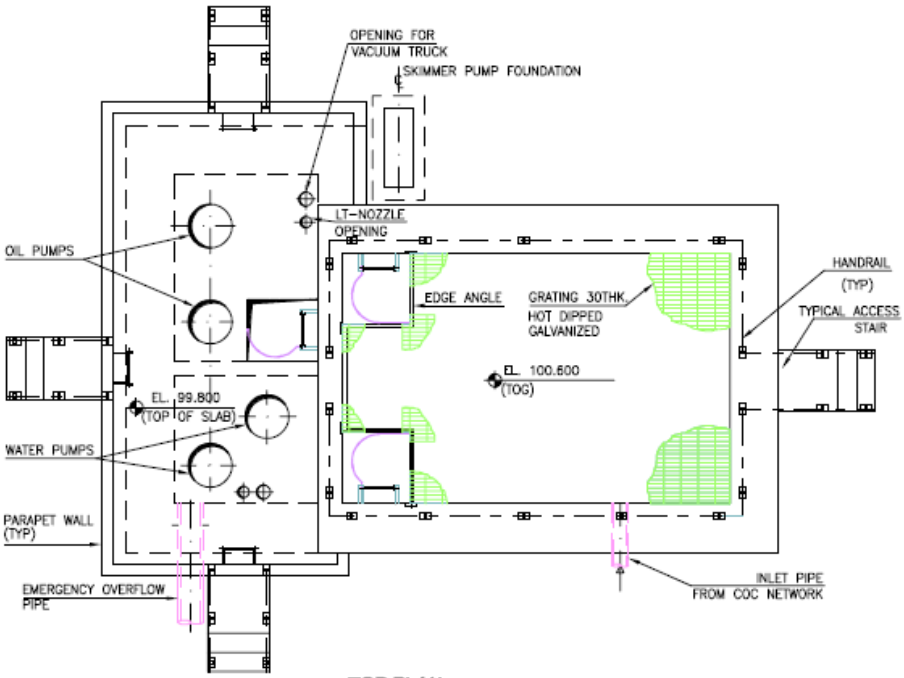
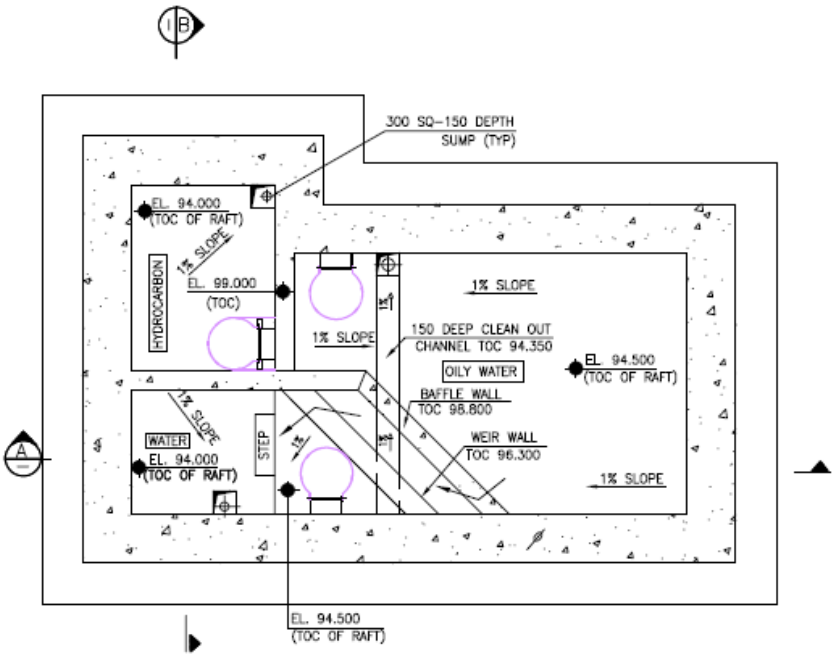
Figure 24 Typical Oily Water Separator 1 of 2



TOP PLAN



BOTTOM PLAN

Note: - The Elevation (TOC, etc) shown are indicative and are for guidance/reference purpose only.

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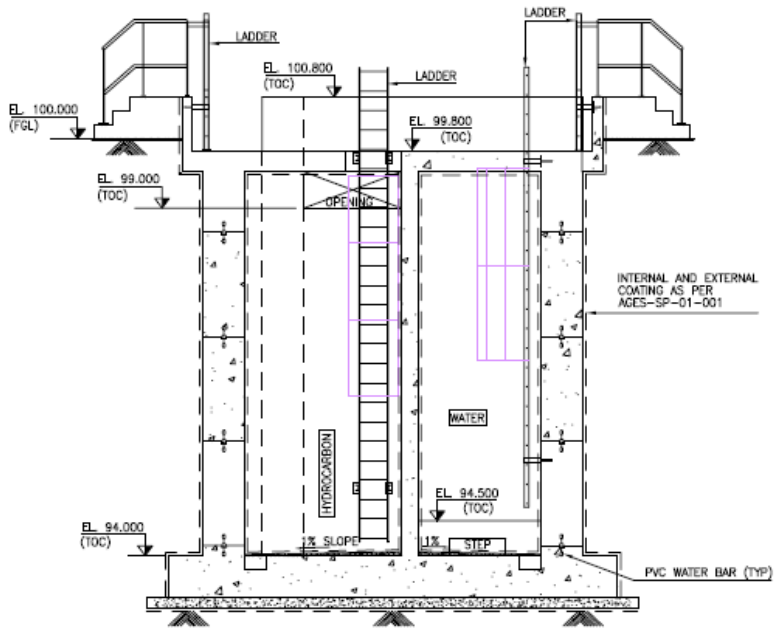
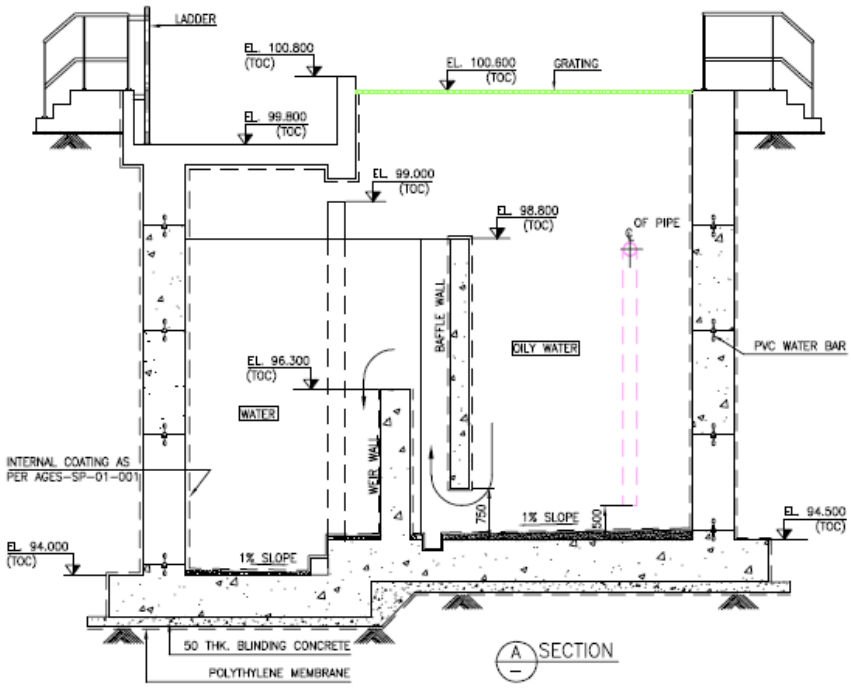
Figure 25 Typical Oily Water Separator 2 of 2



B SECTION



A SECTION

Note: - The Elevation (TOC, etc) shown are indicative and are for guidance/reference purpose only.

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APPENDIX A2. SPECIFIC REQUIREMENTS FOR ADNOC ONSHORE

A2.1. The AOC First Flush Interceptor (AOC Retention basin)

a. The AOC First Flush Interceptor shall be sized to retain greater of:

   i. 30 minute of firefighting water run-off, or

   ii. one hour of the design storm water run-off collected via the AOC drainage system

b. Beyond specified duration or peak flow, the overflow will be diverted to evaporation pond.

A2.2. Evaporation Pond

a. The Evaporation Pond shall be provided with an inlet facility (Inlet Oil Separator) which shall be provided with a baffled outlet, a V-notch weir which is designed to clear free floating oil and facilities to split the flows between the two halves of the Evaporation Pond. The inlet facility shall be provided with narrow weiring penstocks which shall cause the level to rise when the pump is operating such that floating oil is automatically skimmed off via a V-notch weir. The residence time of the inlet facility shall be a minimum of one hour of the capacity of the pump. Skimmed oil shall be returned to the AOC First Flush Interceptor (AOC Retention basin) via the AOC drainage system.

b. Beyond specified peak flow, the overflow will be diverted to desert / low lying area.

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APPENDIX A3. ALTERNATE METHOD FOR CALCULATING CHANNEL AND PIPE SIZE

A3.1. Sizing of Underground Pipe / Channel

a. The size and design capacity of all drainage network shall comply with the Continuity Equation, Manning Formula and Energy Equation including applicable hydraulic losses.

Continuity Equation:

$$Q = AV$$

Where:

Q = flow rate, m^3/s

A = cross-sectional area of flow, m^2

V = flow velocity, m/s, as per Manning's formula as given below.

Manning's formula,

$$V = \frac{R^{2/3} S^{1/2}}{n}$$

Where:

R = Hydraulic radius = area of flow / wetted perimeter, m

S = slope,

n = roughness coefficient of pipe / channel used

A3.2. Roughness Coefficients

a. The roughness coefficients for pipe and channel materials provided in Table A3. 1 can be used in the Manning formula.

Table A3. 1 Roughness Coefficients

Pipe and Open Drain Channel Material	Roughness Coefficient (n)
Ductile Iron Pipe (COATED)	0.013
Concrete, Pre-Cast (RCP)	0.012 - 0.013
Glass Reinforced Epoxy (GRE)	0.011
Poly Vinyl Chloride (PVC) / HDPE	0.010
Lined Channels	n = 0.014 for poured concrete n = 0.017 for shotcrete or gunite n = 0.014 for asphalt n = 0.035 for medium wt. rip-rap
Unlined Channels	n = 0.020 for very fine sand or silt n = 0.025 for sand and gravel n = 0.030 for coarse gravel

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A3.3. Energy Losses

a. Prior to computing the elevation of hydraulic grade line, all applicable energy losses which may occur in pipe/channel runs such as losses at junctions, entrance, exit, bends, manholes, culverts, etc., shall be calculated as per standard industry formula and included in hydraulic design. Unless specified in the project document, the following equations of energy losses can be considered in hydraulic design of drainage system.

A3.4. Pipe/Channel Friction Loss

a. The energy loss due to friction on the wetted perimeter surface of drainage pipe /channel is given by,

$$\mathbf{H_f = S_f L}$$

Where:

H_f = Energy loss due to friction, m

L = Length of pipe/channel of particular run, m

S_f = Friction slope or hydraulic gradient for a particular pipe /channel run.

b. For steady uniform in open channel flow, the friction slope may match with pipe/channel slope flow partially.

c. For the pipe flowing full, the friction slope can be determined by Manning's formula as,

$$S_f = \frac{n^2 Q^2}{A^2 R^{4/3}}$$

A3.5. Outlet / Exit Loss

a. The energy loss at outlet/exit occur due to sudden expansion at outlet and it is function of velocity at outlet. The outlet / exit loss is given by,

$$P = (V_o^2 - V_d^2)/2g$$

Where:

P = Outlet velocity head loss, m

Vo = Average outlet velocity, m/s

Vd = Pipe/Channel velocity at downstream of outlet, m/s

b. When Vd =0, as in reservoir, the exit loss is only velocity head of outlet, i.e. Ho=Vo2/2g, and when a properly aligned pipe flow partially and discharge into a channel with moving water, the exit loss may be reduce to virtually zero.

A3.6. Manhole / Sump Losses and Weir Losses

a. Refer to Section 13.3 of the main body of this specification.

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